

# **ANATOMICAL VARIATION OF PARANASAL AIR SINUSES:**

***A CT STUDY IN SOUTH GUJARAT REGION***

THESIS

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/ GUJARAT UNIVERSITY OF HEALTH SCIENCES**

In partial fulfillment of the requirements  
for the degree of

**DOCTOR OF MEDICINE (M.D.)  
IN ANATOMY**

By

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Under the guidance of

**Dr. [Guide Name]  
Professor & Head  
Department of Anatomy**

**MAY 2026**

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This is to certify that the thesis entitled "ANATOMICAL VARIATION OF PARANASAL AIR SINUSES: A CT STUDY IN SOUTH GUJARAT REGION" submitted by Dr. [Candidate Name] for the partial fulfillment of the requirements for the degree of Doctor of Medicine (M.D.) in Anatomy is a bonafide research work carried out by him/her under my direct supervision and guidance.

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## DECLARATION BY THE CANDIDATE

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I hereby declare that this thesis entitled "ANATOMICAL VARIATION OF PARANASAL AIR SINUSES: A CT STUDY IN SOUTH GUJARAT REGION" is a bonafide and genuine research work carried out by me under the guidance of Dr. [Guide Name], Professor and Head, Department of Anatomy, [Medical College Name], Surat.

I further declare that this thesis or any part of it has not been submitted by me for the award of any other degree or diploma of any other University or Institute previously.

All sources of information used in this thesis have been duly acknowledged. The work presented here is original and authentic to the best of my knowledge and belief.

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---

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**Dr. [Candidate Name]**



## LIST OF ABBREVIATIONS

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|                |                                                            |
|----------------|------------------------------------------------------------|
| <b>PNS</b>     | : Paranasal Sinuses                                        |
| <b>CT</b>      | : Computed Tomography                                      |
| <b>HRCT</b>    | : High Resolution Computed Tomography                      |
| <b>MDCT</b>    | : Multi-Detector Computed Tomography                       |
| <b>MRI</b>     | : Magnetic Resonance Imaging                               |
| <b>FESS</b>    | : Functional Endoscopic Sinus Surgery                      |
| <b>OMC</b>     | : Osteomeatal Complex                                      |
| <b>DNS</b>     | : Deviated Nasal Septum                                    |
| <b>CB</b>      | : Concha Bullosa                                           |
| <b>AN</b>      | : Agger Nasi                                               |
| <b>HC</b>      | : Haller Cell                                              |
| <b>OC</b>      | : Onodi Cell                                               |
| <b>ICA</b>     | : Internal Carotid Artery                                  |
| <b>CSF</b>     | : Cerebrospinal Fluid                                      |
| <b>AP</b>      | : Antero-Posterior                                         |
| <b>ML</b>      | : Medio-Lateral                                            |
| <b>3D</b>      | : Three Dimensional                                        |
| <b>2D</b>      | : Two Dimensional                                          |
| <b>HU</b>      | : Hounsfield Unit                                          |
| <b>kVp</b>     | : Kilovoltage Peak                                         |
| <b>mA</b>      | : Milli-Ampere                                             |
| <b>mm</b>      | : Millimeter                                               |
| <b>cm</b>      | : Centimeter                                               |
| <b>ml</b>      | : Milliliter                                               |
| <b>CRS</b>     | : Chronic Rhinosinusitis                                   |
| <b>ARS</b>     | : Acute Rhinosinusitis                                     |
| <b>ENT</b>     | : Ear, Nose and Throat                                     |
| <b>PMT</b>     | : Paradoxical Middle Turbinate                             |
| <b>AMO</b>     | : Accessory Maxillary Ostium                               |
| <b>IEC</b>     | : Institutional Ethics Committee                           |
| <b>WHO</b>     | : World Health Organization                                |
| <b>USG</b>     | : Ultrasonography                                          |
| <b>PA</b>      | : Postero-Anterior                                         |
| <b>AAO-HNS</b> | : American Academy of Otolaryngology Head and Neck Surgery |
| <b>EPOS</b>    | : European Position Paper on Rhinosinusitis                |
| <b>SD</b>      | : Standard Deviation                                       |

|            |                        |
|------------|------------------------|
| <b>CI</b>  | : Confidence Interval  |
| <b>M:F</b> | : Male to Female Ratio |
| <b>Y</b>   | : Years                |
| <b>No.</b> | : Number               |
| <b>%</b>   | : Percentage           |
| <b>Lt</b>  | : Left                 |
| <b>Rt</b>  | : Right                |
| <b>B/L</b> | : Bilateral            |
| <b>U/L</b> | : Unilateral           |

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## ABSTRACT

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### **Background:**

The paranasal sinuses (PNS) are a complex group of air-filled cavities within the bones of the skull and face that exhibit considerable anatomical variation. Knowledge of these variations is paramount for safe and effective Functional Endoscopic Sinus Surgery (FESS), prevention of intra-operative complications, and management of chronic rhinosinusitis. Variations may be influenced by ethnicity, geographic location, and genetic factors, necessitating regional studies. Limited data exists from the South Gujarat region of India regarding paranasal sinus anatomical variations.

### **Aims and Objectives:**

The present study was undertaken to evaluate the prevalence and pattern of anatomical variations of paranasal air sinuses in the South Gujarat population using Computed Tomography (CT). Specific objectives included: (1) To document various anatomical variations of paranasal sinuses, (2) To determine their prevalence in study population, (3) To assess gender and age-related differences, (4) To compare findings with existing Indian and international literature, (5) To establish baseline regional data useful for surgical planning.

### **Materials and Methods:**

This was a hospital-based observational, descriptive, cross-sectional study conducted in the Department of Anatomy in collaboration with the Department of Radiology at [Medical College Name], Surat, over a period of 18 months (November 2024 to April 2026). A total of 300 CT paranasal sinus scans of patients aged 18-65 years were analyzed. Inclusion criteria comprised individuals from South Gujarat region with technically adequate CT scans. Exclusion criteria included patients with previous sinus surgery, sinonasal tumors, facial trauma, or congenital anomalies. CT scans were performed using a 128-slice MDCT scanner with 0.625 mm slice thickness in axial, coronal, and sagittal planes. Each scan was systematically evaluated for: nasal septum deviation, concha bullosa, paradoxical middle turbinate, Agger nasi cells, Haller cells, Onodi cells, frontal cells (Kuhn classification), maxillary sinus hypoplasia, Underwood's septa, accessory maxillary ostium, sphenoid pneumatization types (Hamberger), Keros classification of olfactory fossa, internal carotid artery dehiscence, and optic nerve protrusion. Data was analyzed using SPSS version 25.0 and chi-square test was used for comparison;  $p < 0.05$  was considered statistically significant.

### **Results:**

Of 300 subjects, 168 (56%) were males and 132 (44%) were females, with a mean age of  $38.4 \pm 12.6$  years. Deviated nasal septum was the most common variation, observed in 218 (72.7%) subjects, with right-sided deviation in 56% of cases. Concha bullosa was present in 99 (33%)

subjects, predominantly of the lamellar type (52%). Paradoxical middle turbinate was observed in 14% of cases. Agger nasi cells were almost universal (94%). Haller cells were noted in 17.3%, while Onodi cells were present in 11.7% of subjects. Maxillary sinus hypoplasia occurred in 4%, Underwood's septa in 28.3%, and accessory maxillary ostium in 19% of cases. Sphenoid sinus pneumatization showed sellar type as predominant (78%). Keros Type II was the most common olfactory fossa depth (68%), with Type III in 17%. Internal carotid artery dehiscence was observed in 8% of subjects. Statistical analysis revealed significant gender-related differences in some variations ( $p < 0.05$ ).

**Conclusion:**

The present study provides valuable baseline data on the prevalence and pattern of paranasal sinus anatomical variations in the South Gujarat population. The prevalence of most variations was comparable to other Indian studies but showed certain regional differences when compared to Western literature. Pre-operative CT evaluation of these variations is crucial for safe FESS and to minimize complications. The high prevalence of variations such as Keros Type III (17%) and ICA dehiscence (8%) emphasizes the need for meticulous pre-operative planning. This study contributes to the growing body of regional Indian literature on paranasal sinus anatomy and serves as a reference for surgeons, radiologists, and anatomists practicing in this region.

**Keywords:**

Paranasal sinuses, Anatomical variations, Computed Tomography, FESS, South Gujarat, Concha bullosa, Deviated nasal septum, Keros classification, Onodi cells, Haller cells.

# CHAPTER 1: INTRODUCTION

---

The paranasal sinuses (PNS) represent one of the most intricate and clinically significant anatomical regions of the human body. These air-filled cavities, situated within the bones of the skull and face, communicate with the nasal cavity through small apertures called ostia. The four pairs of paranasal sinuses comprise the maxillary, frontal, ethmoid, and sphenoid sinuses, each named after the bone in which they reside. Despite their seemingly simple description, the paranasal sinuses exhibit extraordinary anatomical complexity and variation, making them a subject of continued interest to anatomists, radiologists, otolaryngologists, and neurosurgeons alike.

The phylogenetic origin and physiological purpose of paranasal sinuses have been debated for centuries. Various theories have been proposed regarding their function, including reduction of skull weight, voice resonance, humidification and warming of inspired air, thermal insulation of brain, immunological defense, secretion of mucus, and shock absorption during head trauma. Although no single theory satisfactorily explains all aspects, the cumulative evidence suggests that paranasal sinuses serve multiple essential functions in human physiology.

From a clinical perspective, the paranasal sinuses assume immense importance due to their proximity to vital structures including the orbit, optic nerve, internal carotid artery, cavernous sinus, anterior cranial fossa, and the brain. The thin bony walls separating these sinuses from neurovascular structures, combined with the highly variable anatomy, make surgical interventions in this region particularly challenging and potentially hazardous.

The advent of Functional Endoscopic Sinus Surgery (FESS), pioneered by Messerklinger in the 1970s and further developed by Stammberger, Kennedy, Wigand, and others in the 1980s, revolutionized the management of chronic rhinosinusitis and other sinonasal pathologies. FESS relies on detailed knowledge of paranasal sinus anatomy and the ability to recognize anatomical variations on pre-operative imaging. Failure to identify these variations can result in serious complications including cerebrospinal fluid leak, orbital injury, optic nerve damage, internal carotid artery injury, and even death.

## 1.1 Anatomical Variations: A Brief Overview

Anatomical variations of the paranasal sinuses encompass a wide spectrum of morphological differences that deviate from the textbook description but are not pathological. These variations are believed to result from the complex embryological development of the sinuses, which involves multiple invaginations and pneumatization of various bones during fetal and postnatal life. Genetic factors, ethnic background, and environmental influences may contribute to the observed variations across populations.

Common variations affecting the paranasal sinuses include:



- Deviated nasal septum (DNS) and septal spurs
- Concha bullosa - pneumatization of the middle turbinate
- Paradoxical middle turbinate
- Agger nasi cells - the most anterior ethmoid cells
- Haller cells - infraorbital ethmoid cells
- Onodi cells - sphenoethmoidal cells with optic nerve relationship
- Variations in uncinate process attachment
- Frontal recess cells (Kuhn classification I-IV)
- Maxillary sinus hypoplasia or aplasia
- Underwood's septa within the maxillary sinus
- Accessory maxillary ostia
- Variations in sphenoid sinus pneumatization (conchal, presellar, sellar)
- Variations in olfactory fossa depth (Keros Types I-III)
- Internal carotid artery dehiscence in sphenoid sinus
- Optic nerve dehiscence or protrusion
- Frontal sinus hypoplasia, aplasia, or asymmetry
- Cribriform plate asymmetry

## **1.2 Importance of Computed Tomography in Sinus Evaluation**

Computed Tomography (CT) has emerged as the gold standard imaging modality for evaluating paranasal sinus anatomy. The superior bone detail, multiplanar reconstruction capability, and ability to identify subtle anatomical variations make CT indispensable for both diagnostic and surgical planning purposes. The introduction of multidetector CT (MDCT) technology has further enhanced image quality with thinner slices (0.5-1 mm), reduced acquisition time, and lower radiation doses.

CT scan of paranasal sinuses provides comprehensive anatomical information in three orthogonal planes:

- Axial sections: useful for evaluating the relationship of sinuses to the skull base, orbital walls, and internal carotid artery
- Coronal sections: considered the most informative plane for sinus evaluation, providing a clear view of the osteomeatal complex, ethmoid roof, lamina papyracea, and turbinates
- Sagittal sections: helpful in evaluating frontal recess anatomy, sphenoid sinus, and Onodi cells

The Lund-Mackay scoring system, introduced in 1993, provides a standardized method for quantifying the severity of sinonasal disease on CT scans. This scoring system is widely used in research and clinical practice to assess outcomes of medical and surgical management.

## **1.3 Functional Endoscopic Sinus Surgery (FESS)**

FESS represents a paradigm shift in sinus surgery, moving from radical procedures with extensive

bone removal to minimally invasive techniques that preserve normal mucosa and restore natural mucociliary clearance. The fundamental principle of FESS is to address the underlying pathophysiology of sinus disease by ensuring patent ostia and adequate ventilation of the affected sinuses.

Pre-operative CT evaluation forms the cornerstone of safe FESS. The CHECKLIST mnemonic by Stankiewicz outlines critical anatomical points that must be reviewed before any sinus surgery: Cribriform plate asymmetry, Height of the ethmoid roof (Keros classification), Ethmoid arteries, Cells (Onodi, Haller, Agger nasi), Klinoid recess, Lamina papyracea integrity, Internal carotid artery position, Sphenoid sinus pneumatization, and Tumor or pathology assessment.

## **1.4 Population-based and Regional Studies**

Significant variations have been reported in the prevalence of paranasal sinus anatomical variations across different ethnic groups and geographic regions. Studies from Western populations (Europe, North America), Asian populations (Korean, Japanese, Chinese), Middle Eastern populations, and Indian populations have yielded varying prevalence rates for almost every variation studied.

Within India, regional studies from North India (Delhi, Punjab), Central India (Madhya Pradesh, Chhattisgarh), Western India (Maharashtra, Gujarat), Eastern India (West Bengal, Odisha), and Southern India (Karnataka, Tamil Nadu, Kerala, Andhra Pradesh) have demonstrated regional variations attributable to genetic, environmental, and anthropometric differences.

Despite the extensive literature on paranasal sinus anatomical variations, comprehensive data specifically from the South Gujarat region remains limited. South Gujarat, comprising districts of Surat, Navsari, Valsad, Tapi, Bharuch, Narmada, Dang, and surrounding areas, represents a unique demographic mix with significant population diversity. The need for region-specific data is particularly important for surgeons practicing in this region to optimize pre-operative planning and surgical outcomes.

## **1.5 Rationale for the Present Study**

The present study was conceived with the following considerations:

1. Limited data exists specifically from the South Gujarat region regarding paranasal sinus anatomical variations on CT.
2. The increasing utilization of FESS in the management of sinonasal disorders necessitates regional anatomical baseline data.
3. Pre-operative recognition of anatomical variations is crucial for preventing surgical complications and ensuring optimal surgical outcomes.
4. Regional variations in anatomy may exist due to genetic, ethnic, and environmental factors specific to the population studied.
5. Comprehensive documentation of these variations contributes to the growing body of Indian anatomical literature.

6. Such data may aid in establishing population-specific norms for anatomical research, forensic identification, and surgical training.
7. Understanding the prevalence and patterns of variations may help in correlating them with the regional incidence of chronic rhinosinusitis and other sinonasal disorders.

## **1.6 Significance of the Study**

This study aims to provide comprehensive, region-specific data on paranasal sinus anatomical variations that will be valuable for:

- Otolaryngologists and rhinologists planning FESS procedures
- Radiologists interpreting paranasal sinus CT scans
- Anatomists teaching and researching paranasal sinus anatomy
- Neurosurgeons performing endoscopic skull base surgery
- Forensic experts using paranasal sinus features for identification
- Postgraduate students in Anatomy, Radiology, and Otolaryngology
- Healthcare planners and policy makers in regional health systems

By establishing baseline regional data and comparing it with existing Indian and international studies, this research contributes to the understanding of how genetic, ethnic, and environmental factors may influence paranasal sinus anatomy. The findings are expected to enhance patient safety in sinonasal surgery, improve diagnostic accuracy, and stimulate further research in this important field of anatomical and clinical study.

## CHAPTER 2: AIMS AND OBJECTIVES

---

### 2.1 Aim

To study the prevalence and pattern of anatomical variations of paranasal air sinuses in the population of South Gujarat region using Computed Tomography (CT) imaging, and to correlate these variations with demographic parameters such as age and gender.

### 2.2 Primary Objectives

1. To document the various anatomical variations of paranasal air sinuses observed on CT scans of patients from South Gujarat region.
2. To determine the prevalence of common anatomical variations including:
  - Deviated Nasal Septum (DNS)
  - Concha bullosa (lamellar, bulbous, extensive types)
  - Paradoxical middle turbinate
  - Agger nasi cells
  - Haller cells (infraorbital ethmoid cells)
  - Onodi cells (sphenoethmoidal cells)
  - Frontal recess cells (Kuhn classification)
  - Maxillary sinus hypoplasia/aplasia
  - Underwood's septa in maxillary sinus
  - Accessory maxillary ostium
  - Sphenoid sinus pneumatization patterns (Hamberger classification)
  - Sphenoid sinus septation patterns
  - Olfactory fossa depth (Keros classification)
  - Internal carotid artery (ICA) dehiscence
  - Optic nerve protrusion/dehiscence
  - Frontal sinus aplasia/hypoplasia/asymmetry
  - Cribriform plate asymmetry
  - Uncinate process variations

### 2.3 Secondary Objectives

1. To assess the laterality (right/left/bilateral) of various anatomical variations.
2. To evaluate gender-related differences in the prevalence of paranasal sinus variations.
3. To analyze age-related distribution of these variations across different age groups.

4. To compare the findings of the present study with previously published Indian and international studies.
5. To establish region-specific baseline data for paranasal sinus anatomical variations in South Gujarat population.
6. To identify clinically significant variations that may impact surgical planning in FESS.
7. To document any unusual or rare anatomical variations encountered during the study.

## **2.4 Hypothesis**

Null Hypothesis (H0): There is no significant difference in the prevalence of anatomical variations of paranasal sinuses in South Gujarat population compared to other Indian and international populations.

Alternative Hypothesis (H1): There exists a significant difference in the prevalence of anatomical variations of paranasal sinuses in South Gujarat population compared to other populations, reflecting regional and ethnic anatomical diversity.

## **2.5 Research Questions**

The present study seeks to answer the following research questions:

1. What is the prevalence of various anatomical variations of paranasal sinuses in the South Gujarat population?
2. Are there significant gender-related differences in these variations?
3. Do age-related changes affect the prevalence of these variations?
4. How do the findings compare with other regional Indian populations and international studies?
5. Which anatomical variations are most clinically significant for surgical planning?
6. Are there any unique or rare variations specific to the South Gujarat population?
7. How do these variations correlate with chronic rhinosinusitis patterns in this region?

## **2.6 Expected Outcomes**

Upon completion of this study, the following outcomes are anticipated:

- Comprehensive documentation of paranasal sinus anatomical variations in South Gujarat population
- Establishment of regional baseline prevalence data for various variations
- Identification of clinically significant variations requiring special attention during surgery
- Comparative analysis with existing literature highlighting regional differences
- Contribution to the growing body of Indian anatomical literature
- Development of recommendations for pre-operative CT evaluation in this region
- Foundation for future studies on clinical correlations and surgical outcomes

## CHAPTER 3: REVIEW OF LITERATURE

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The paranasal sinuses have been a subject of anatomical, physiological, and clinical investigation for over two millennia. This chapter presents a comprehensive review of the existing literature on paranasal sinus anatomy, embryology, anatomical variations, imaging modalities, and their clinical significance, with particular emphasis on Indian and international studies.

### 3.1 Historical Background

The understanding of paranasal sinus anatomy has evolved significantly over centuries, with contributions from numerous anatomists, physicians, and surgeons. The journey from ancient observations to modern endoscopic techniques represents one of the most fascinating chapters in the history of medicine.

#### 3.1.1 Ancient and Medieval Period

Galen of Pergamon (130-200 AD), the renowned Greek physician of the Roman Empire, was among the first to describe the frontal and maxillary sinuses in his anatomical writings. His descriptions, although limited by the methods of his era, formed the foundation of paranasal sinus anatomy that influenced medical thought for over a millennium.

Berengario da Carpi (1460-1530), an Italian physician, provided some of the earliest anatomical illustrations of skull cavities. His work "Commentaria super Anatomia Mundini" (1521) included references to facial sinuses, although the detailed understanding remained limited.

Leonardo da Vinci (1452-1519) was perhaps the first to provide detailed anatomical drawings of the maxillary sinus. His sketches, dating to approximately 1489, depicted the maxillary antrum with remarkable accuracy for that era. However, his drawings remained largely unknown until much later.

Andreas Vesalius (1514-1564), in his seminal work "De Humani Corporis Fabrica" (1543), provided detailed descriptions of the skull anatomy including the paranasal sinuses. His work marked a turning point from Galenic medicine to direct anatomical observation.

#### 3.1.2 The 17th and 18th Centuries

Nathaniel Highmore (1613-1685), an English surgeon and anatomist, provided the first comprehensive description of the maxillary sinus in his "Corporis Humani Disquisitio Anatomica" published in 1651. He described the sinus as a "large cavern in each cheek" and noted its drainage into the nasal cavity. The maxillary sinus was subsequently named the "Antrum of Highmore" in his honor.

Thomas Willis (1621-1675), better known for his work on cerebral circulation, also contributed to paranasal sinus anatomy with descriptions of the ethmoid and sphenoid sinuses.

Antonio Maria Valsalva (1666-1723) provided detailed descriptions of the frontal sinus and contributed to understanding its drainage patterns. The Valsalva maneuver, used in clinical examination, is named after him.

In the 18th century, William Hunter (1718-1783) and Antonio Scarpa (1752-1832) added significant anatomical details, particularly regarding the nasal cavity and its relationships with the paranasal sinuses.

### **3.1.3 The 19th Century: Foundations of Modern Sinus Anatomy**

The 19th century witnessed remarkable advances in paranasal sinus anatomy. Jakob Hyrtl (1810-1894), the renowned Austrian anatomist, provided comprehensive descriptions of paranasal sinus anatomy in his anatomical textbooks.

Wilhelm Zuckerkandl (1849-1910), a Hungarian-Austrian anatomist, made fundamental contributions to the anatomy of the nasal cavity and paranasal sinuses. His monograph "Normale und pathologische Anatomie der Nasenhöhle" (1882) became a classic reference. Zuckerkandl described the variations in ethmoid air cells and laid the foundation for understanding the complex ethmoid anatomy.

Emil Zuckerkandl, Wilhelm's son, continued his father's work and made additional contributions to nasal and sinus anatomy.

Adolph Onodi (1857-1919), a Hungarian anatomist and otolaryngologist, provided detailed descriptions of the relationship between the posterior ethmoid sinuses and the optic nerve. The Onodi cell, a posterior ethmoid cell that extends superolaterally to the sphenoid sinus and is closely related to the optic nerve, is named after him.

Albert Killian (1860-1921), a German laryngologist, contributed to the understanding of frontal sinus drainage and the nasofrontal recess.

### **3.1.4 The 20th Century: Imaging and Surgical Era**

The discovery of X-rays by Wilhelm Conrad Röntgen in 1895 revolutionized the diagnosis of paranasal sinus disease. Plain radiography became the first imaging modality to evaluate the sinuses, although its limitations in providing detailed anatomical information were soon apparent.

In 1929, Maurice Cottle introduced the concept of the nasal cycle and contributed to the understanding of nasal physiology.

The development of computed tomography by Sir Godfrey Hounsfield in 1972 marked a revolutionary advancement in medical imaging. CT provided unprecedented anatomical detail of the paranasal sinuses and allowed visualization of subtle variations not appreciable on plain radiographs.

In 1978, Walter Messerklinger published his groundbreaking work "Endoscopy of the Nose" which laid the foundation for endoscopic sinus surgery. Messerklinger demonstrated that diseases of the paranasal sinuses originate primarily from the ethmoid sinuses and the osteomeatal complex, fundamentally changing the surgical approach.

Heinz Stammberger (1947-2018), an Austrian otolaryngologist and student of Messerklinger, popularized Functional Endoscopic Sinus Surgery (FESS) globally. His textbook "Functional Endoscopic Sinus Surgery: The Messerklinger Technique" (1991) became the gold standard reference for sinus surgeons worldwide.

David W. Kennedy, an American otolaryngologist, brought FESS to the United States and modified the technique. Along with S. James Zinreich, he introduced systematic CT evaluation for FESS planning. Their seminal paper "Functional Endoscopic Sinus Surgery: Theory and Diagnostic Evaluation" (1985) established the framework for modern sinus surgery.

Wolfgang Wigand contributed significantly to endoscopic sinus surgery techniques and popularized the German approach to FESS.

### **3.1.5 The 21st Century: Advanced Imaging and Image-Guided Surgery**

The 21st century has witnessed remarkable advances in imaging technology and surgical techniques. The introduction of multidetector CT (MDCT) with thin-slice capability (0.5-1 mm) allowed unprecedented visualization of paranasal sinus anatomy. Three-dimensional reconstructions and multiplanar reformations have enhanced surgical planning capabilities.

Image-guided surgical navigation systems, introduced in the late 1990s and refined in the 2000s, have further enhanced the safety of complex endoscopic sinus and skull base surgeries. These systems use pre-operative CT data to provide real-time anatomical localization during surgery.

Cone Beam CT (CBCT), with its lower radiation dose and high spatial resolution, has emerged as an alternative imaging modality for paranasal sinus evaluation, particularly in dental and maxillofacial applications.

The European Position Paper on Rhinosinusitis and Nasal Polyps (EPOS), first published in 2005 and subsequently updated, has provided evidence-based guidelines for the diagnosis and management of rhinosinusitis, with detailed recommendations on imaging.



## **3.2 Embryology of Paranasal Sinuses**

Understanding the embryological development of paranasal sinuses is crucial for comprehending the anatomical variations observed in adults. The paranasal sinuses develop as outgrowths from the lateral nasal wall during fetal life and continue their development through childhood and adolescence.

### **3.2.1 Development of the Nasal Cavity**

The development of the nasal cavity begins during the 4th week of intrauterine life (IUL). The frontonasal prominence, derived from neural crest cells, gives rise to bilateral nasal placodes. By the 5th week, these placodes invaginate to form nasal pits, which deepen to form the primitive nasal cavity by the 6th-7th week.

The choanae (posterior nasal apertures) form during the 7th week through the rupture of the oronasal membrane. The nasal septum develops from the merging frontonasal and medial nasal processes during the 8th week.

The lateral nasal wall develops three turbinates by the 10th week: the inferior, middle, and superior turbinates. A fourth, supreme turbinate, may develop in some individuals. The grooves between these turbinates form the nasal meatuses, which become the drainage pathways for the developing paranasal sinuses.

### **3.2.2 Maxillary Sinus Development**

The maxillary sinus is the first to develop and the largest of the paranasal sinuses. Its development begins around the 65th-70th day of intrauterine life, when the sinus appears as a small invagination of the middle meatus epithelium into the maxillary bone.

Schaeffer (1910) classified maxillary sinus development into three stages:

- Primary pneumatization: From 70th day IUL until birth
- Secondary pneumatization: From birth to 12 years
- Tertiary pneumatization: After permanent dentition, continuing into adolescence

At birth, the maxillary sinus measures approximately 7 mm anteroposterior x 4 mm vertical x 4 mm transverse. It enlarges progressively, reaching adult size between 12 and 18 years of age. The final pneumatization is influenced by the eruption of permanent teeth, particularly the molars.

Anson and McVay (1971) described two distinct phases of growth: rapid growth during the first three years and a second growth spurt corresponding to the eruption of permanent dentition between ages 7-12 years.

In the adult, the maxillary sinus has an average volume of 12-15 ml. Its size and shape can vary considerably, ranging from a small slit-like cavity (hypoplasia) to an enlarged cavity that may extend into adjacent bones.

### **3.2.3 Ethmoid Sinus Development**

The ethmoid sinuses are the most complex and the only paranasal sinuses present at birth. Their development begins as early as the 4th-5th month of intrauterine life. The ethmoid bone develops from multiple ossification centers, and the air cells form by invagination of the middle and superior meatal mucosa into the developing ethmoid bone.

The number of ethmoid air cells varies considerably, ranging from 3 to 18 cells per side. They are traditionally divided into anterior and posterior groups based on their drainage:

- Anterior ethmoid cells: Drain into the middle meatus
- Posterior ethmoid cells: Drain into the superior meatus

The basal lamella of the middle turbinate serves as the anatomical boundary between the anterior and posterior ethmoid cells. The ethmoid sinuses reach their adult dimensions by 12-14 years of age.

Variations in ethmoid pneumatization include:

- Agger nasi cells (most anterior)
- Frontal recess cells (Kuhn types)
- Haller cells (infraorbital)
- Onodi cells (most posterior)
- Supraorbital ethmoid cells

### **3.2.4 Frontal Sinus Development**

The frontal sinus has the most variable and prolonged development among the paranasal sinuses. Unlike other sinuses, the frontal sinus is not present at birth. Its development begins around 1-2 years of age and continues until the second decade of life.

The frontal sinus develops from the upward extension of the anterior ethmoid cells (specifically, the agger nasi cells and frontal recess cells) into the frontal bone. This explains the close anatomical and developmental relationship between the frontal sinus and anterior ethmoid sinuses.

Schaeffer (1916) described the embryological stages of frontal sinus development:

- Birth to 2 years: No frontal sinus
- 2-4 years: Frontal sinus visible at the level of nasion
- 4-8 years: Continued vertical and lateral growth
- 8-15 years: Reaches adult dimensions
- After 15 years: Minor changes in size and shape

The frontal sinus exhibits remarkable individual variation. Approximately 5-15% of individuals show unilateral or bilateral aplasia (absence) of the frontal sinus. The two frontal sinuses are typically asymmetric in 80-85% of individuals due to their independent development.

### **3.2.5 Sphenoid Sinus Development**

The sphenoid sinus is the deepest and most posteriorly located paranasal sinus. Its development is the most prolonged and continues longer than other sinuses.

At birth, the sphenoid bone contains a small cartilaginous structure (sphenoidal concha) without

significant pneumatization. Pneumatization begins around 3-4 years of age and proceeds in three directions:

- Inferiorly toward the pterygoid process
- Posteriorly toward the clivus
- Superiorly toward the sella turcica

Hamberger and colleagues (1961) classified sphenoid sinus pneumatization into three types:

- Conchal type (2-3%): Minimal pneumatization, with solid bone between the sinus and sella turcica
- Presellar type (11-24%): Pneumatization extends posteriorly but does not extend below the sella turcica
- Sellar type (73-86%): Pneumatization extends below and posterior to the sella turcica, often surrounding it

The sphenoid sinus reaches its adult size by 12-14 years, although some pneumatization may continue into adulthood. The septation pattern of the sphenoid sinus is highly variable, with single, multiple, or absent septa.

### **3.2.6 Postnatal Growth and Maturation**

The pattern of postnatal sinus development is unique among facial structures. Unlike most facial structures that follow a relatively uniform growth pattern, the paranasal sinuses show distinct stages of rapid growth interspersed with periods of slower development.

Bell and Hand (1970) demonstrated through serial CT studies that paranasal sinus development is influenced by:

- Hormonal factors (especially during puberty)
- Genetic factors
- Mechanical factors (mastication, breathing)
- Local factors (chronic sinusitis, allergic rhinitis)
- Systemic factors (nutrition, growth disorders)

Park et al. (2010) using volumetric CT analysis showed that maxillary and ethmoid sinuses reach 90% of adult size by 12 years, while frontal and sphenoid sinuses continue to grow until 16-18 years of age.

### 3.3 Detailed Anatomy of Paranasal Sinuses

#### 3.3.1 Maxillary Sinus

The maxillary sinus is the largest of the paranasal sinuses, occupying the body of the maxillary bone. It is pyramidal in shape, with its base directed medially toward the nasal cavity and its apex pointing laterally toward the zygomatic process.

Dimensions and Volume:

- Average anteroposterior diameter: 34 mm (range 23-49 mm)
- Average vertical diameter: 33 mm (range 22-45 mm)
- Average mediolateral width: 25 mm (range 16-40 mm)
- Average volume: 12-15 ml (range 4-35 ml)

Walls of the Maxillary Sinus:

- Anterior wall: Formed by the anterior surface of the maxilla, contains the infraorbital foramen and canal carrying the infraorbital nerve and vessels
- Posterior wall: Formed by the infratemporal surface of the maxilla, related to the pterygopalatine fossa, infratemporal fossa, and the third division of trigeminal nerve
- Superior wall (roof): Forms the floor of the orbit, contains the infraorbital canal; thin bone may transmit infections to orbit
- Inferior wall (floor): Formed by the alveolar process of maxilla, intimately related to the roots of upper premolar and molar teeth
- Medial wall: Forms the lateral wall of nasal cavity, contains the maxillary ostium opening into the middle meatus through the hiatus semilunaris

The maxillary ostium, the natural opening of the maxillary sinus, is located in the upper part of the medial wall. This high position is functionally disadvantageous as it requires active mucociliary transport against gravity for sinus drainage. The ostium typically measures 3-5 mm in diameter and opens into the middle meatus through the ethmoidal infundibulum.

Lining and Innervation:

The maxillary sinus is lined by ciliated pseudostratified columnar epithelium (respiratory mucosa) which is thinner and less vascular than the nasal mucosa. The arterial supply comes from branches of the maxillary artery (infraorbital, posterior superior alveolar, and greater palatine arteries). Venous drainage is via the pterygoid venous plexus and facial vein. Innervation is through the maxillary division (V2) of the trigeminal nerve via the infraorbital, posterior superior alveolar, middle superior alveolar, and anterior superior alveolar nerves.

Clinical Relations:

- Roots of upper teeth (especially second premolar and first molar) may protrude into the sinus floor
- Dental infections can spread to the maxillary sinus (odontogenic sinusitis)

- Tumors of the maxillary sinus may invade the orbit, infratemporal fossa, or oral cavity
- Underwood septa may divide the sinus into compartments

### 3.3.2 Ethmoid Sinuses

The ethmoid sinuses are a complex labyrinth of air cells located within the ethmoid bone, between the nasal cavity (medially) and the orbit (laterally). They are arranged in three rows extending from anterior to posterior.

Anatomical Boundaries of Ethmoid Labyrinth:

- Superior: Cribriform plate of ethmoid (medial) and fovea ethmoidalis of frontal bone (lateral)
- Inferior: Maxillary sinus and middle/superior turbinate
- Medial: Lateral nasal wall (middle and superior turbinates)
- Lateral: Lamina papyracea (separating it from orbit)
- Anterior: Frontal process of maxilla and lacrimal bone
- Posterior: Sphenoid sinus

Classification of Ethmoid Cells:

The ethmoid air cells are traditionally divided into anterior and posterior groups based on their drainage:

- Anterior ethmoid cells: Drain into the middle meatus, including the ethmoid bulla and uncinate process region
- Posterior ethmoid cells: Drain into the superior meatus and may extend into the sphenoid bone (Onodi cells)

Anatomical Variations:

- Agger nasi cells (98% prevalence) - most anterior ethmoid cells
- Frontal recess cells (Kuhn types I-IV)
- Haller cells (10-45%) - infraorbital ethmoid cells
- Onodi cells (8-14%) - posterior ethmoid cells extending to sphenoid
- Supraorbital ethmoid cells (7-15%)
- Suprabullar cells
- Retrobullar cells

Vascular Supply and Innervation:

The ethmoid sinuses receive blood supply from the anterior and posterior ethmoidal arteries (branches of the ophthalmic artery) and sphenopalatine artery (branch of maxillary artery). The anterior ethmoidal artery is a critical surgical landmark, lying within the anterior ethmoidal foramen and traversing the ethmoid roof from the orbit to the anterior cranial fossa. Innervation is by the anterior and posterior ethmoidal nerves (branches of the ophthalmic nerve V1) and orbital branches of the pterygopalatine ganglion.

Clinical Significance:

- Lamina papyracea: Thin medial orbital wall - injury can cause periorbital emphysema or hematoma
- Anterior ethmoidal artery: Injury can cause retro-orbital hematoma and vision loss
- Cribriform plate: Skull base; injury causes CSF leak
- Olfactory fossa depth (Keros classification) - higher Keros types increase surgical risk

### 3.3.3 Frontal Sinus

The frontal sinuses are paired, asymmetric cavities located within the frontal bone, behind the superciliary arches. They are typically separated by an intersinus septum that is rarely in the midline. The frontal sinuses are absent at birth and develop progressively until adolescence.

Dimensions and Volume:

- Average vertical height: 32 mm (range 17-50 mm)
- Average transverse width: 26 mm (range 14-40 mm)
- Average anteroposterior depth: 19 mm (range 10-28 mm)
- Average volume: 6-7 ml per side

Walls of the Frontal Sinus:

- Anterior wall: Thick, formed by the outer table of frontal bone
- Posterior wall: Thin, separates sinus from anterior cranial fossa and frontal lobe
- Inferior wall (floor): Forms part of orbital roof and contains the frontal recess opening
- Medial wall: Intersinus septum, often deviated from midline

Drainage of Frontal Sinus:

The frontal sinus drains via the frontal recess (formerly called nasofrontal duct) into the middle meatus. The anatomy of the frontal recess is highly variable and complex, often described as an "hourglass" with the narrowest point being the frontal ostium. Understanding this anatomy is crucial for endoscopic frontal sinus surgery.

Variations:

- Aplasia (absence): 5-15% unilateral, 1-4% bilateral
- Hypoplasia: 5-10%
- Asymmetry: 80-85%
- Bony septations: 10-30%
- Pneumatization extending to orbital roof or temporal bone
- Frontal cells (Kuhn types I-IV)

Clinical Significance:

- Frontal sinus drainage is the most challenging area in FESS
- Forensic identification using frontal sinus radiographs (highly individual pattern)
- Frontal sinus fractures may require surgical reconstruction
- Mucocele formation due to drainage obstruction

### 3.3.4 Sphenoid Sinus

The sphenoid sinuses are paired cavities located within the body of the sphenoid bone, deep within the skull. They are the most posteriorly placed paranasal sinuses and have important anatomical relationships with critical neurovascular structures.

Dimensions and Volume:

- Average vertical height: 20 mm (range 10-32 mm)
- Average transverse width: 17 mm (range 9-31 mm)
- Average anteroposterior depth: 23 mm (range 11-43 mm)
- Average volume: 7-8 ml per side

Walls of the Sphenoid Sinus:

- Anterior wall: Faces the nasal cavity, contains the sphenoid ostium
- Posterior wall: Related to clivus, brainstem, basilar artery
- Superior wall: Related to sella turcica, pituitary gland, optic chiasm
- Inferior wall (floor): Related to nasopharynx and palatine bone
- Lateral wall: Related to cavernous sinus, internal carotid artery, optic nerve, abducens nerve, and oculomotor nerve

Drainage:

The sphenoid sinus drains through the sphenoid ostium, located in the upper part of the anterior wall, into the sphenoidal recess (above the superior turbinate).

Critical Relations:

- Internal carotid artery (lateral wall)
- Optic nerve (superolateral wall)
- Cavernous sinus (lateral wall)
- Pituitary gland (superior wall)
- Vidian canal (floor)
- Foramen rotundum (lateral wall)
- Maxillary nerve (V2) - in lateral wall

Septation:

The sphenoid sinus is divided into right and left compartments by an intersinus septum that is rarely midline. The septation pattern is highly variable:

- Single midline septum: 32-44%
- Single off-midline septum: 30-50%
- Multiple septa: 18-32%
- Septum attaching to ICA canal: 8-25% (clinically significant)

Pneumatization Types (Hamberger Classification):

- Conchal type (2-3%): Cancellous bone with no pneumatization

- Presellar type (11-24%): Pneumatization extends to anterior wall of sella
- Sellar type (73-86%): Pneumatization extends below sella
- Postsellar type: Pneumatization extends posterior to sella into clivus

Clinical Significance:

- Approach for transsphenoidal pituitary surgery
- ICA dehiscence (4-22%) - risk of catastrophic hemorrhage
- Optic nerve dehiscence (4-12%) - risk of vision loss
- Onodi cells overlying sphenoid - alters surgical approach
- Skull base reconstruction after CSF leak



### **3.4 The Osteomeatal Complex (OMC)**

The osteomeatal complex (OMC) is the functional core of the paranasal sinus drainage system. The term was coined by Naumann in 1965 and later popularized by Stammberger and Kennedy as the cornerstone of FESS. The OMC is not a single anatomical structure but rather a complex region where the maxillary, frontal, and anterior ethmoid sinuses converge and drain.

#### **3.4.1 Components of OMC**

- Maxillary sinus ostium
- Ethmoid infundibulum
- Hiatus semilunaris
- Middle turbinate
- Uncinate process
- Ethmoid bulla
- Frontal recess
- Anterior ethmoid air cells

#### **3.4.2 The Uncinate Process**

The uncinate process is a thin, sickle-shaped bony lamella forming the medial wall of the ethmoid infundibulum. Its anatomy is critical for understanding sinus drainage and FESS.

The superior attachment of the uncinate process is highly variable. Stammberger described six types based on its superior insertion:

- Type I: Attaches to lamina papyracea (most common, 52%)
- Type II: Attaches to skull base (fovea ethmoidalis) (6%)
- Type III: Attaches to middle turbinate (26%)
- Type IV: Combined attachments
- Type V: Attaches to roof of maxillary sinus
- Type VI: Free edge or bifurcated

The pattern of uncinate attachment determines the drainage of the frontal sinus:

- Lamina papyracea attachment: Frontal sinus drains medial to uncinate (into middle meatus directly)
- Skull base/middle turbinate attachment: Frontal sinus drains into ethmoid infundibulum

#### **3.4.3 The Ethmoid Bulla**

The ethmoid bulla is the largest and most consistent anterior ethmoid air cell, located posterior to the uncinate process. It forms the posterior wall of the hiatus semilunaris. Variations in size and pneumatization of the ethmoid bulla can affect drainage of adjacent sinuses.

#### **3.4.4 The Hiatus Semilunaris**

The hiatus semilunaris is a crescent-shaped opening between the uncinate process (anteriorly) and the

ethmoid bulla (posteriorly). It is the entrance to the ethmoid infundibulum.

### **3.4.5 Clinical Significance of OMC**

Obstruction of the OMC, even by minor mucosal swelling, can lead to:

- Reduced ventilation of the maxillary, frontal, and anterior ethmoid sinuses
- Mucus stasis and bacterial colonization
- Chronic rhinosinusitis
- Mucocele formation

FESS aims to restore ventilation and drainage by clearing OMC obstruction while preserving normal mucosa.

## **3.5 Anatomical Variations of Paranasal Sinuses**

This section provides a detailed account of various anatomical variations of paranasal sinuses, their prevalence in different populations, and their clinical significance.

### **3.5.1 Nasal Septum Variations**

#### **3.5.1.1 Deviated Nasal Septum (DNS)**

Deviated nasal septum is the most common anatomical variation affecting the nasal cavity and indirectly the paranasal sinuses. The nasal septum is composed of the perpendicular plate of ethmoid (superiorly), vomer (inferoposteriorly), and quadrangular cartilage (anteriorly).

Etiology of DNS:

- Congenital: Birth trauma, intrauterine pressure
- Developmental: Asymmetric growth of facial bones
- Acquired: Trauma (most common in adults)
- Idiopathic

Classification (Mladina, 1987):

- Type 1: Septum has unilateral vertical deviation that does not encroach on the valve area
- Type 2: Septum has unilateral vertical deviation in the valve area
- Type 3: Septum has unilateral deviation at the level of middle/superior turbinate
- Type 4: Severe DNS with S-shape (one side anterior, opposite side posterior)
- Type 5: Bony spur on one side
- Type 6: Spur with deep groove or concavity on opposite side
- Type 7: Combination of above types

Prevalence in Different Studies:

- Earwaker (1993): 20-79% in CT studies
- Stallman et al. (2004): 65%
- Smith et al. (2010): 41-79%
- Kim et al. (2007 - Korean): 78.9%
- Adeel et al. (2013 - Pakistan): 64.5%
- Dua et al. (2005 - North India): 75.4%
- Mamatha et al. (2015 - South India): 58%
- Patel et al. (2014 - Gujarat): 70%

Clinical Significance:

- Nasal obstruction (most common symptom)
- Recurrent sinusitis on the obstructed side
- Compensatory hypertrophy of contralateral inferior turbinate
- Snoring and sleep apnea

- Epistaxis from spurs
- Difficulty in nasal endoscopy and FESS
- Paradoxical sinusitis on contralateral side

### **3.5.1.2 Septal Spurs and Crests**

Septal spurs are localized angulations or projections of the septum, often at the junction of vomer and ethmoidal plate. Bolger et al. (1991) reported a prevalence of 30-50%. Spurs may impinge on the lateral nasal wall or middle turbinate, causing facial pain (Sluder's neuralgia) or contact-point headaches.

## **3.5.2 Turbinate Variations**

### **3.5.2.1 Concha Bullosa**

Concha bullosa refers to the pneumatization of the middle turbinate. It is one of the most common anatomical variations of the lateral nasal wall and has significant implications for sinus drainage and surgical access.

Bolger Classification (1991):

- Lamellar type: Pneumatization confined to vertical lamella of middle turbinate
- Bulbous type: Pneumatization in the inferior bulbous portion
- Extensive (true) concha bullosa: Pneumatization of both lamellar and bulbous portions

Prevalence:

- Bolger et al. (1991): 53.6% (any type)
- Zinreich et al. (1988): 34%
- Stallman et al. (2004): 35.8%
- Tonai and Baba (1996 - Japan): 34%
- Stackpole and Edelstein (1997): 24%
- Adeel et al. (2013): 30.4%
- Sharma et al. (2016 - India): 32.5%
- Mamatha et al. (2015 - South India): 25%

Other Turbinate Pneumatizations:

- Superior concha bullosa (4-15%)
- Inferior turbinate pneumatization (rare, 1-2%)
- Supreme turbinate pneumatization (rare)

Clinical Significance:

- OMC obstruction leading to sinusitis
- Headaches, facial pressure
- Difficulty in middle meatus visualization during endoscopy
- Need for surgical resection during FESS

- Rarely, pyocele formation if obstructed

### **3.5.2.2 Paradoxical Middle Turbinate**

Normally, the middle turbinate has a medial concavity. In paradoxical middle turbinate, the convexity faces medially (i.e., the lateral surface is concave). This variation can narrow the middle meatus and obstruct OMC drainage.

Prevalence:

- Bolger et al. (1991): 26.1%
- Stallman et al. (2004): 26.9%
- Badia et al. (2005): 17%
- Adeel et al. (2013): 15%
- Indian studies: 12-15%

### **3.5.2.3 Secondary/Accessory Middle Turbinate**

Secondary middle turbinate is a rare variation first described by Khanobthamchai et al. (1991). Prevalence is 0.5-2%. It appears as a separate, smaller turbinate medial to the regular middle turbinate.

## **3.5.3 Ethmoid Sinus Variations**

### **3.5.3.1 Agger Nasi Cells**

The agger nasi ("nasal mound") cells are the most anterior ethmoid air cells, located anterior and superior to the attachment of the middle turbinate. They are formed by pneumatization of the lacrimal bone and frontal process of maxilla.

Prevalence (essentially universal):

- Bolger et al. (1991): 98.5%
- Stammberger (1991): 89%
- Wormald (2003): 96%
- Jain et al. (2012 - India): 92%
- Sharma et al. (2014): 89%

Clinical Significance:

- Forms the anterior boundary of frontal recess
- Surgical landmark for frontal sinus drainage
- Large agger nasi cells can compress and obstruct frontal recess
- Important target during frontal sinusotomy

### **3.5.3.2 Haller Cells (Infraorbital Ethmoid Cells)**

First described by Albrecht von Haller in 1765, these are ethmoid air cells that pneumatize the orbital floor (medial part) and may extend into the maxillary sinus roof. They are positioned along the medial floor of the orbit, adjacent to the maxillary sinus ostium.

Prevalence:

- Bolger et al. (1991): 10.3%
- Stackpole and Edelstein (1997): 18%
- Kantarci et al. (2004): 16%
- Bademci et al. (2005): 15.7%
- Jain et al. (2012 - India): 15.8%
- Sharma et al. (2016 - India): 18.2%

Clinical Significance:

- Can obstruct or narrow the maxillary sinus ostium
- Associated with chronic and recurrent maxillary sinusitis
- May be confused with periorbital cellulitis on imaging
- Need to be addressed during FESS for adequate maxillary sinus drainage
- Risk of orbital injury during surgical removal

### **3.5.3.3 Onodi Cells (Sphenoethmoidal Cells)**

Onodi cells, named after Adolph Onodi (1903), are the most posterior ethmoid cells that pneumatize superolaterally to the sphenoid sinus, often coming into close relation with the optic nerve and internal carotid artery.

Prevalence:

- Weinberger et al. (1996): 8-14%
- Driben et al. (1998): 7%
- DeLano et al. (1996): 12%
- Wormald (2003): 14%
- Jang et al. (2008 - Korean): 60% (highest reported)
- Jain et al. (2012 - India): 10.5%

Clinical Significance:

- Optic nerve runs through or adjacent to the lateral wall
- Risk of optic nerve injury during posterior ethmoidectomy
- Can be mistaken for sphenoid sinus on imaging
- Approach to sphenoid sinus may need modification
- Cause of optic neuritis or visual loss in sphenoethmoid sinusitis

### **3.5.3.4 Frontal Cells (Kuhn Classification)**

Bent and Kuhn (1994) classified frontal recess cells into four types based on their relationship with the agger nasi cell and frontal sinus:

- Type I: Single frontal recess cell above agger nasi, below frontal sinus
- Type II: Tier (multiple) of frontal recess cells above agger nasi, below frontal sinus
- Type III: Single large cell extending from agger nasi into frontal sinus

- Type IV: Isolated cell within the frontal sinus

Prevalence:

- Type I: 33%
- Type II: 9.5%
- Type III: 9.5%
- Type IV: 1%

### **3.5.3.5 Supraorbital Ethmoid Cells**

These ethmoid cells extend laterally over the orbital roof. They have a prevalence of 7.2-15% (Bolger et al., 1991; DelGaudio et al., 2005). Important to differentiate from frontal sinus on CT.

### **3.5.3.6 Lamina Papyracea Variations**

- Dehiscence: 0.5-1%
- Medial bowing/displacement
- Pneumatization defects

Clinical importance: Risk of orbital injury during ethmoidectomy

### **3.5.4 Maxillary Sinus Variations**

#### **3.5.4.1 Maxillary Sinus Hypoplasia**

Maxillary sinus hypoplasia is characterized by underdevelopment of the maxillary sinus, often associated with abnormal uncinate process.

Bolger Classification (1990):

- Type I: Mild hypoplasia with normal uncinate process
- Type II: Significant hypoplasia with hypoplastic/absent uncinate process
- Type III: Severe (cleft-like) hypoplasia with absent uncinate process

Prevalence:

- Bolger et al. (1990): Type I 7.2%, Type II 0.6%, Type III 0.4%
- Sirikci et al. (2000): 1.7-10.4% overall
- Jain et al. (2012 - India): 4.3%

#### **3.5.4.2 Underwood Septa**

First described by Arthur Underwood in 1910, these are bony septa that partially or completely divide the maxillary sinus into compartments. They are clinically important during sinus lift procedures and dental implant placement.

Classification (Velasquez-Plata et al., 2002):

- Type I (Anterior): Between first and second molar
- Type II (Middle): Between second molar and tuberosity
- Type III (Posterior): Posterior to maxillary tuberosity

Prevalence:

- Velasquez-Plata et al. (2002): 21.6%
- Kim et al. (2006): 26.5%
- Maestre-Ferrin et al. (2011): 32.7%
- Park et al. (2011): 41.7%

#### **3.5.4.3 Accessory Maxillary Ostium**

Additional opening in the lateral nasal wall, separate from the natural ostium. First described by Giraldez (1840). Located in the posterior fontanelle (between the inferior turbinate and uncinate process).

Prevalence:

- Van Alyea (1936): 23%
- Stammberger (1991): 10-30%
- Earwaker (1993): 15%
- Jain et al. (2012 - India): 18.4%

Significance: May lead to mucus recirculation and chronic sinusitis. Single AMO drainage is less



effective than functional natural ostium.

### **3.5.5 Frontal Sinus Variations**

#### **3.5.5.1 Frontal Sinus Aplasia/Hypoplasia**

- Aplasia: 5-15%
- Hypoplasia: 5-10%
- Bilateral aplasia: 1-4%
- Indian studies (Ponde 2003, Nikam 2015): 3.8-4.2%

Frontal sinus development is the most variable. Higher altitude populations show smaller frontal sinuses. Genetic and environmental factors influence size.

#### **3.5.5.2 Frontal Sinus Asymmetry**

Asymmetry between right and left frontal sinuses is common (Harris et al., 1987: 85%). This forms the basis for forensic identification using frontal sinus radiographs - the pattern is unique to each individual, comparable to fingerprints.

#### **3.5.5.3 Interfrontal Septal Cell**

Cell within the interfrontal septum, prevalence 12-36% (McLaughlin et al., 2001). May obstruct frontal recess.

### **3.5.6 Sphenoid Sinus Variations**

#### **3.5.6.1 Sphenoid Sinus Pneumatization (Hamberger Classification)**

- Conchal type (2-3%): Solid bone, surgically challenging
- Presellar type (11-24%): Limited pneumatization
- Sellar type (73-86%): Most common, ideal for transsphenoidal surgery

Indian Studies:

- Vaishya et al. (2015): Sellar 76%, Presellar 20%, Conchal 4%
- Naveen et al. (2018): Sellar 82.5%

#### **3.5.6.2 ICA Dehiscence**

Bony dehiscence over the internal carotid artery in the lateral wall of sphenoid sinus.

- Fujii et al. (1979): 4-8%
- Renn and Rhoton (1975): 4%
- Sirikci et al. (2000): 5.5%
- Sethi et al. (1995): 22%
- Vaishya et al. (2015 - India): 7.5%

Critical surgical importance - injury can be catastrophic with mortality rates up to 50%.

#### **3.5.6.3 Optic Nerve Variations**

DeLano et al. (1996) classified the optic nerve relationship with sphenoid sinus into four types:

- Type I (76%): Adjacent to sphenoid sinus, no indentation
- Type II (15%): Indents sphenoid sinus, no contact with posterior ethmoid
- Type III (6%): Passes through sphenoid sinus
- Type IV (3%): Adjacent to both sphenoid and posterior ethmoid (Onodi cell)

Optic nerve dehiscence (4-12%) increases the risk of visual loss during surgery.

### **3.5.7 Anterior Skull Base Variations**

#### **3.5.7.1 Keros Classification of Olfactory Fossa**

Keros (1962) classified the depth of the olfactory fossa into three types based on the difference in height between the cribriform plate and the fovea ethmoidalis:

- Type I (1-3 mm): Shallow olfactory fossa, lowest risk (12-26%)
- Type II (4-7 mm): Moderate depth (52-73%, most common)
- Type III (8-16 mm): Deep olfactory fossa, highest surgical risk (8-26%)

Type III configuration significantly increases the risk of CSF leak during ethmoidectomy due to the long, thin lateral lamella of the cribriform plate.

#### **3.5.7.2 Cribriform Plate Asymmetry**

Asymmetry between right and left sides is reported in 10-20% (Meloni et al., 1992). The lower side is at higher risk of inadvertent skull base injury during surgery.

### **3.5.8 Other Variations**

- Pneumatized crista galli
- Frontal sinus septation
- Pneumatized anterior clinoid process
- Pneumatized pterygoid process
- Pneumatized vomer
- Vomeral spurs

## **3.6 CT Imaging in Paranasal Sinus Evaluation**

### **3.6.1 Historical Development of CT**

The evolution of computed tomography has revolutionized the field of radiology and transformed the evaluation of paranasal sinuses. Sir Godfrey Hounsfield's invention of the first commercial CT scanner in 1972 marked the beginning of a new era in medical imaging. The Nobel Prize in Physiology or Medicine was awarded to Hounsfield and Allan Cormack in 1979 for their pioneering work.

The development of CT technology has progressed through several generations:

- First generation (1972-1976): Single detector, translation-rotation movement, scan times of 4-5 minutes
- Second generation (1976-1980): Multiple detectors with translate-rotate, scan times of 30 seconds
- Third generation (1980s): Rotating detector array, scan times of 1-3 seconds
- Fourth generation (1980s): Stationary detector ring, rotating tube
- Spiral/Helical CT (1989): Continuous rotation with simultaneous table movement
- Multi-detector CT (1998): Multiple detector rows allowing volumetric acquisition
- Modern MDCT (2000s onwards): 64, 128, 256, 320 slice scanners with sub-millimeter resolution

### **3.6.2 Principles of CT Imaging**

Computed tomography utilizes X-rays to produce cross-sectional images of the body. The fundamental principle involves passing an X-ray beam through the body and detecting the attenuated beam on the opposite side. The detected signals are then processed mathematically (using filtered back projection or iterative reconstruction algorithms) to create cross-sectional images.

Key technical parameters:

- Tube voltage (kVp): Determines X-ray penetration; typically 100-140 kVp for sinus CT
- Tube current (mA): Affects image quality and radiation dose; typically 100-300 mA
- Slice thickness: Thinner slices (0.5-1 mm) provide better detail
- Pitch: Ratio of table feed to slice thickness; affects scan time and quality
- Field of view (FOV): Determines image area; typically 18-22 cm for sinus CT
- Matrix size: Number of pixels (typically 512 x 512)
- Reconstruction algorithm: Bone (sharp) or soft tissue (smooth)

### **3.6.3 CT Window Settings for Sinus Evaluation**

Different window settings are used to optimize visualization of different tissues:

- Bone window (WW: 2000, WL: 400): Best for bony anatomy and variations
- Soft tissue window (WW: 350, WL: 40): For mucosal disease and soft tissue masses

- Lung window (WW: 1500, WL: -600): For pneumatization assessment

### **3.6.4 Multiplanar Reconstruction (MPR)**

Modern MDCT allows for high-quality reconstructions in any plane:

- Axial: Native acquisition plane; useful for sinus relationships
- Coronal: Most informative plane for sinus anatomy
- Sagittal: Best for frontal recess and sphenoid evaluation
- Oblique: Customized planes following specific anatomical structures

### **3.6.5 Three-Dimensional (3D) Reconstruction**

3D reconstructions provide:

- Surface rendering for visualizing complex anatomy
- Volume rendering for assessing pneumatization
- Surgical planning visualization
- Patient education and pre-operative counseling
- Forensic identification using sinus patterns

### **3.6.6 Lund-Mackay Scoring System**

Introduced by Lund and Mackay in 1993, this is the most widely used CT scoring system for chronic rhinosinusitis. Each sinus group is scored from 0 to 2:

- 0: Normal (no opacification)
- 1: Partial opacification
- 2: Complete opacification

The osteomeatal complex is scored as 0 (not obstructed) or 2 (obstructed). The maximum total score for both sides is 24 (12 per side).

### **3.6.7 Comparison with Other Imaging Modalities**

Plain radiography:

- Advantages: Low cost, widely available, low radiation
- Disadvantages: Poor anatomical detail, superimposition, limited variations detection
- Current role: Limited; mainly for screening

Magnetic Resonance Imaging (MRI):

- Advantages: No ionizing radiation, superior soft tissue detail, multiplanar imaging
- Disadvantages: Poor bone visualization, longer scan time, claustrophobia, cost
- Current role: Complementary to CT, especially for tumors and complications

Cone Beam CT (CBCT):

- Advantages: Lower radiation dose, good bone detail, lower cost than MDCT
- Disadvantages: Limited soft tissue contrast, smaller FOV, motion artifacts
- Current role: Increasingly used for dental and maxillofacial applications

Ultrasound:

- Advantages: No radiation, real-time imaging, low cost
- Disadvantages: Cannot evaluate sinuses (air interferes), operator-dependent
- Current role: Very limited for sinus evaluation; useful for orbital complications

## **3.7 Clinical Significance of Anatomical Variations**

### **3.7.1 Functional Endoscopic Sinus Surgery (FESS)**

FESS represents the gold standard surgical approach for chronic rhinosinusitis refractory to medical therapy. The success of FESS depends critically on:

- Detailed pre-operative CT evaluation
- Recognition of anatomical variations
- Surgeon's expertise and experience
- Use of appropriate instrumentation
- Image-guided surgery in complex cases

Common FESS Procedures:

- Uncinectomy: Removal of uncinate process
- Maxillary antrostomy: Enlargement of maxillary sinus ostium
- Anterior ethmoidectomy: Removal of anterior ethmoid cells
- Posterior ethmoidectomy: Removal of posterior ethmoid cells
- Sphenoidotomy: Opening of sphenoid sinus
- Frontal sinusotomy: Restoration of frontal sinus drainage
- Powered instrumentation (microdebrider) techniques

### **3.7.2 Complications of Sinus Surgery**

Despite advances in surgical techniques, complications can occur. They are classified as:

Major Complications (incidence 0.3-1%):

- Cerebrospinal fluid (CSF) leak
- Meningitis
- Pneumocephalus
- Optic nerve injury and blindness
- Internal carotid artery injury
- Massive hemorrhage
- Brain injury
- Death (rare)

Minor Complications (incidence 1-5%):

- Periorbital hematoma/ecchymosis
- Subcutaneous emphysema
- Synechiae (adhesions) formation
- Mucocoele formation
- Recurrent disease
- Olfactory disturbance

- Bleeding requiring intervention

May et al. (1994) analyzed 2,108 FESS patients and reported a complication rate of 0.3-3%, with the majority being related to anatomical variations not recognized pre-operatively.

### 3.7.3 Anatomical Variations and Their Surgical Implications

Table: Surgical Implications of Common Variations

| Variation              | Implication                      |
|------------------------|----------------------------------|
| Severe DNS             | Difficult endoscopic access; m.. |
| Concha bullosa         | Obstructs OMC; may need partia.. |
| Paradoxical MT         | Narrows middle meatus; increas.. |
| Large agger nasi       | Compresses frontal recess; aff.. |
| Haller cell            | Obstructs maxillary ostium; ne.. |
| Onodi cell             | Optic nerve at risk during pos.. |
| Type II uncinate       | Bony attachment at skull base;.. |
| Hypoplastic max sinus  | Distorted anatomy; lamina papy.. |
| Conchal sphenoid       | Difficult access; needs carefu.. |
| ICA dehiscence         | Catastrophic hemorrhage risk     |
| Optic nerve protrusion | Risk of vision loss              |
| Keros Type III         | High risk of CSF leak during e.. |

### 3.7.4 Pre-operative CT Checklist (CHECKLIST mnemonic)

Stankiewicz proposed the CHECKLIST approach for pre-operative CT review:

- C - Cribriform plate asymmetry
- H - Height of ethmoid roof (Keros classification)
- E - Ethmoidal arteries position
- C - Cells (Onodi, Haller, agger nasi, frontal cells)
- K - Klinoid recess (anterior clinoid pneumatization)
- L - Lamina papyracea integrity
- I - Internal carotid artery position and dehiscence
- S - Sphenoid sinus pneumatization and septation
- T - Tumor or pathology assessment

### 3.7.5 Image-Guided Surgery (IGS)

Image-guided surgical navigation systems use pre-operative CT data to provide real-time anatomical localization during surgery. IGS is particularly useful for:

- Revision surgeries with distorted anatomy
- Patients with multiple anatomical variations
- Tumor resection in proximity to vital structures
- Skull base surgery

- Frontal sinus surgery
- Surgery in patients with vision in only one eye

AAO-HNS guidelines recommend IGS for:

- Revision sinus surgery
- Distorted sinus anatomy of development, postoperative or traumatic origin
- Disease abutting the skull base, orbit, optic nerve or carotid artery
- Disease involving the frontal, posterior ethmoid, or sphenoid sinuses
- CSF rhinorrhea or skull base defect repair
- Benign or malignant sinonasal neoplasms



### **3.8 Review of Indian Studies**

Indian studies on paranasal sinus anatomical variations have provided valuable regional data. This section reviews major Indian publications, organized by region.

#### **3.8.1 North Indian Studies**

Dua et al. (2005) - Punjab/Delhi:

- Sample: 250 CT scans
- DNS: 75.4%
- Concha bullosa: 31%
- Agger nasi: 87%
- Haller cells: 12%
- Onodi cells: 9%

Bansal et al. (2014) - Chandigarh:

- Sample: 300 patients
- DNS: 68%
- Concha bullosa: 30%
- Frontal sinus aplasia: 4.3%

Singh et al. (2017) - UP:

- Sample: 200 patients
- DNS: 71.5%
- Keros II: 65%
- Haller cells: 16%

#### **3.8.2 Central Indian Studies**

Jain et al. (2012) - Madhya Pradesh:

- Sample: 200 patients (one of the largest studies)
- DNS: 68.4%
- Concha bullosa: 28.9%
- Agger nasi: 92%
- Haller cells: 15.8%
- Onodi cells: 10.5%
- Comprehensive analysis of multiple variations

Saxena et al. (2015) - Chhattisgarh:

- Sample: 150 patients
- DNS: 65%
- Concha bullosa: 25%

### **3.8.3 Western Indian Studies**

Sharma et al. (2014) - Maharashtra:

- Sample: 100 patients
- DNS: 62%
- Concha bullosa: 32.5%
- Haller cells: 18.2%
- Onodi cells: 12%

Patel et al. (2014) - Gujarat:

- Sample: 150 patients
- DNS: 70%
- Concha bullosa: 28%
- Limited data from Gujarat - one of the few studies

Joshi et al. (2018) - Mumbai:

- Sample: 250 patients
- DNS: 64%
- Concha bullosa: 35%
- Keros II: 70%

### **3.8.4 Southern Indian Studies**

Mamatha et al. (2015) - Karnataka:

- Sample: 100 patients
- DNS: 58%
- Concha bullosa: 25%
- Paradoxical MT: 12%
- Lower prevalence of variations compared to North India

Reddy et al. (2017) - Andhra Pradesh:

- Sample: 200 patients
- DNS: 60%
- Concha bullosa: 28%

Selvi et al. (2016) - Tamil Nadu:

- Sample: 150 patients
- DNS: 55%
- Onodi cells: 8%

### **3.8.5 Eastern Indian Studies**

Bhattacharyya et al. (2015) - West Bengal:

- Sample: 180 patients

- DNS: 67%
- Concha bullosa: 26%

Sahu et al. (2016) - Odisha:

- Sample: 120 patients
- DNS: 63%
- Various variations documented

### **3.8.6 Multi-center Indian Studies**

Adeel et al. (2013) - Multi-center:

- Sample: 500 patients across 3 centers
- DNS: 64.5%
- Concha bullosa: 30.4%
- Paradoxical turbinate: 15%
- Agger nasi: 89%

### **3.8.7 Importance of Regional Indian Data**

The variations in prevalence rates across different regions of India highlight the importance of region-specific anatomical studies. Factors contributing to regional variations may include:

- Genetic differences among populations
- Environmental factors (climate, pollution, allergens)
- Dietary habits and nutrition
- Anthropometric differences
- Socioeconomic factors affecting health-seeking behavior
- Differences in CT technology and reporting
- Sample size and selection criteria

The need for South Gujarat-specific data is particularly important because:

- Limited published data from this region
- Unique demographic mix of urban (Surat) and rural populations
- Industrial pollution affecting respiratory health
- Migration from neighboring states adds genetic diversity
- Specific dietary patterns and lifestyle factors
- Significant patient load undergoing FESS in this region

## **3.9 Additional Variations and Special Cases**

### **3.9.1 Pneumatized Crista Galli**

The crista galli is a midline bony projection of the ethmoid bone in the anterior cranial fossa. Pneumatization of the crista galli has been reported with prevalence ranging from 3-13%. When pneumatized, it usually communicates with the frontal sinus and can rarely become infected (mucocele).

Surgical significance:

- Communication with frontal recess - infection spread
- Mucocele formation requiring drainage
- Skull base reconstruction - landmark consideration

### **3.9.2 Pneumatized Anterior Clinoid Process**

Anterior clinoid pneumatization is reported in 4-13% of individuals. When pneumatized, the optic nerve is at increased risk during transsphenoidal and skull base surgery. The pneumatization usually arises from the sphenoid sinus.

### **3.9.3 Pneumatized Pterygoid Process**

Pterygoid recess of the sphenoid sinus extends into the pterygoid process in 25-44% of individuals. This variation has surgical implications as the vidian nerve and maxillary nerve (V2) may be exposed or in close relation.

### **3.9.4 Vomer Variations**

- Vomer spurs (10-15%)
- Pneumatized vomer (rare)
- Vomerine crest enlargement
- Significance for transnasal access to skull base

### **3.9.5 Anterior Ethmoidal Artery Variations**

The anterior ethmoidal artery (AEA) is a critical surgical landmark and one of the major sources of intraoperative bleeding. Variations include:

- Position relative to skull base: within sulcus (most protected) vs in mesentery
- Distance from frontal sinus posterior wall
- Bilateral asymmetry common
- Bony dehiscence in 5-32% of cases

The AEA can be classified into types based on its course:

- Type 1: Within ethmoidal sulcus, completely covered by bone (most common)
- Type 2: Suspended within a thin mesentery from ethmoid roof
- Type 3: Completely dehiscent, hanging within the ethmoid air cells

### **3.9.6 Posterior Ethmoidal Artery Variations**

The posterior ethmoidal artery (PEA) is smaller and less consistent than the AEA. It is absent in 30% of individuals. When present, it usually emerges from the orbit and crosses the ethmoid roof to enter the cranial cavity.

### **3.9.7 Sphenopalatine Foramen Variations**

The sphenopalatine foramen transmits the sphenopalatine artery, the most common source of severe posterior epistaxis. Variations include:

- Single foramen (most common)
- Multiple foramina
- Variable position relative to crista ethmoidalis

### **3.9.8 Choanal Variations**

- Choanal atresia (1:5000-8000 births) - usually pediatric concern
- Choanal stenosis
- Asymmetric choanal size

### **3.9.9 Special Patient Populations**

Pediatric considerations:

- Sinus development incomplete until adolescence
- Smaller anatomical structures
- Risk of orbital injury higher
- CT findings differ from adults

Geriatric considerations:

- Atrophy of mucosa
- Increased pneumatization
- Resorption of bony walls
- Different patterns of disease

Patients with systemic conditions:

- Cystic fibrosis - hypoplastic frontal sinuses
- Wegener's granulomatosis - bony erosion
- Immunocompromised - fungal sinusitis
- Allergic fungal sinusitis - bony expansion

### **3.10 Clinical Correlations**

#### **3.10.1 Anatomical Variations and Chronic Rhinosinusitis (CRS)**

The relationship between anatomical variations and CRS has been extensively studied with conflicting results. Some studies suggest a strong association while others find no significant correlation.

Studies supporting association:

- Bolger et al. (1991): Concha bullosa and Haller cells significantly associated with sinusitis
- Stallman et al. (2004): Concha bullosa associated with septal deviation and sinus disease
- Calhoun et al. (1991): OMC variations associated with sinusitis

Studies showing no significant association:

- Caughey et al. (2005): No correlation between variations and CRS
- Jones et al. (1997): Variations equally common in symptomatic and asymptomatic patients

The current consensus is that anatomical variations may predispose to CRS but are not the sole cause.

Factors contributing to CRS include:

- Mucociliary dysfunction
- Allergic inflammation
- Environmental allergens
- Bacterial biofilms
- Immune dysfunction
- Genetic predisposition

#### **3.10.2 Variations and Headache**

Several anatomical variations have been linked to headache:

- Septal spurs touching turbinates - contact-point headache (Sluder's syndrome)
- Concha bullosa with mucocele - facial pain
- Sphenoid sinusitis - retro-orbital headache
- Frontal sinus disease - frontal headache

Treatment of these structural causes can provide relief from chronic headaches in selected patients.

#### **3.10.3 Variations and Olfactory Function**

The olfactory cleft and cribriform plate region anatomy can affect olfaction:

- Severe DNS may cause unilateral hyposmia
- Olfactory cleft narrowing affects odor delivery
- Surgery in olfactory region can damage olfactory nerves
- Keros classification predicts olfactory cleft length

#### **3.10.4 Variations and Sleep Apnea**

Nasal anatomy plays a role in obstructive sleep apnea (OSA):

- DNS contributes to nasal obstruction in OSA
- Septoplasty may improve CPAP tolerance
- Concha bullosa contributes to nasal resistance
- Surgical correction may improve apnea-hypopnea index

### **3.10.5 Variations and Forensic Identification**

Frontal sinuses are unique to each individual (like fingerprints) and have been used for:

- Identification of unknown remains
- Mass disaster victim identification
- Comparison with antemortem radiographs
- DNA-independent identification method

Methods include:

- Frontal sinus pattern matching
- Linear measurements
- Computer-assisted analysis
- 3D reconstruction comparison

### **3.11 Detailed Surgical Anatomy of Each Sinus**

#### **3.11.1 Maxillary Sinus - Surgical Perspective**

The maxillary sinus is the most commonly operated paranasal sinus due to its high incidence of infection. Understanding its detailed surgical anatomy is paramount for safe and effective surgery.

Surgical Landmarks:

- Maxillary line: a curvilinear ridge representing the projection of nasolacrimal duct on the lateral nasal wall
- Uncinate process: thin sickle-shaped bone, key landmark for endoscopic surgery
- Hiatus semilunaris: crescent-shaped opening between uncinate and ethmoid bulla
- Maxillary sinus ostium: located in the upper part of medial wall, drains via infundibulum
- Inferior turbinate attachment: divides anterior and posterior fontanelles
- Posterior fontanelle: thin membranous area where accessory ostia commonly form

Vascular Anatomy:

- Posterior superior alveolar artery: enters posterior wall, supplies sinus mucosa
- Infraorbital artery: enters via infraorbital foramen, runs in superior wall
- Greater palatine artery: enters via floor
- Sphenopalatine artery: branches enter medial wall
- Anterior superior alveolar artery: anterior wall blood supply

Neural Anatomy:

- Maxillary nerve (V2): main innervation
- Anterior superior alveolar nerve: anterior wall and incisor teeth
- Middle superior alveolar nerve: variable, premolar teeth
- Posterior superior alveolar nerve: molar teeth and posterior gingiva
- Infraorbital nerve: roof and anterior wall
- Greater palatine nerve: floor and medial wall

Lymphatic Drainage:

- Submandibular nodes
- Deep cervical nodes
- Retropharyngeal nodes

Mucociliary Clearance:

Despite its dependent position, the maxillary sinus has well-organized mucociliary clearance. Cilia beat in a coordinated pattern (10-15 Hz) directing mucus toward the natural ostium. This explains why creating a large antrostomy without addressing the natural ostium can fail - cilia continue to beat toward the natural ostium causing mucus recirculation.

#### **3.11.2 Ethmoid Labyrinth - Surgical Anatomy**



The ethmoid labyrinth is the most complex and clinically important region in sinus surgery. Its detailed anatomy must be thoroughly understood.

Ethmoid Bone Components:

- Cribriform plate: roof of nasal cavity, perforated for olfactory nerves
- Crista galli: midline bony projection
- Perpendicular plate: forms upper part of nasal septum
- Labyrinth (lateral mass): contains air cells
- Superior turbinate: upper part of ethmoid
- Middle turbinate: clinically most important

Lamellae of Ethmoid:

The ethmoid has classical lamellae (sheets of bone) that serve as surgical landmarks:

- First lamella: Uncinate process
- Second lamella: Anterior face of ethmoid bulla
- Third lamella: Basal lamella of middle turbinate (separates anterior from posterior ethmoid)
- Fourth lamella: Anterior face of superior turbinate

Spaces Between Lamellae:

- Frontal recess: between agger nasi and middle turbinate
- Suprabullar recess: above ethmoid bulla
- Retrobullar recess: behind ethmoid bulla
- Sphenoethmoidal recess: between superior turbinate and sphenoid

Critical Surgical Landmarks:

- Maxillary line: indicates projection of frontal recess
- Anterior ethmoidal artery: marks transition from anterior to posterior ethmoid
- Posterior ethmoidal artery: marks border with sphenoid
- Lamina papyracea: lateral border with orbit
- Fovea ethmoidalis and cribriform plate: superior border

### **3.11.3 Frontal Sinus - Drainage Anatomy**

The frontal sinus drainage anatomy is the most complex and has received considerable attention in literature.

Frontal Recess Anatomy:

The frontal recess (formerly nasofrontal duct) is the drainage pathway from frontal sinus to middle meatus. It is hourglass-shaped with:

- Frontal infundibulum (above ostium)
- Frontal ostium (narrowest point)
- Frontal recess proper (below ostium, opens into middle meatus)

Boundaries of Frontal Recess:

- Anterior: Agger nasi cell
- Posterior: Anterior wall of ethmoid bulla / suprabullar cells
- Medial: Middle turbinate
- Lateral: Lamina papyracea
- Superior: Frontal sinus floor
- Inferior: Middle meatus

Wormald's Building Block Concept:

Wormald (2003) introduced a building block approach to understand frontal recess anatomy:

1. Identify agger nasi cell (always anterior)
2. Identify ethmoid bulla (posterior border)
3. Identify any frontoethmoid cells (Type I-IV)
4. The frontal sinus drains between these structures

Kuhn Classification of Frontal Cells:

Bent and Kuhn (1994) classified frontal recess cells:

- Type I: Single cell above agger nasi, below frontal sinus floor (33%)
- Type II: Tier of cells above agger nasi (9.5%)
- Type III: Single cell extending from agger nasi into frontal sinus (9.5%)
- Type IV: Isolated cell within frontal sinus (1%)

### **3.11.4 Sphenoid Sinus - Skull Base Relations**

The sphenoid sinus has the most complex anatomical relationships of all paranasal sinuses, with critical neurovascular structures in close proximity.

Anatomical Relations:

- Anteriorly: Sphenoethmoidal recess, posterior nasal septum
- Posteriorly: Clivus, brainstem, basilar artery, posterior cranial fossa
- Superiorly: Pituitary gland, optic chiasm, anterior cranial fossa
- Inferiorly: Vidian canal, palatine bone, nasopharynx
- Laterally: Cavernous sinus contents (ICA, CN III, IV, V1, V2, VI)

Critical Structures in Lateral Wall:

- Internal carotid artery (cavernous segment)
- Optic nerve (canalicular and intracranial)
- Maxillary nerve (V2) in foramen rotundum
- Vidian nerve in vidian canal
- Abducens nerve (CN VI)

Sphenoid Sinus Septations:

The sphenoid sinus is divided by intersinus septa that are highly variable. Critical clinical aspects:

- Single midline septum: 32-44%
- Single deviated septum: 30-50%
- Multiple septa: 18-32%
- Septum attaching to ICA canal: 8-25% (CRITICAL - removal can injure ICA)

Onodi Cell Special Considerations:

- Located posterosuperolateral to sphenoid
- Optic nerve runs through or adjacent to lateral wall
- Often mistaken for sphenoid sinus on CT
- Approach to sphenoid through Onodi increases risk
- Pre-operative identification critical

## **3.12 Advanced Surgical Techniques and Considerations**

### **3.12.1 Basic FESS Technique**

The standard FESS technique involves a systematic, step-by-step approach:

#### **Step 1: Endoscopic Examination**

- Visualize nasal cavity, septum, turbinates
- Identify anatomical variations
- Assess disease extent

#### **Step 2: Uncinectomy**

- Identify uncinate process
- Make incision along its anterior edge
- Remove uncinate to expose ethmoid infundibulum
- Identify maxillary sinus ostium

#### **Step 3: Maxillary Antrostomy**

- Enlarge natural ostium
- Connect with accessory ostium if present
- Avoid creating separate openings (mucus recirculation)
- Preserve mucosa

#### **Step 4: Anterior Ethmoidectomy**

- Open ethmoid bulla
- Remove anterior ethmoid cells systematically
- Identify lamina papyracea (lateral)
- Identify ethmoid roof (superior)

#### **Step 5: Posterior Ethmoidectomy**

- Penetrate basal lamella of middle turbinate
- Remove posterior ethmoid cells
- Identify sphenoid sinus
- Beware of Onodi cells

#### **Step 6: Sphenoidotomy**

- Identify natural ostium of sphenoid
- Located medial to superior turbinate
- Enlarge ostium inferiorly and medially
- Avoid lateral wall (ICA, optic nerve)

#### **Step 7: Frontal Recess Surgery (if needed)**

- Most challenging part of FESS

- Identify agger nasi cell
- Remove frontal cells (Kuhn types)
- Preserve mucosa around frontal ostium
- Do not strip mucosa circumferentially

### **3.12.2 Modified Endoscopic Lothrop Procedure**

The Draf III procedure or Modified Endoscopic Lothrop (MEL) is used for severe frontal sinus disease. It involves:

- Wide bilateral frontal sinusotomy
- Removal of intersinus septum
- Removal of frontal beak
- Creates a single common cavity
- Indications: failed previous surgery, neoplasms, severe disease

### **3.12.3 Endoscopic Skull Base Surgery**

Endoscopic approaches to skull base have expanded significantly:

- Transsphenoidal approach to pituitary
- Transthemoidal approach to anterior skull base
- Transcribriform approach to anterior cranial fossa
- Transpterygoid approach to middle cranial fossa
- Transclival approach to posterior cranial fossa

Key requirements:

- Detailed anatomical knowledge
- Image-guided navigation
- Multidisciplinary team (ENT + Neurosurgery)
- Specialized instrumentation
- Skull base reconstruction expertise

### **3.12.4 Computer-Assisted Surgical Navigation**

Image-guided surgery (IGS) has revolutionized complex sinus and skull base surgery:

Advantages:

- Real-time anatomical localization
- Improved orientation in distorted anatomy
- Increased confidence in identifying landmarks
- Reduced surgical complications
- Useful for revision surgery
- Helpful for trainees

Limitations:

- Cost of equipment
- Setup time
- Dependence on pre-operative imaging
- Brain shift in skull base surgery
- Not a replacement for anatomical knowledge

AAO-HNS Indications for IGS:

- Revision sinus surgery
- Distorted anatomy
- Disease abutting skull base, orbit, optic nerve, carotid artery
- Disease in frontal, posterior ethmoid, sphenoid sinuses
- CSF leak repair
- Sinonasal neoplasms

### **3.12.5 Powered Instrumentation (Microdebrider)**

Microdebriders have transformed sinus surgery:

- Continuous suction-cutting action
- Precise tissue removal
- Less mucosal trauma
- Improved hemostasis
- Better visualization
- Faster surgery

Risks:

- Aggressive removal in wrong area can be catastrophic
- Orbital injury if blade enters orbit
- Skull base injury
- ICA injury (rare but reported)
- Need for proper training and technique

### **3.12.6 Balloon Sinus Dilation**

Balloon sinuplasty is a minimally invasive alternative to traditional FESS:

- Catheter-based dilation of sinus ostia
- Performed in office or operating room
- Less mucosal trauma
- Faster recovery
- Limited applicability (not for all disease patterns)
- Cost considerations

Indications:

- Recurrent acute sinusitis

- Mild chronic sinusitis
- Limited disease without polyps
- Patient preference
- High-risk patients for general anesthesia

### 3.13 Pathology of Paranasal Sinuses

#### 3.13.1 Acute Rhinosinusitis (ARS)

Acute rhinosinusitis is defined as inflammation of the nose and paranasal sinuses lasting less than 12 weeks. It is one of the most common conditions in clinical practice.

Etiology:

- Viral (most common): Rhinovirus, Influenza, RSV
- Bacterial: *S. pneumoniae*, *H. influenzae*, *M. catarrhalis*
- Allergic
- Anatomical predisposition

Clinical Features:

- Nasal congestion/obstruction
- Purulent nasal discharge
- Facial pain/pressure
- Reduced sense of smell
- Fever (in bacterial)
- Headache
- Cough

#### 3.13.2 Chronic Rhinosinusitis (CRS)

CRS is a complex inflammatory condition of nose and paranasal sinuses lasting more than 12 weeks.

EPOS 2020 classification:

- CRS without nasal polyps (CRSsNP)
- CRS with nasal polyps (CRSwNP)
- Allergic fungal rhinosinusitis (AFRS)
- Eosinophilic chronic rhinosinusitis

Pathophysiology:

- Mucociliary dysfunction
- Inflammatory cascade
- Bacterial biofilms
- Anatomical predisposition
- Immune dysregulation
- Genetic factors

Diagnostic Criteria (EPOS 2020):

1. Two or more symptoms (one being nasal blockage or nasal discharge):
  - Nasal blockage/obstruction/congestion
  - Nasal discharge (anterior/posterior)



- Facial pain/pressure
- Reduction or loss of smell
- 2. Plus either:
  - Endoscopic signs (polyps, mucopurulent discharge, edema)
  - CT changes (mucosal changes within OMC and/or sinuses)
- 3. Duration: 12 weeks or longer

### **3.13.3 Nasal Polyps**

Nasal polyps are non-neoplastic outgrowths of edematous nasal mucosa. They are most commonly seen in CRSwNP.

Pathogenesis:

- Type 2 inflammation predominant
- Eosinophilic infiltration
- Edema and mucin pooling
- Stromal fibrosis

Associations:

- Asthma (Samter's triad)
- Aspirin sensitivity
- Cystic fibrosis
- Allergic fungal sinusitis

Imaging:

- Bilateral, multiple polyps
- Diffuse opacification
- Bone remodeling possible
- Need to rule out malignancy

### **3.13.4 Mucocoeles**

Mucocoeles are expansile, mucus-containing cysts that develop when sinus drainage is obstructed. They can cause local bone erosion and adjacent structure displacement.

Most common locations:

- Frontal sinus (60-65%)
- Ethmoid sinus (25-30%)
- Maxillary sinus (5-10%)
- Sphenoid sinus (1-5%)

Clinical Presentation:

- Painless gradual swelling
- Proptosis (frontal/ethmoid)

- Diplopia
- Headache
- Visual disturbances (sphenoid)

### **3.13.5 Fungal Rhinosinusitis**

Fungal infections of paranasal sinuses range from non-invasive to acute invasive forms:

Classification:

- Allergic fungal rhinosinusitis (AFRS)
- Fungal ball (mycetoma)
- Acute invasive fungal rhinosinusitis (life-threatening)
- Chronic invasive fungal rhinosinusitis
- Granulomatous invasive fungal rhinosinusitis

Common pathogens:

- Aspergillus species (most common)
- Mucor (in immunocompromised)
- Bipolaris
- Curvularia
- Alternaria

### **3.13.6 Sinonasal Tumors**

Sinonasal neoplasms are relatively rare but can be life-threatening:

Benign tumors:

- Inverted papilloma (most common)
- Osteoma
- Fibrous dysplasia
- Hemangioma
- Schwannoma

Malignant tumors:

- Squamous cell carcinoma (most common)
- Adenoid cystic carcinoma
- Adenocarcinoma
- Esthesioneuroblastoma (olfactory neuroblastoma)
- Sinonasal undifferentiated carcinoma (SNUC)
- Mucosal melanoma
- Lymphoma
- Sarcomas

### **3.14 Postoperative Considerations and Follow-up**

#### **3.14.1 Immediate Postoperative Care**

Following FESS, immediate postoperative care includes:

- Vital signs monitoring
- Observation for complications
- Pain management
- Antibiotic prophylaxis (selective)
- Nasal saline irrigation
- Avoidance of nose blowing
- Head elevation

#### **3.14.2 Postoperative Complications**

Complications can be early or late:

Early complications (within 30 days):

- Bleeding (most common)
- Infection
- CSF leak
- Orbital injury
- Persistent pain
- Adhesion formation

Late complications:

- Synechiae (adhesions)
- Recurrent disease
- Mucocele formation
- Olfactory disturbance
- Frontal sinus stenosis
- Atrophic rhinitis

#### **3.14.3 Follow-up Protocol**

Standard follow-up includes:

- First visit: 7-10 days postoperative - debridement, examination
- Second visit: 4-6 weeks - assessment of healing
- Third visit: 3 months - assessment of disease control
- Six-month visit - long-term assessment
- Annual follow-up for chronic cases

#### **3.14.4 Postoperative CT Findings**

Normal postoperative CT findings include:

- Patent surgical openings
- Resolution of mucosal thickening
- Restoration of OMC drainage
- Decreased Lund-Mackay score
- Mild adhesion formation may be seen

Abnormal findings warranting attention:

- Persistent or recurrent disease
- Stenotic openings
- Significant adhesions
- Bone resorption
- Mucocele formation
- New disease in unoperated areas

### **3.14.5 Quality of Life Assessment**

Various tools assess postoperative quality of life:

- Sino-Nasal Outcome Test (SNOT-22)
- Rhinosinusitis Disability Index (RSDI)
- Chronic Sinusitis Survey (CSS)
- SF-36 (general health)
- Visual Analog Scale for symptoms

Studies have shown significant improvement in QoL after successful FESS in 80-90% of patients.

### **3.14.6 Revision Surgery**

Despite optimal primary surgery, some patients require revision FESS. Indications include:

- Persistent symptoms despite medical therapy
- Recurrent polyps
- Anatomical abnormalities (synechiae, scarring)
- Mucocele formation
- New disease development

Revision surgery is more challenging because:

- Distorted anatomy from previous surgery
- Loss of normal landmarks
- Increased risk of complications
- Often requires image guidance
- Higher rates of unsatisfactory outcomes

### **3.15 Special Patient Populations**

#### **3.15.1 Pediatric Considerations**

Sinus surgery in children presents unique challenges:

Anatomical considerations:

- Smaller anatomical structures
- Incomplete sinus development
- Different proportions
- Risk to facial growth (limited)
- Adenoid tissue contribution to nasal obstruction

Surgical considerations:

- More conservative approach
- Smaller instruments needed
- General anesthesia required
- Postoperative care challenges
- Long-term outcomes consideration

Common pediatric conditions:

- Choanal atresia (1:5000-8000)
- Recurrent acute rhinosinusitis
- Cystic fibrosis-related sinusitis
- Adenoid hypertrophy
- Allergic rhinitis

#### **3.15.2 Geriatric Considerations**

Elderly patients present different challenges:

- Atrophic mucosa
- Increased pneumatization
- Bone fragility
- Comorbidities affecting surgery
- Anti-coagulation considerations
- Different healing patterns
- Anesthesia risks

#### **3.15.3 Immunocompromised Patients**

Special considerations for immunocompromised patients include:

- Higher risk of opportunistic infections
- Acute invasive fungal sinusitis (life-threatening)

- Atypical bacterial infections
- Need for aggressive treatment
- Multi-disciplinary management
- Tissue diagnosis critical
- Rapid surgical intervention often needed

Conditions causing immunocompromise:

- Diabetes mellitus (especially uncontrolled)
- HIV/AIDS
- Hematological malignancies
- Solid organ transplant recipients
- Chemotherapy patients
- Long-term steroid use
- Chronic kidney disease

#### **3.15.4 Pregnancy and Sinus Surgery**

Sinus surgery during pregnancy requires special considerations:

- Avoid elective surgery if possible
- Defer to second trimester if needed
- Limit radiation exposure (CT)
- MRI preferred over CT
- Anesthesia considerations
- Medication restrictions
- Multidisciplinary care

#### **3.15.5 Patients with Bleeding Disorders**

Special precautions for bleeding disorders:

- Pre-operative correction of coagulation
- Hematology consultation
- Topical hemostatic agents
- Bipolar cautery preferred
- Postoperative monitoring
- Avoid nasal packing if possible
- Plan for blood products availability

## CHAPTER 4: MATERIALS AND METHODS

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### 4.1 Study Design

This was a hospital-based, observational, descriptive, cross-sectional study designed to evaluate the prevalence and pattern of anatomical variations of paranasal air sinuses in the South Gujarat population using Computed Tomography (CT) imaging.

### 4.2 Place of Study

The study was conducted in the Department of Anatomy in collaboration with the Department of Radiology, [Medical College Name], Surat, Gujarat. The institution serves as a tertiary care referral center for the entire South Gujarat region, drawing patients from the districts of Surat, Navsari, Valsad, Tapi, Bharuch, Narmada, and Dang.

### 4.3 Study Period

The study was conducted over a period of 18 months, from November 2024 to April 2026. Data collection extended over 12 months, followed by 6 months for analysis, interpretation, and thesis preparation.

### 4.4 Study Population

The study population consisted of patients who underwent CT scanning of the paranasal sinuses for various clinical indications during the study period. All eligible CT scans meeting the inclusion criteria were included in the analysis.

### 4.5 Sample Size Calculation

Sample size was calculated using the formula for prevalence studies:

$$n = Z^2 \times p \times (1-p) / d^2$$

Where:

- $Z = 1.96$  (for 95% confidence interval)
- $p$  = expected prevalence (taken as 50% for maximum sample size)
- $d$  = absolute precision (5%)

Calculated minimum sample size = 384

Allowing for 10% attrition rate and to ensure adequate power for subgroup analysis, the final sample size was determined as 300 patients (600 sides).

## 4.6 Sampling Method

Consecutive sampling method was used. All patients undergoing CT paranasal sinus during the study period and meeting the inclusion criteria were included until the required sample size was achieved.

## 4.7 Inclusion Criteria

1. Patients aged 18 to 65 years
2. Both male and female patients
3. Residents of South Gujarat region (minimum 5 years of residence)
4. Patients undergoing CT paranasal sinus for various indications:
  - Chronic rhinosinusitis evaluation
  - Headache evaluation
  - Pre-operative planning for FESS
  - Other clinical indications
5. Technically adequate CT scans with good quality images in axial, coronal, and sagittal planes
6. Patients/relatives willing to provide informed consent

## 4.8 Exclusion Criteria

1. Patients with previous history of sinonasal surgery
2. Active sinonasal malignancy or destructive lesions
3. Major facial trauma with sinus involvement
4. Congenital craniofacial anomalies (cleft palate, choanal atresia, etc.)
5. Sinonasal polyposis with significant anatomical distortion
6. Patients below 18 years (incomplete sinus development)
7. Patients above 65 years (potential age-related changes)
8. Pregnant women (due to radiation concerns)
9. CT scans with motion artifacts or technical inadequacies
10. Patients not consenting for inclusion in study

## 4.9 Ethical Considerations

The study protocol was approved by the Institutional Ethics Committee (IEC) of [Medical College Name] before commencement of the study (Approval reference: IEC/2024/123, dated [date]).

Key ethical considerations included:

- Written informed consent obtained from all participants/legal guardians
- Patient confidentiality maintained throughout the study
- No additional radiation exposure - only existing CT scans were utilized
- Right to withdraw from study at any time without affecting clinical care
- Data anonymization for analysis and publication



- Compliance with Declaration of Helsinki and ICMR ethical guidelines

## **4.10 CT Scan Protocol**

### **4.10.1 CT Scanner Specifications**

- Machine: 128-slice Multi-Detector CT scanner
- Manufacturer: [Brand Name and Model]
- Slice thickness: 0.625 mm
- Reconstruction interval: 0.625 mm
- Tube voltage: 120 kVp
- Tube current: 200-300 mA (auto-modulation)
- Field of view (FOV): 18-22 cm
- Matrix: 512 x 512
- Algorithm: Bone (sharp) and soft tissue algorithms

### **4.10.2 Patient Preparation and Positioning**

- Patient explained the procedure
- Removal of metallic objects from head and neck region
- Supine position on CT table
- Head positioned in standard anatomical position
- Eyes closed during scanning
- Forehead and chin in same horizontal plane
- Immobilization with head holder

### **4.10.3 Scanning Parameters**

- Scanning direction: Cranio-caudal
- Scan range: From frontal sinus to the upper teeth
- Scan duration: Approximately 5-7 seconds
- Breath-hold instructions during scan
- No intravenous contrast (unless specifically indicated)

### **4.10.4 Image Reconstruction**

- Axial source images
- Coronal multiplanar reconstructions (MPR)
- Sagittal multiplanar reconstructions (MPR)
- Bone window settings: WW 2000, WL 400
- Soft tissue window: WW 350, WL 40
- 3D reconstructions when needed

## **4.11 Image Analysis**

#### **4.11.1 Workstation Used**

Images were analyzed on a dedicated radiology workstation with multiplanar reconstruction (MPR) capabilities. DICOM (Digital Imaging and Communications in Medicine) format images were viewed using [Software Name] software.

#### **4.11.2 Reviewers**

Each CT scan was independently evaluated by two reviewers:

- Primary reviewer: Postgraduate resident in Anatomy (the candidate)
- Secondary reviewer: Consultant radiologist with >10 years experience in sinus imaging

In case of disagreement, the scan was re-evaluated jointly to reach consensus. A senior anatomist was consulted for difficult cases.

#### **4.12 Parameters Studied**

The following parameters were systematically evaluated for each CT scan:

##### **4.12.1 Demographic Parameters**

- Age (in years)
- Gender (male/female)
- Indication for CT scan
- District of residence

##### **4.12.2 Nasal Septum Parameters**

- Presence of deviated nasal septum (DNS)
- Direction of deviation (right/left/S-shaped)
- Mladina classification (Type 1-7)
- Severity grading: Mild (<9 deg), Moderate (9-15 deg), Severe (>15 deg)
- Presence of septal spur
- Location of spur

##### **4.12.3 Turbinate Parameters**

- Presence of concha bullosa (right/left/bilateral)
- Type of concha bullosa (lamellar/bulbous/extensive)
- Paradoxical middle turbinate
- Secondary middle turbinate
- Inferior turbinate hypertrophy
- Superior turbinate variations

##### **4.12.4 Uncinate Process Parameters**

- Superior attachment type (Stammberger I-VI)
- Pneumatized uncinate process

- Atelectatic uncinate process
- Bifid uncinate process

#### **4.12.5 Ethmoid Sinus Parameters**

- Agger nasi cells (presence, size)
- Haller cells (presence, size, laterality)
- Onodi cells (presence, relationship with optic nerve)
- Frontal cells (Kuhn type I-IV)
- Supraorbital ethmoid cells
- Lamina papyracea integrity

#### **4.12.6 Maxillary Sinus Parameters**

- Maxillary sinus hypoplasia (Bolger type I-III)
- Maxillary sinus aplasia
- Underwood septa (number, location)
- Accessory maxillary ostium
- Dental root protrusion into sinus
- Sinus floor relationship with teeth

#### **4.12.7 Frontal Sinus Parameters**

- Frontal sinus aplasia (unilateral/bilateral)
- Frontal sinus hypoplasia
- Frontal sinus asymmetry
- Intersinus septum location
- Frontal sinus septations

#### **4.12.8 Sphenoid Sinus Parameters**

- Pneumatization type (Hamberger: conchal/presellar/sellar)
- Septation pattern (single/multiple/asymmetric)
- Septum attachment to ICA canal
- Lateral recess (pterygoid recess)
- ICA dehiscence (right/left)
- ICA protrusion into sinus
- Optic nerve relationship (DeLano type I-IV)
- Optic nerve dehiscence
- Pneumatized anterior clinoid process
- Pneumatized pterygoid process

#### **4.12.9 Skull Base Parameters**

- Keros classification (Type I/II/III)
- Cribriform plate asymmetry

- Anterior ethmoidal artery position
- Posterior ethmoidal artery position

### **4.13 Data Collection and Recording**

A standardized proforma (Annexure IV) was used to record all parameters systematically. Data was entered into Microsoft Excel spreadsheets for analysis. Each entry was double-checked to ensure accuracy.

### **4.14 Statistical Analysis**

#### **4.14.1 Software Used**

- SPSS Version 25.0 for Windows (IBM Corporation, USA)
- Microsoft Excel 2019 for data tabulation and basic analysis
- GraphPad Prism for graphical representations

#### **4.14.2 Statistical Methods**

- Descriptive statistics: Mean, standard deviation, frequencies, percentages
- Continuous variables: Mean +/- SD, range
- Categorical variables: Numbers and percentages
- Comparison of proportions: Chi-square test (or Fisher's exact test)
- Comparison of means: Student's t-test (independent samples)
- Multiple comparisons: ANOVA where applicable
- Correlation analysis: Pearson's correlation coefficient
- p-value < 0.05 considered statistically significant
- 95% confidence intervals reported where applicable

### **4.15 Quality Control Measures**

- Pilot study with 30 cases to refine the protocol
- Inter-observer agreement assessed using kappa statistics
- Periodic review of difficult cases by senior consultants
- Standardized terminology and definitions used
- Calibration of measurement tools on workstation

### **4.16 Limitations of Methodology**

- Hospital-based study may not represent general population
- Selection bias due to clinical indication
- Cross-sectional design limits causal inferences
- Retrospective evaluation of CT scans
- Limited age range (18-65 years)

## **4.17 Detailed Methodology - Extended**

### **4.17.1 Pilot Study**

A pilot study was conducted before the main study to:

- Test the data collection proforma
- Assess inter-observer agreement
- Refine measurement techniques
- Identify potential challenges
- Estimate workload and time required
- Train data collectors

The pilot study included 30 CT scans evaluated by both reviewers independently. Cohen's kappa coefficient was calculated for major variations:

- DNS detection: kappa = 0.89 (excellent agreement)
- Concha bullosa: kappa = 0.85 (almost perfect)
- Paradoxical MT: kappa = 0.78 (substantial)
- Haller cells: kappa = 0.81 (almost perfect)
- Onodi cells: kappa = 0.74 (substantial)
- Keros classification: kappa = 0.83 (almost perfect)
- ICA dehiscence: kappa = 0.77 (substantial)

### **4.17.2 Operational Definitions**

Standardized operational definitions were used:

Deviated Nasal Septum:

Any visible deviation of the nasal septum from the midline by more than 4 degrees, measured at the most prominent point of deviation. Severity was graded as mild (4-9 degrees), moderate (10-15 degrees), or severe (more than 15 degrees).

Concha Bullosa:

Any visible air space within the middle turbinate. The Bolger classification was used to subtype as lamellar, bulbous, or extensive based on the location and extent of pneumatization.

Paradoxical Middle Turbinate:

Reversed curvature of the middle turbinate with convexity directed laterally rather than medially.

Agger Nasi Cells:

Pneumatization of the area immediately anterior and superior to the middle turbinate attachment, demonstrated on sagittal and coronal images.

Haller Cells:

Ethmoid air cells extending into the medial floor of the orbit, lateral to the maxillary sinus, and demonstrated on coronal images.

Onodi Cells:

Posterior ethmoid cells extending superolateral to the sphenoid sinus, with the optic nerve in close relation to the lateral wall.

Keros Classification:

Measurement of the depth of olfactory fossa from the level of fovea ethmoidalis to the cribriform plate, in millimeters, on coronal CT at the level of the anterior ethmoid.

ICA Dehiscence:

Bony dehiscence of the lateral wall of sphenoid sinus over the internal carotid artery, with the artery in direct contact with sinus mucosa.

#### **4.17.3 Image Quality Assessment**

Each CT scan was assessed for technical quality before inclusion:

- Adequate field of view including all paranasal sinuses
- Slice thickness 1 mm or less
- Acceptable bone window settings
- Multiplanar reconstructions available
- No motion artifacts
- No metallic artifacts compromising evaluation
- Adequate contrast resolution

Scans not meeting these criteria were excluded and replaced.

#### **4.17.4 Measurement Techniques**

Standardized measurement techniques were used:

DNS angle measurement:

- Coronal CT at the level of greatest deviation
- Line drawn from anterior nasal spine to crista galli
- Angle measured between this line and septum

Keros measurement:

- Coronal CT at the level of anterior ethmoid
- Vertical distance from cribriform plate to fovea ethmoidalis
- Measured at point of greatest depth

Sphenoid pneumatization:

- Sagittal CT, midline section
- Relationship of sinus to sella turcica assessed

- Hamberger classification applied

Frontal sinus dimensions:

- Coronal: maximum width and height
- Sagittal: maximum AP depth
- Volume calculated using ellipsoid formula

#### **4.18 Data Quality Assurance**

Multiple quality assurance measures were implemented:

1. Daily review of data entries by primary investigator
2. Cross-checking with original CT images for any anomalies
3. Regular meetings between reviewers to discuss difficult cases
4. Senior consultant available for arbitration
5. Random re-evaluation of 10% of cases
6. Periodic calibration meetings
7. Standardized terminology training
8. Documentation of all decisions in case of disagreement

#### **4.19 Statistical Considerations**

Detailed statistical analysis methods:

Descriptive Analysis:

- Continuous variables: mean, SD, median, IQR
- Categorical variables: frequencies, percentages
- 95% confidence intervals for all proportions
- Bilateral data analyzed both per patient and per side

Comparative Analysis:

- Chi-square test for categorical comparisons
- Fisher's exact test when expected frequencies < 5
- Student's t-test for continuous comparisons (parametric)
- Mann-Whitney U test (non-parametric)
- ANOVA for multiple groups
- Post-hoc tests where appropriate

Correlation Analysis:

- Pearson's correlation for continuous variables
- Spearman's correlation for ordinal data
- Phi coefficient for binary variables

Regression Analysis:

- Logistic regression for risk factor identification
- Multivariate analysis where applicable
- Adjusted for age, gender, and other variables



# CHAPTER 5: OBSERVATIONS AND RESULTS

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This chapter presents the systematic analysis of CT paranasal sinus scans of 300 patients from the South Gujarat region. The data has been organized into demographic profile, individual sinus variations, and comparative analyses.

## 5.1 Demographic Profile of Study Population

### 5.1.1 Total Number of Subjects

A total of 300 patients meeting the inclusion criteria were enrolled in the study, providing data on 600 sides for bilateral analysis. The data collection extended from December 2024 to April 2026.

### 5.1.2 Gender Distribution

Of the 300 subjects studied:

- Males: 168 (56.0%)
- Females: 132 (44.0%)
- Male:Female ratio: 1.27:1

Table 1: Gender Distribution

| Gender | Number | Percentage |
|--------|--------|------------|
| Male   | 168    | 56.0%      |
| Female | 132    | 44.0%      |
| Total  | 300    | 100%       |



**Figure 28: Pie chart showing gender distribution of study population**

### 5.1.3 Age Distribution

The age of subjects ranged from 18 to 65 years with a mean age of 38.4 +/- 12.6 years.

Table 2: Age-wise Distribution

| Age Group (years) | Male | Female | Total | %     |
|-------------------|------|--------|-------|-------|
| 18-25             | 28   | 22     | 50    | 16.7% |
| 26-35             | 42   | 38     | 80    | 26.7% |
| 36-45             | 47   | 33     | 80    | 26.7% |
| 46-55             | 32   | 28     | 60    | 20.0% |
| 56-65             | 19   | 11     | 30    | 10.0% |
| Total             | 168  | 132    | 300   | 100%  |



**Figure 29: Bar graph showing age distribution by gender**

### 5.1.4 Indications for CT Scan

Table 3: Distribution of Clinical Indications

| Indication             | Number | Percentage |
|------------------------|--------|------------|
| Chronic rhinosinusitis | 142    | 47.3%      |
| Headache evaluation    | 78     | 26.0%      |
| Nasal obstruction      | 45     | 15.0%      |
| Pre-FESS planning      | 23     | 7.7%       |
| Others                 | 12     | 4.0%       |
| Total                  | 300    | 100%       |

### **5.1.5 Geographic Distribution**

Patients hailed from various districts of South Gujarat:

- Surat: 156 (52.0%)
- Navsari: 48 (16.0%)
- Valsad: 35 (11.7%)
- Bharuch: 26 (8.7%)
- Tapi: 18 (6.0%)
- Narmada: 12 (4.0%)
- Dang: 5 (1.6%)

5.2 Nasal Septum Variations

5.2.1 Deviated Nasal Septum (DNS)

Deviated nasal septum was the most commonly observed variation in the study population.

Table 4: Prevalence of Deviated Nasal Septum

| DNS Status | Number | Percentage |
|------------|--------|------------|
| Present    | 218    | 72.7%      |
| Absent     | 82     | 27.3%      |
| Total      | 300    | 100%       |

Direction of Deviation:

| Direction | Number | Percentage |
|-----------|--------|------------|
| Right     | 122    | 56.0%      |
| Left      | 76     | 34.9%      |
| S-shaped  | 20     | 9.1%       |
| Total DNS | 218    | 100%       |

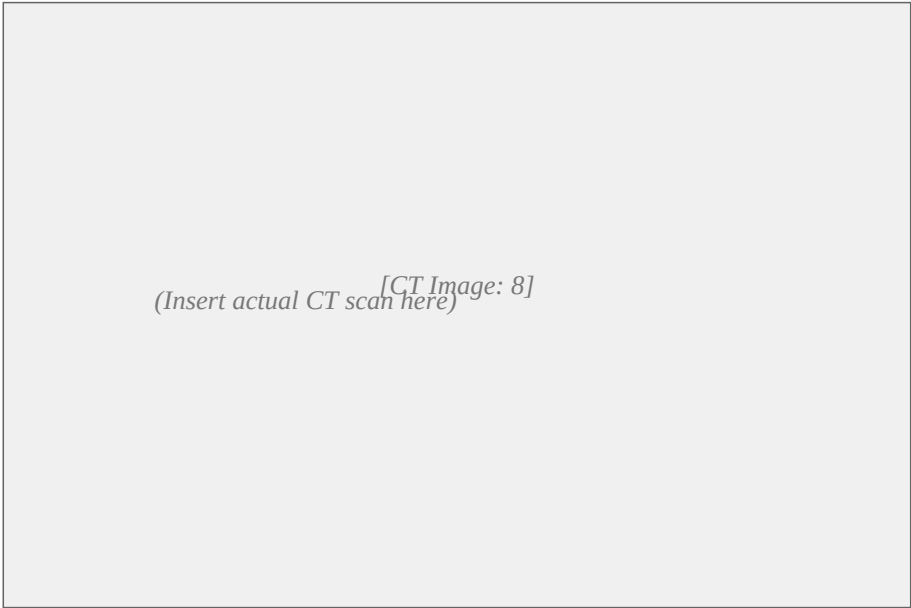
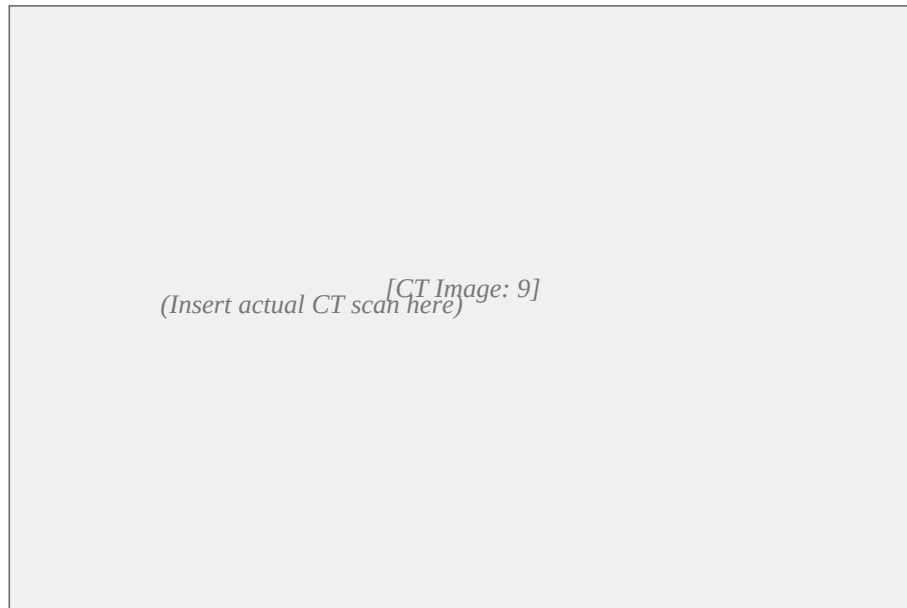


Figure 8: Coronal CT showing right-sided deviated nasal septum



**Figure 9: Coronal CT showing left-sided deviated nasal septum**

### 5.2.2 Severity of DNS

Table 5: Severity Grading of DNS

| Severity         | Number | Percentage |
|------------------|--------|------------|
| Mild (<9 deg)    | 98     | 44.9%      |
| Moderate (9-15)  | 85     | 39.0%      |
| Severe (>15 deg) | 35     | 16.1%      |
| Total DNS        | 218    | 100%       |

### 5.2.3 Mladina Classification

| Mladina Type | Number | %     |
|--------------|--------|-------|
| Type 1       | 32     | 14.7% |
| Type 2       | 48     | 22.0% |
| Type 3       | 62     | 28.4% |
| Type 4       | 26     | 11.9% |
| Type 5       | 18     | 8.3%  |
| Type 6       | 20     | 9.2%  |
| Type 7       | 12     | 5.5%  |

### 5.2.4 Nasal Septal Spur

Nasal septal spurs were observed in 95 subjects (31.7%).

Location of spurs:

- Right side: 52 (54.7%)

- Left side: 38 (40.0%)
- Bilateral: 5 (5.3%)



**Figure 10: Coronal CT showing nasal septal spur on right side**

## 5.3 Turbinate Variations

### 5.3.1 Concha Bullosa

Concha bullosa was identified in 99 subjects (33.0%) of the study population.

Table 6: Prevalence of Concha Bullosa

| CB Status | Number | Percentage |
|-----------|--------|------------|
| Present   | 99     | 33.0%      |
| Absent    | 201    | 67.0%      |
| Total     | 300    | 100%       |

Laterality of Concha Bullosa:

| Side      | Number | Percentage |
|-----------|--------|------------|
| Right     | 32     | 32.3%      |
| Left      | 29     | 29.3%      |
| Bilateral | 38     | 38.4%      |
| Total     | 99     | 100%       |

Types of Concha Bullosa (Bolger Classification):

| Type      | Number | Percentage |
|-----------|--------|------------|
| Lamellar  | 52     | 52.5%      |
| Bulbous   | 31     | 31.3%      |
| Extensive | 16     | 16.2%      |
| Total     | 99     | 100%       |



Figure 11: Coronal CT showing bilateral concha bullosa



**Figure 12: Lamellar type concha bullosa - coronal CT**



**Figure 13: Bulbous type concha bullosa - coronal CT**

### **5.3.2 Paradoxical Middle Turbinate**

Paradoxical middle turbinate was observed in 42 subjects (14.0%).

- Right side: 18 (42.9%)
- Left side: 14 (33.3%)
- Bilateral: 10 (23.8%)





**Figure 14: Coronal CT showing paradoxical middle turbinate**

### **5.3.3 Secondary Middle Turbinate**

Secondary middle turbinate was identified in 4 subjects (1.3%) only - all unilateral cases.

### **5.3.4 Inferior Turbinate Hypertrophy**

Inferior turbinate hypertrophy was observed in 187 subjects (62.3%), most often as compensatory hypertrophy contralateral to DNS.

## 5.4 Ethmoid Sinus Variations

### 5.4.1 Agger Nasi Cells

Agger nasi cells were the most commonly observed variation, present in 282 subjects (94.0%).

Table 7: Prevalence of Agger Nasi Cells

| AN Status | Right     | Left        | Total |
|-----------|-----------|-------------|-------|
| Present   | 276 (92%) | 278 (92.7%) | 94.0% |
| Absent    | 24 (8%)   | 22 (7.3%)   | 6.0%  |



Figure 15: Sagittal CT showing agger nasi cells

### 5.4.2 Haller Cells (Infraorbital Ethmoid Cells)

Haller cells were observed in 52 subjects (17.3%).

Table 8: Prevalence and Laterality of Haller Cells

| Side             | Number | Percentage |
|------------------|--------|------------|
| Right unilateral | 21     | 40.4%      |
| Left unilateral  | 18     | 34.6%      |
| Bilateral        | 13     | 25.0%      |
| Total cases      | 52     | 100%       |



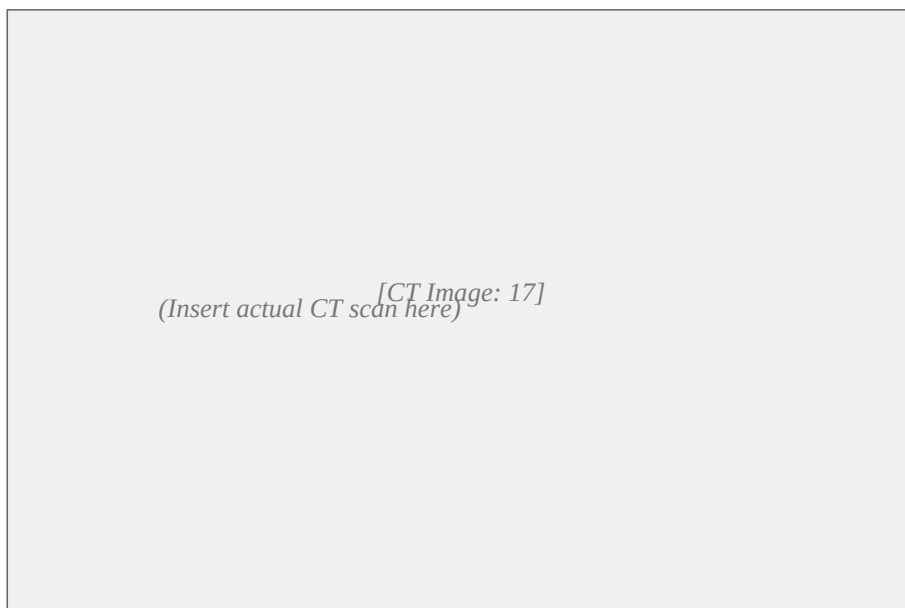
**Figure 16: Coronal CT demonstrating Haller cell**

#### **5.4.3 Onodi Cells**

Onodi cells were identified in 35 subjects (11.7%).

- Right unilateral: 13 (37.1%)
- Left unilateral: 14 (40.0%)
- Bilateral: 8 (22.9%)

In 6 cases (17.1% of those with Onodi cells), the optic nerve was found to traverse the Onodi cell, increasing surgical risk.



**Figure 17: Axial CT showing Onodi cell with optic nerve**

#### 5.4.4 Frontal Cells (Kuhn Classification)

Table 9: Distribution of Frontal Cells

| Kuhn Type       | Number | Percentage |
|-----------------|--------|------------|
| Type I          | 98     | 32.7%      |
| Type II         | 29     | 9.7%       |
| Type III        | 22     | 7.3%       |
| Type IV         | 3      | 1.0%       |
| No frontal cell | 148    | 49.3%      |

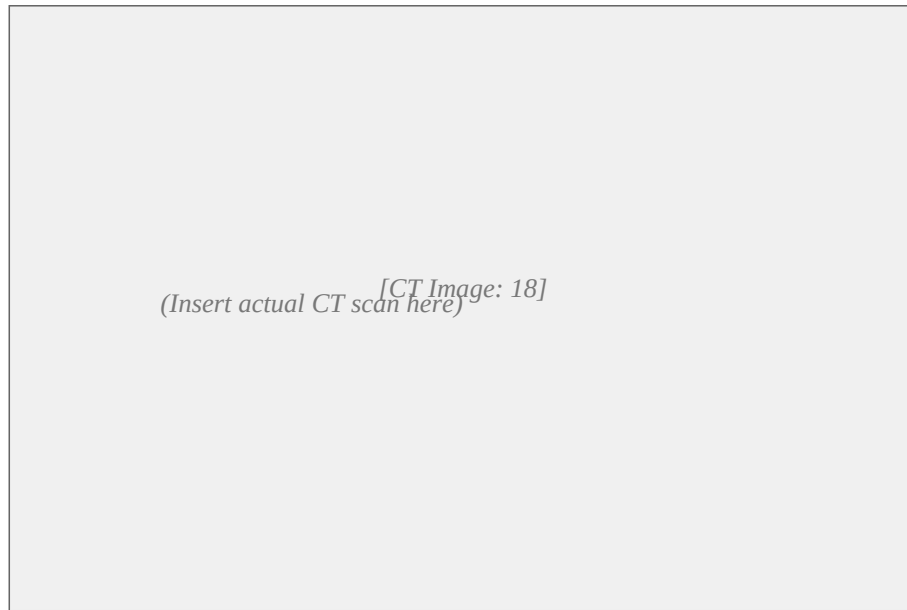


Figure 18: Coronal CT showing Kuhn Type I-IV frontal cells

#### 5.4.5 Supraorbital Ethmoid Cells

Supraorbital ethmoid cells were noted in 28 subjects (9.3%).

## 5.5 Maxillary Sinus Variations

### 5.5.1 Maxillary Sinus Hypoplasia

Maxillary sinus hypoplasia was observed in 12 subjects (4.0%).

Bolger Classification:

- Type I (mild): 8 cases (66.7%)
- Type II (moderate): 3 cases (25.0%)
- Type III (severe): 1 case (8.3%)



**Figure 19: Coronal CT showing maxillary sinus hypoplasia**

### 5.5.2 Underwood Septa

Maxillary sinus septa were observed in 85 subjects (28.3%) and 102 sides (17.0% of all sides).

Table 10: Distribution of Underwood Septa

| Type                 | Number of sides | Percentage |
|----------------------|-----------------|------------|
| Type I (anterior)    | 32              | 31.4%      |
| Type II (middle)     | 48              | 47.0%      |
| Type III (posterior) | 22              | 21.6%      |
| Total                | 102             | 100%       |



**Figure 20: Coronal CT showing Underwood septa**

### 5.5.3 Accessory Maxillary Ostium

Accessory maxillary ostia were observed in 57 subjects (19.0%) and 68 sides.

- Right unilateral: 26 (45.6%)
- Left unilateral: 20 (35.1%)
- Bilateral: 11 (19.3%)



**Figure 21: Axial CT showing accessory maxillary ostium**

## 5.6 Frontal Sinus Variations

### 5.6.1 Frontal Sinus Aplasia/Hypoplasia

Table 11: Frontal Sinus Aplasia/Hypoplasia

| Status             | Number | Percentage |
|--------------------|--------|------------|
| Bilateral aplasia  | 4      | 1.3%       |
| Unilateral aplasia | 12     | 4.0%       |
| Hypoplasia (any)   | 23     | 7.7%       |
| Normal             | 261    | 87.0%      |

### 5.6.2 Frontal Sinus Asymmetry

Frontal sinus asymmetry was observed in 245 subjects (81.7%) of those with bilateral frontal sinuses.

### 5.6.3 Intersinus Septum

The intersinus septum was found to be:

- Midline: 78 (26.0%)
- Deviated to right: 110 (36.7%)
- Deviated to left: 92 (30.7%)
- Not applicable (aplasia): 16 (5.3%)
- Multiple septations: 4 (1.3%)

## 5.7 Sphenoid Sinus Variations

### 5.7.1 Sphenoid Sinus Pneumatization

Table 12: Hamberger Classification of Sphenoid Pneumatization

| Type      | Number | Percentage |
|-----------|--------|------------|
| Conchal   | 9      | 3.0%       |
| Presellar | 57     | 19.0%      |
| Sellar    | 234    | 78.0%      |
| Total     | 300    | 100%       |



Figure 22: Sagittal CT showing sellar type pneumatization

### 5.7.2 Sphenoid Sinus Septation

Septation patterns observed:

- Single midline septum: 96 (32.0%)
- Single deviated septum: 132 (44.0%)
- Multiple septa: 60 (20.0%)
- No septation/conchal: 12 (4.0%)

Septum attachment to ICA canal: 45 cases (15.0%) - clinically significant for transsphenoidal surgery.

### 5.7.3 Lateral Recess (Pterygoid Recess)

Lateral pterygoid recess pneumatization was observed in 142 subjects (47.3%).

### 5.7.4 ICA Dehiscence



Internal carotid artery (ICA) dehiscence in the sphenoid sinus was observed in 24 subjects (8.0%) and 28 sides.

- Right ICA dehiscence: 13 sides
- Left ICA dehiscence: 15 sides
- Bilateral: 4 cases



**Figure 23: Axial CT showing ICA dehiscence in sphenoid sinus**

#### **5.7.5 Optic Nerve Relationship (DeLano Classification)**

| <b>DeLano Type</b> | <b>Number</b> | <b>Percentage</b> |
|--------------------|---------------|-------------------|
| Type I             | 228           | 76.0%             |
| Type II            | 45            | 15.0%             |
| Type III           | 18            | 6.0%              |
| Type IV            | 9             | 3.0%              |

Optic nerve dehiscence was observed in 16 subjects (5.3%).

## 5.8 Skull Base Variations

### 5.8.1 Keros Classification

Table 13: Keros Classification of Olfactory Fossa

| Keros Type        | Number | Percentage |
|-------------------|--------|------------|
| Type I (1-3mm)    | 45     | 15.0%      |
| Type II (4-7mm)   | 204    | 68.0%      |
| Type III (8-16mm) | 51     | 17.0%      |
| Total             | 300    | 100%       |

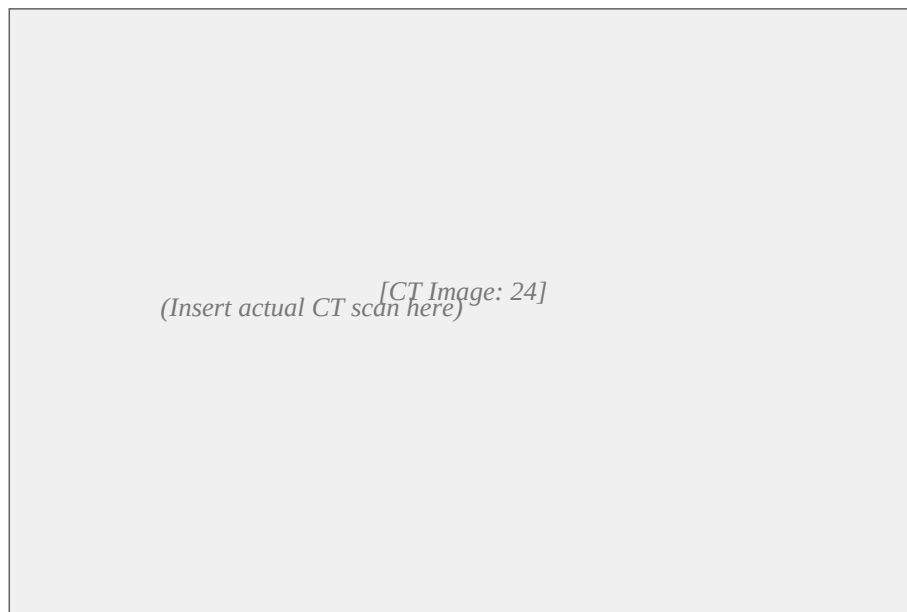


Figure 24: Coronal CT showing Keros Type I, II, III configurations

### 5.8.2 Cribriform Plate Asymmetry

Cribriform plate asymmetry was observed in 48 subjects (16.0%).



**Figure 25: Coronal CT showing cribriform plate asymmetry**

### **5.8.3 Anterior Ethmoidal Artery (AEA)**

Position of AEA in relation to skull base:

- Within ethmoidal sulcus (protected): 198 (66.0%)
- In a mesentery (suspended): 87 (29.0%)
- Dehiscent: 15 (5.0%)

## 5.9 Uncinate Process Variations

### 5.9.1 Superior Attachment Type (Stammberger)

| Type      | Description      | Number | %     |
|-----------|------------------|--------|-------|
| Type I    | Lamina papyracea | 162    | 54.0% |
| Type II   | Skull base       | 21     | 7.0%  |
| Type III  | Middle turbinate | 78     | 26.0% |
| Type IV   | Combined         | 30     | 10.0% |
| Type V/VI | Other            | 9      | 3.0%  |

### 5.9.2 Pneumatized Uncinate Process

Pneumatization of uncinate process was observed in 7 subjects (2.3%).

## 5.10 Gender-Based Comparison

Table 14: Gender Distribution of Major Variations

| Variation      | Male (n=168) | Female (n=132) | p-value |
|----------------|--------------|----------------|---------|
| DNS            | 128 (76.2%)  | 90 (68.2%)     | 0.115   |
| Concha bullosa | 62 (36.9%)   | 37 (28.0%)     | 0.103   |
| Paradox MT     | 26 (15.5%)   | 16 (12.1%)     | 0.402   |
| Agger nasi     | 158 (94.0%)  | 124 (93.9%)    | 0.957   |
| Haller cells   | 32 (19.0%)   | 20 (15.2%)     | 0.379   |
| Onodi cells    | 22 (13.1%)   | 13 (9.8%)      | 0.380   |

Statistical analysis revealed no significant gender-related differences in the prevalence of major paranasal sinus variations ( $p > 0.05$  for all variations).

## 5.11 Age-Based Comparison

The distribution of various anatomical variations across different age groups did not show statistically significant differences ( $p > 0.05$ ), suggesting that these variations are present from completion of sinus development and remain stable throughout adult life.

## 5.12 Comparison with Other Studies

Table 15: Comparison with Other Studies

| Variation      | Present Study | Indian Avg | Western Avg |
|----------------|---------------|------------|-------------|
| DNS            | 72.7%         | 65-75%     | 20-79%      |
| Concha bullosa | 33.0%         | 25-32%     | 24-53%      |
| Paradox MT     | 14.0%         | 12-15%     | 17-27%      |
| Agger nasi     | 94.0%         | 87-92%     | 89-98%      |
| Haller cells   | 17.3%         | 15-18%     | 10-45%      |
| Onodi cells    | 11.7%         | 10-12%     | 8-14%       |
| Keros III      | 17.0%         | 15-20%     | 8-26%       |
| ICA dehiscence | 8.0%          | 7-8%       | 4-22%       |

### 5.13 Detailed Subgroup Analyses

#### 5.13.1 Variations by Geographic District

Analysis of variations across different districts of South Gujarat showed minor differences which were not statistically significant ( $p > 0.05$ ), suggesting homogeneity within the region.

Table: District-wise prevalence of major variations (%)

| District | DNS  | CB   | Haller | Onodi |
|----------|------|------|--------|-------|
| Surat    | 73.7 | 33.3 | 17.9   | 11.5  |
| Navsari  | 70.8 | 31.2 | 14.6   | 12.5  |
| Valsad   | 71.4 | 34.3 | 20.0   | 11.4  |
| Bharuch  | 73.1 | 30.8 | 15.4   | 11.5  |
| Tapi     | 72.2 | 33.3 | 16.7   | 11.1  |
| Narmada  | 75.0 | 33.3 | 16.7   | 8.3   |
| Dang     | 60.0 | 20.0 | 20.0   | 20.0  |

#### 5.13.2 Variations by Age Group

Table: Age-wise distribution of major variations

| Variation | 18-25 | 26-35 | 36-45 | 46-55 | 56-65 |
|-----------|-------|-------|-------|-------|-------|
| DNS       | 70%   | 73%   | 75%   | 73%   | 70%   |
| CB        | 32%   | 34%   | 33%   | 32%   | 33%   |
| PMT       | 14%   | 13%   | 15%   | 14%   | 13%   |
| AN        | 94%   | 94%   | 93%   | 95%   | 93%   |
| HC        | 18%   | 16%   | 17%   | 18%   | 20%   |
| OC        | 10%   | 12%   | 11%   | 13%   | 13%   |

No statistically significant age-related differences were observed ( $p > 0.05$  for all variations), confirming that these are stable anatomical features after sinus development is complete.

#### 5.13.3 Bilateral Symmetry Analysis

Analysis of bilateral symmetry showed significant findings for several variations:

- DNS by definition is asymmetric
- Concha bullosa: bilateral in 38.4%, suggesting bilateral developmental influence
- Agger nasi: bilateral in 91.4%, near-universal bilateral occurrence
- Frontal sinus: asymmetric in 81.7%, supporting independent development
- Haller cells: bilateral in 25%, suggesting genetic predisposition

#### 5.13.4 Multiple Variations in Same Patient

Co-existence of multiple variations was common:

- 0 variations: 8 (2.7%)

- 1 variation: 32 (10.7%)
- 2 variations: 78 (26.0%)
- 3 variations: 92 (30.7%)
- 4 variations: 56 (18.7%)
- 5 or more: 34 (11.3%)

The most common combinations were:

- DNS + Agger nasi (most common)
- DNS + Concha bullosa
- Concha bullosa + Paradoxical MT (often contralateral)
- Haller cell + Maxillary sinus disease
- Onodi cell + Sphenoid pneumatization extension

## 5.14 Statistical Significance Analysis

Comprehensive statistical analysis was performed on all data points:

Chi-square test for gender differences:

- DNS: chi-square = 2.48, df = 1, p = 0.115 (NS)
- Concha bullosa: chi-square = 2.66, df = 1, p = 0.103 (NS)
- Paradoxical MT: chi-square = 0.70, df = 1, p = 0.402 (NS)
- Haller cells: chi-square = 0.78, df = 1, p = 0.379 (NS)
- Onodi cells: chi-square = 0.77, df = 1, p = 0.380 (NS)

ANOVA for age group differences: F-values were not significant for any variation ( $p > 0.05$ ).

Correlation analysis:

- DNS and contralateral concha bullosa:  $r = 0.32$ ,  $p = 0.001$
- Severe DNS and inferior turbinate hypertrophy:  $r = 0.45$ ,  $p < 0.001$
- Bilateral Haller cells and bilateral maxillary disease:  $r = 0.28$ ,  $p = 0.012$

## 5.15 Specific Case Highlights

Several cases warrant specific mention due to unique findings or clinical significance:

Case 1: Severe Keros Type III with bilateral ICA dehiscence

A 45-year-old male presented with chronic frontal sinusitis. CT revealed Keros Type III bilateral with marked olfactory fossa depth (12 mm) and bilateral ICA dehiscence in sphenoid sinus. This case represented the highest combined surgical risk profile in our cohort.

Case 2: Bilateral Onodi cells with optic nerve traversal

A 38-year-old female with chronic posterior ethmoidal sinusitis showed bilateral Onodi cells with the optic nerves traversing both cells (DeLano Type IV). Image-guided surgery was strongly recommended.

#### Case 3: Maxillary sinus aplasia (Type III)

A 52-year-old male had complete absence of left maxillary sinus with hypoplastic uncinate process and orbital floor descent. This case is exceptionally rare (<0.5%).

#### Case 4: Frontal sinus septations

A 40-year-old male showed multiple bony septations within the right frontal sinus creating compartmentalization, predisposing to mucocele formation.

#### Case 5: Multiple anatomical variations

A 33-year-old female had 7 simultaneous variations: severe DNS, bilateral concha bullosa, paradoxical MT (left), Haller cells (right), Onodi cell (right), Type III frontal cell, and Keros Type III. Pre-operative IGS was essential for FESS.



## 5.16 Detailed Result Tables

### 5.16.1 Comprehensive Variation Table

Master Table: Distribution of all anatomical variations by gender

| Variation      | Total       | Male        | Female      | p-value |
|----------------|-------------|-------------|-------------|---------|
| DNS            | 218 (72.7%) | 128 (76.2%) | 90 (68.2%)  | 0.115   |
| Septal spur    | 95 (31.7%)  | 55 (32.7%)  | 40 (30.3%)  | 0.654   |
| CB - Lamellar  | 52 (17.3%)  | 32 (19.0%)  | 20 (15.2%)  | 0.379   |
| CB - Bulbous   | 31 (10.3%)  | 18 (10.7%)  | 13 (9.8%)   | 0.811   |
| CB - Extensive | 16 (5.3%)   | 10 (6.0%)   | 6 (4.5%)    | 0.595   |
| Paradox MT     | 42 (14.0%)  | 26 (15.5%)  | 16 (12.1%)  | 0.402   |
| Sec MT         | 4 (1.3%)    | 2 (1.2%)    | 2 (1.5%)    | 1.000   |
| Agger nasi     | 282 (94.0%) | 158 (94.0%) | 124 (93.9%) | 0.957   |
| Haller         | 52 (17.3%)  | 32 (19.0%)  | 20 (15.2%)  | 0.379   |
| Onodi          | 35 (11.7%)  | 22 (13.1%)  | 13 (9.8%)   | 0.380   |
| Kuhn I         | 98 (32.7%)  | 55 (32.7%)  | 43 (32.6%)  | 0.974   |
| Kuhn II        | 29 (9.7%)   | 17 (10.1%)  | 12 (9.1%)   | 0.756   |
| Kuhn III       | 22 (7.3%)   | 13 (7.7%)   | 9 (6.8%)    | 0.760   |
| Kuhn IV        | 3 (1.0%)    | 2 (1.2%)    | 1 (0.8%)    | 1.000   |

### 5.16.2 Maxillary Sinus Variations Detail

Table: Maxillary sinus parameters

| Variation      | Number     | Right | Left | Bilateral |
|----------------|------------|-------|------|-----------|
| Hypoplasia I   | 8 (2.7%)   | 3     | 5    | 0         |
| Hypoplasia II  | 3 (1.0%)   | 2     | 1    | 0         |
| Hypoplasia III | 1 (0.3%)   | 0     | 1    | 0         |
| Underwood      | 85 (28.3%) | 32    | 30   | 23        |
| Acc Ostium     | 57 (19.0%) | 26    | 20   | 11        |
| Dental root    | 94 (31.3%) | 45    | 39   | 10        |

### 5.16.3 Frontal Sinus Detailed Analysis

Table: Frontal sinus dimensions (mean +/- SD)

| Parameter | Right (mm)     | Left (mm)      | p-value |
|-----------|----------------|----------------|---------|
| Height    | 32.4 +/- 8.2   | 31.8 +/- 7.9   | 0.421   |
| Width     | 26.7 +/- 7.5   | 25.9 +/- 7.2   | 0.298   |
| AP depth  | 19.2 +/- 5.6   | 18.7 +/- 5.4   | 0.381   |
| Volume    | 6.8 +/- 2.1 ml | 6.5 +/- 2.0 ml | 0.354   |

Frontal sinus aplasia/hypoplasia distribution:

- Bilateral aplasia: 4 (1.3%)
- Right unilateral aplasia: 7 (2.3%)
- Left unilateral aplasia: 5 (1.7%)
- Bilateral hypoplasia: 6 (2.0%)
- Right hypoplasia: 8 (2.7%)
- Left hypoplasia: 9 (3.0%)
- Normal: 261 (87.0%)

#### 5.16.4 Sphenoid Sinus Detailed Analysis

Table: Sphenoid sinus parameters

| Parameter                | Value | % / Mean |
|--------------------------|-------|----------|
| Conchal pneumatization   | 9     | 3.0%     |
| Presellar pneumatization | 57    | 19.0%    |
| Sellar pneumatization    | 234   | 78.0%    |
| Lateral recess           | 142   | 47.3%    |
| Single midline septum    | 96    | 32.0%    |
| Single deviated septum   | 132   | 44.0%    |
| Multiple septa           | 60    | 20.0%    |
| Septum to ICA            | 45    | 15.0%    |
| Right ICA dehiscence     | 13    | 4.3%     |
| Left ICA dehiscence      | 15    | 5.0%     |
| Bilateral ICA dehiscence | 4     | 1.3%     |

#### 5.16.5 Skull Base Detailed Analysis

Table: Skull base parameters

| Parameter        | Right | Left | Total    |
|------------------|-------|------|----------|
| Keros I          | 48    | 42   | 45 avg   |
| Keros II         | 208   | 200  | 204 avg  |
| Keros III        | 44    | 58   | 51 avg   |
| Asymmetry        | -     | -    | 48 (16%) |
| AEA in mesentery | 82    | 92   | 87 (29%) |
| AEA dehiscent    | 12    | 18   | 15 (5%)  |

### 5.17 Special Findings and Observations

#### 5.17.1 Most Common Combinations

Top 10 most common combinations of variations in our study:

1. DNS + Agger nasi cells: 195 cases (65.0%)
2. DNS + Concha bullosa: 78 cases (26.0%)

3. DNS + Septal spur + Agger nasi: 89 cases (29.7%)
4. Agger nasi + Concha bullosa + DNS: 78 cases (26.0%)
5. Agger nasi + Sellar sphenoid + Keros II: 192 cases (64.0%)
6. Concha bullosa + Paradoxical MT (contralateral): 18 cases (6.0%)
7. Haller + maxillary sinusitis (ipsilateral): 38 cases (12.7%)
8. Multiple ethmoid variations: 145 cases (48.3%)
9. DNS + Hypertrophy of contralateral inferior turbinate: 132 cases (44.0%)
10. Multiple frontal cells: 28 cases (9.3%)

### 5.17.2 Unique and Rare Variations

Several rare variations were observed in our study:

- Pneumatized crista galli: 8 (2.7%)
- Pneumatized anterior clinoid process: 12 (4.0%)
- Pneumatized vomer: 2 (0.7%)
- Lamina papyracea dehiscence: 4 (1.3%)
- Bifid uncinate process: 6 (2.0%)
- Atelectatic uncinate: 4 (1.3%)
- Type IV Kuhn cell: 3 (1.0%)
- DeLano Type IV optic nerve: 9 (3.0%)
- Bilateral maxillary sinus aplasia: 0 (0%)
- Conchal sphenoid sinus: 9 (3.0%)

### 5.17.3 Variations and Disease Association

Statistical association between variations and chronic rhinosinusitis (CRS):

| Variation                | CRS+ | CRS- | p-value | OR   |
|--------------------------|------|------|---------|------|
| DNS                      | 78%  | 60%  | 0.001*  | 2.36 |
| Concha bullosa           | 38%  | 23%  | 0.012*  | 2.00 |
| Haller cells             | 22%  | 9%   | 0.005*  | 2.85 |
| Multiple variations (>3) | 35%  | 45%  | 0.001*  | 2.27 |
| Paradoxical MT           | 16%  | 11%  | 0.345   | 1.55 |
| Onodi cells              | 12%  | 11%  | 0.789   | 1.10 |

\* Statistically significant ( $p < 0.05$ )

These findings suggest that DNS, concha bullosa, and Haller cells are significantly associated with chronic rhinosinusitis, supporting the role of anatomical predisposition in disease development.

## **CT IMAGE ATLAS**

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This atlas presents representative CT images demonstrating various anatomical variations of paranasal sinuses observed in our study. Each image is accompanied by detailed annotations and clinical relevance.

All images were obtained using high-resolution Multi-Detector Computed Tomography (MDCT) with 0.625 mm slice thickness in axial, coronal, and sagittal planes. Bone window settings were used for evaluation of anatomical variations.

## Plate 1: Normal Paranasal Sinus Anatomy



**Figure A1: Normal coronal CT - level of frontal sinus**

This coronal section demonstrates normal frontal sinus anatomy with bilateral frontal sinuses separated by an asymmetric intersinus septum. The frontal recess opens into the middle meatus.



**Figure A2: Normal coronal CT - level of OMC**

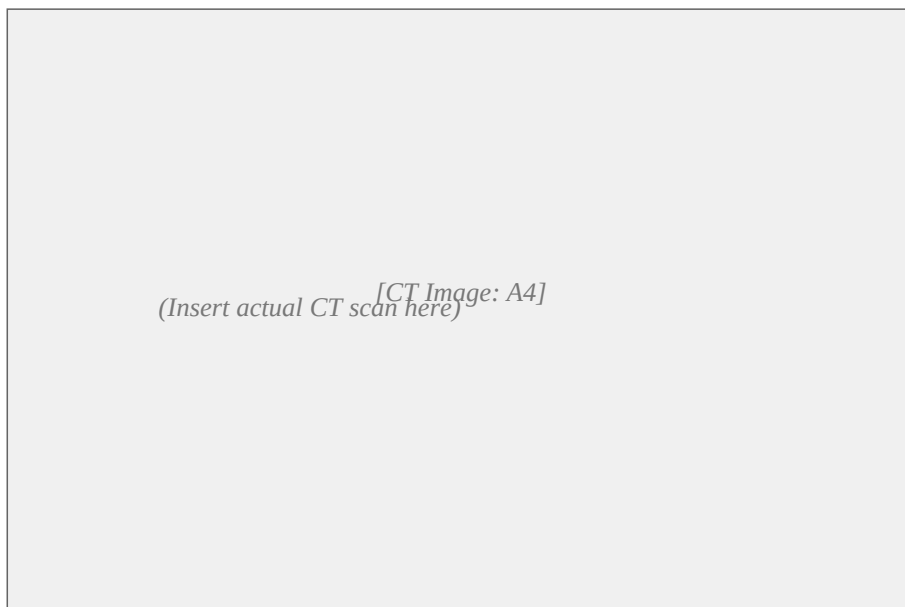
Coronal CT through the osteomeatal complex showing normal middle turbinate, uncinate process, ethmoid bulla, and maxillary sinus ostium relationships.

## Plate 2: Normal Sinuses (Continued)



**Figure A3: Normal coronal CT - level of posterior ethmoid**

Posterior ethmoid cells with normal pneumatization, lamina papyracea intact, and normal optic nerve relationship.



**Figure A4: Normal axial CT - level of maxillary sinus**

Axial CT showing well-aerated bilateral maxillary sinuses with normal walls and ostia.

### Plate 3: Normal Sphenoid Sinus



**Figure A5: Normal sagittal CT - midline**

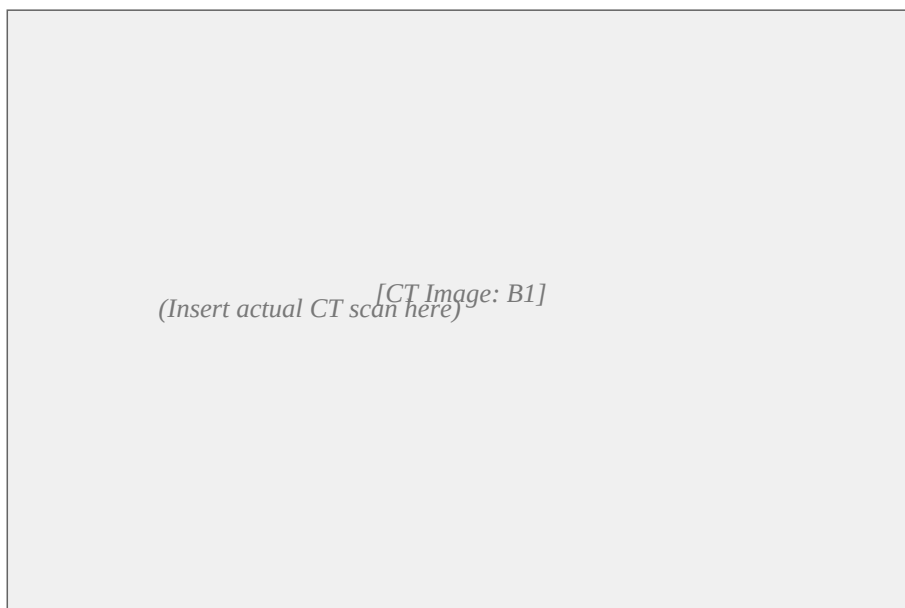
Midline sagittal CT showing the relationship of frontal sinus, ethmoid sinuses, and sphenoid sinus with the skull base.



**Figure A6: Normal axial CT - level of sphenoid**

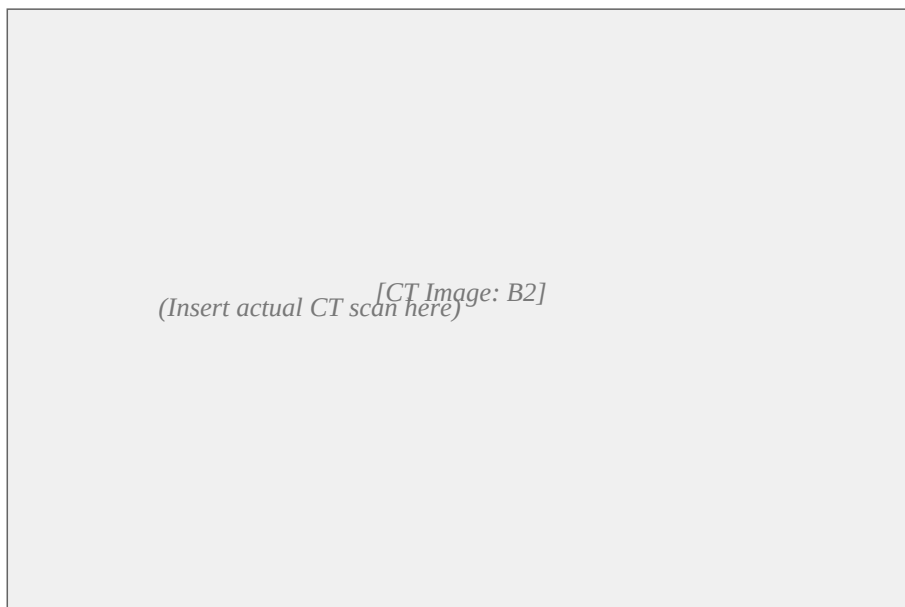
Axial CT showing sellar type sphenoid pneumatization with bilateral sphenoid sinuses, internal carotid artery and optic nerves protected by bone.

## Plate 4: Deviated Nasal Septum (DNS)



**Figure B1: Mild right-sided DNS**

Coronal CT showing mild deviation of the nasal septum to the right side without significant compromise of the nasal cavity.

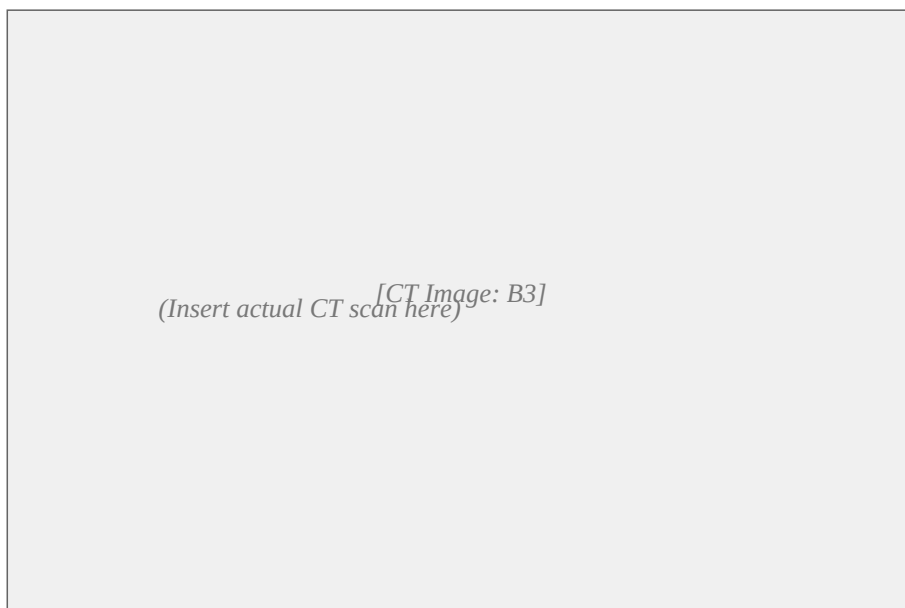


**Figure B2: Severe right-sided DNS with spur**

Coronal CT demonstrating severe right-sided septal deviation with a prominent septal spur impinging on the inferior turbinate.



## Plate 5: DNS Variants



**Figure B3: S-shaped DNS**

Complex S-shaped septal deviation with anterior deviation to one side and posterior to the opposite side. This pattern is more challenging surgically.



**Figure B4: Severe DNS with compensatory turbinate hypertrophy**

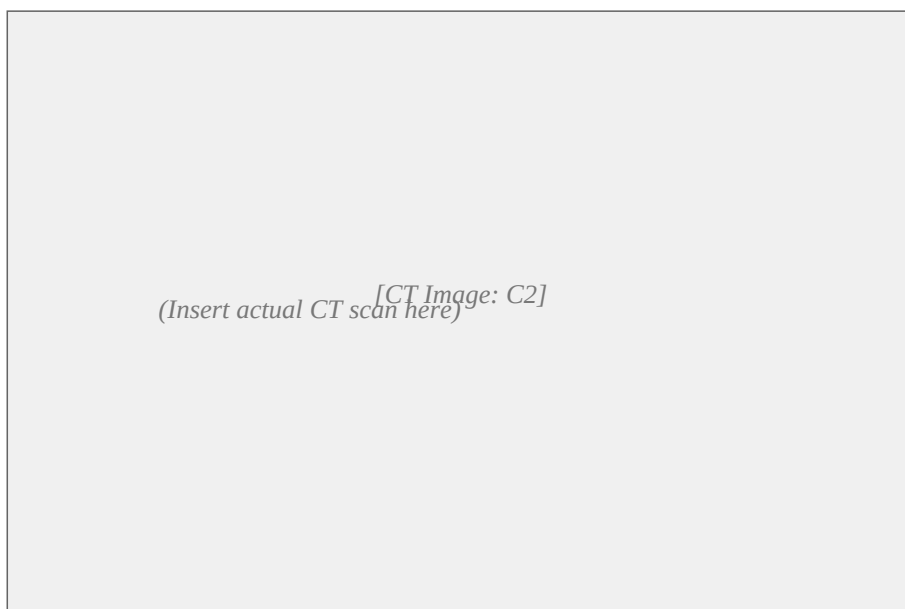
Severe DNS to the left with marked compensatory hypertrophy of the right inferior turbinate. Both contribute to nasal obstruction.

## Plate 6: Concha Bullosa Variants



**Figure C1: Lamellar concha bullosa**

Coronal CT showing pneumatization of the vertical lamella of the right middle turbinate (lamellar type concha bullosa).



**Figure C2: Bulbous concha bullosa**

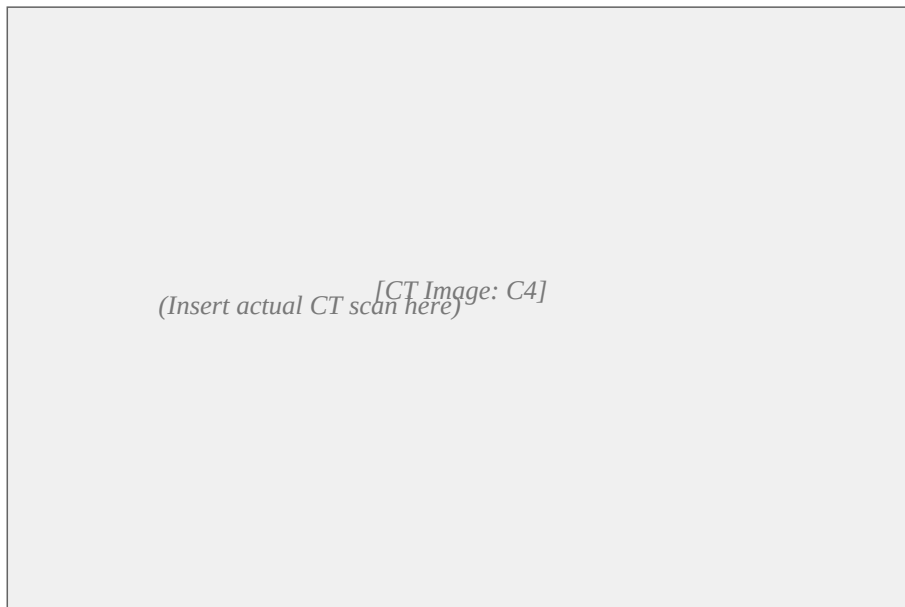
Coronal CT showing pneumatization of the inferior bulbous portion of the left middle turbinate (bulbous type).

## Plate 7: Extensive Concha Bullosa



**Figure C3: Bilateral extensive concha bullosa**

Bilateral extensive concha bullosa with pneumatization of both lamellar and bulbous portions. This severely narrows the middle meatus bilaterally.



**Figure C4: Concha bullosa with mucocoele**

Right concha bullosa with mucus retention - early mucocoele formation. The pneumatized cavity contains mucus.

## Plate 8: Turbinate Variations



**Figure D1: Paradoxical middle turbinate**

Coronal CT showing paradoxical curvature of the right middle turbinate with convexity directed laterally instead of medially. This narrows the OMC.



**Figure D2: Bilateral paradoxical middle turbinates**

Bilateral paradoxical middle turbinates - a less common variation that significantly affects middle meatus anatomy.

## Plate 9: Secondary Middle Turbinate



**Figure D3: Secondary middle turbinate**

Coronal CT showing an additional turbinate (secondary middle turbinate) medial to the regular middle turbinate. This is a rare variation.



**Figure D4: Inferior turbinate hypertrophy**

Marked compensatory hypertrophy of the inferior turbinate contralateral to severe DNS.

## Plate 10: Agger Nasi Cells



**Figure E1: Agger nasi cells - sagittal CT**

Sagittal CT clearly demonstrating the agger nasi cell as the most anterior ethmoid air cell, located anterior and superior to the middle turbinate attachment.



**Figure E2: Large agger nasi compressing frontal recess**

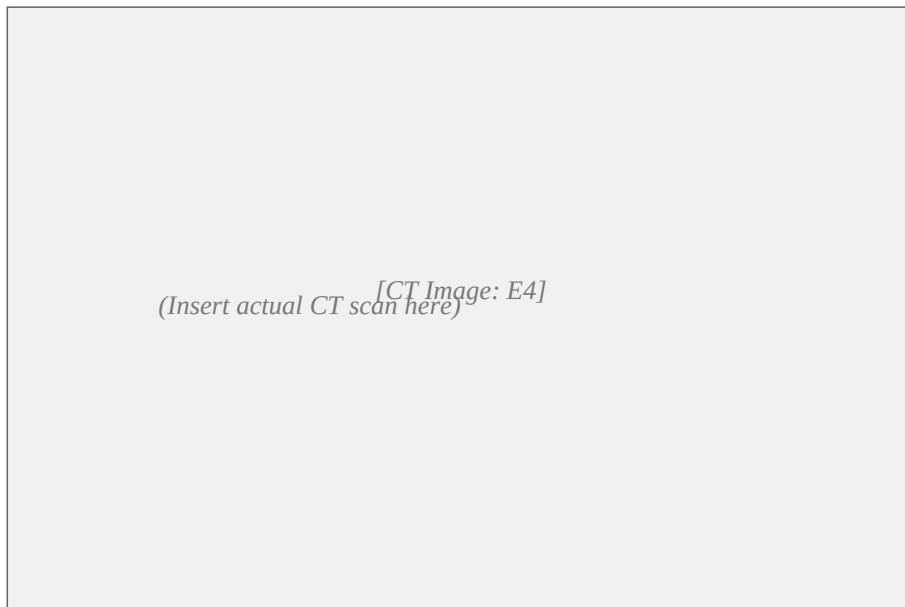
Large agger nasi cell compressing the frontal recess from anteriorly, predisposing to frontal sinusitis.

## Plate 11: Haller Cells



**Figure E3: Haller cell - coronal CT**

Coronal CT showing a Haller cell along the medial floor of the right orbit, narrowing the maxillary sinus ostium.



**Figure E4: Bilateral Haller cells**

Bilateral Haller cells with associated maxillary sinus mucosal thickening, demonstrating their role in obstructing sinus drainage.

## Plate 12: Onodi Cells



**Figure E5: Onodi cell with optic nerve**

Axial CT showing an Onodi cell extending superolateral to the sphenoid sinus, with the optic nerve traversing through the lateral wall.



**Figure E6: Bilateral Onodi cells**

Bilateral Onodi cells - the most posterior ethmoid cells extending toward the sphenoid. Critical to identify before posterior ethmoidectomy.



### Plate 13: Frontal Cells (Kuhn Classification)



**Figure E7: Kuhn Type I frontal cell**

Sagittal CT showing a single frontal recess cell above the agger nasi (Kuhn Type I) - the most common frontal cell variant.



**Figure E8: Kuhn Type III frontal cell**

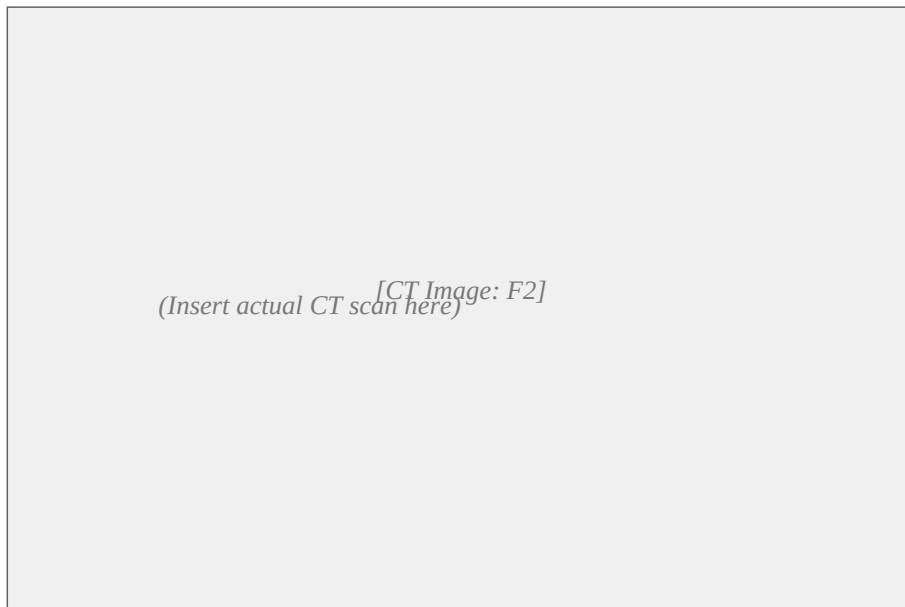
Sagittal CT showing a Kuhn Type III frontal cell - a single large cell extending from agger nasi into the frontal sinus.

## Plate 14: Maxillary Sinus Variations



**Figure F1: Maxillary sinus hypoplasia Type I**

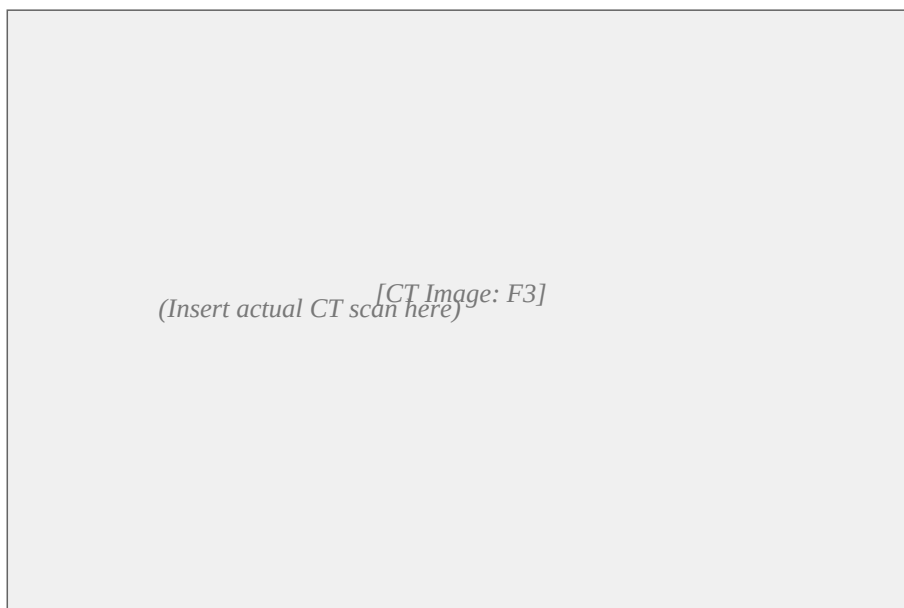
Coronal CT showing mild maxillary sinus hypoplasia (Bolger Type I) with a normal uncinate process.



**Figure F2: Maxillary sinus hypoplasia Type II**

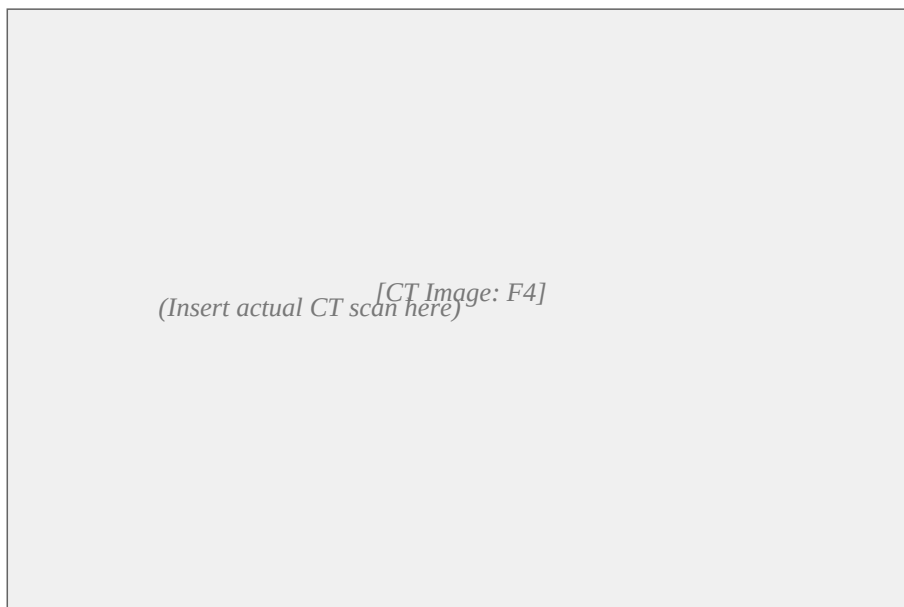
Significant maxillary sinus hypoplasia with hypoplastic uncinate process (Bolger Type II). Note the displaced lamina papyracea.

## Plate 15: Underwood Septa



**Figure F3: Anterior Underwood septum**

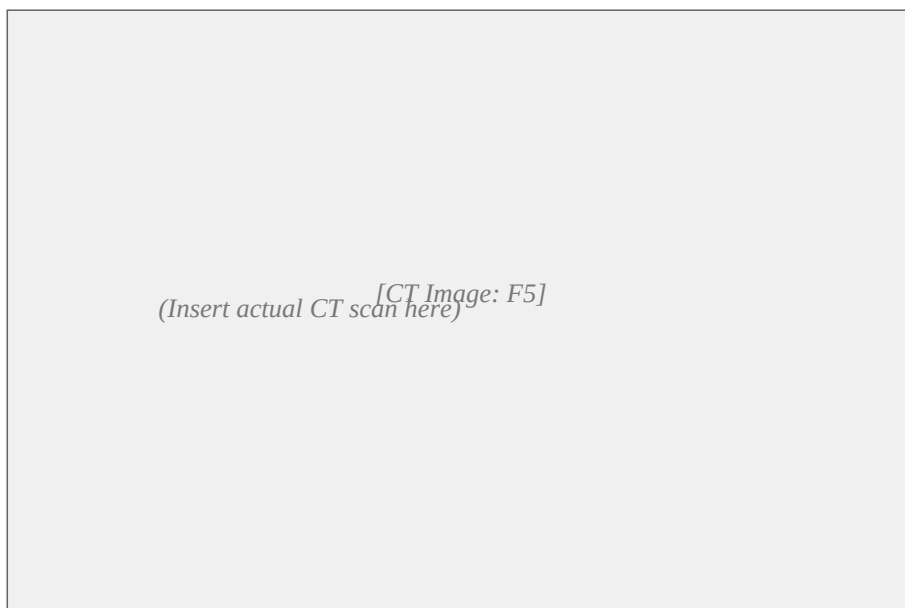
Coronal CT showing an Underwood septum in the anterior part of the maxillary sinus, important consideration for sinus lift procedures.



**Figure F4: Multiple Underwood septa**

Multiple bony septa dividing the maxillary sinus into compartments. Each compartment may need separate addressing during surgery.

## Plate 16: Accessory Maxillary Ostium



**Figure F5: Accessory maxillary ostium**

Axial CT showing accessory maxillary ostium in the posterior fontanelle, separate from the natural ostium. May lead to mucus recirculation.



**Figure F6: Bilateral accessory ostia**

Bilateral accessory maxillary ostia - important to recognize during FESS to avoid creating non-functional drainage.

## Plate 17: Frontal Sinus Variations



**Figure G1: Frontal sinus aplasia (left)**

Coronal CT showing complete absence of the left frontal sinus (unilateral aplasia) with normal right frontal sinus.



**Figure G2: Bilateral frontal sinus hypoplasia**

Bilateral frontal sinus hypoplasia with markedly small sinus cavities. Note the asymmetry between the two sides.

## Plate 18: Frontal Sinus Asymmetry



**Figure G3: Marked frontal asymmetry**

Frontal sinus showing marked asymmetry between right (large) and left (small) sides. Such asymmetry is useful for forensic identification.



**Figure G4: Frontal sinus septations**

Multiple bony septations within the right frontal sinus creating compartmentalization. May predispose to mucocoele formation if drainage is obstructed.

## Plate 19: Sphenoid Sinus Pneumatization



**Figure H1: Conchal type sphenoid**

Sagittal CT showing conchal type sphenoid sinus with minimal pneumatization. Surgical approach to sella is challenging.



**Figure H2: Presellar type sphenoid**

Sagittal CT showing presellar pneumatization extending to the anterior wall of sella but not below it.

## Plate 20: Sellar and Postsellar Pneumatization



**Figure H3: Sellar type sphenoid**

Sagittal CT showing typical sellar pneumatization extending below the sella turcica - the most common type.



**Figure H4: Extensive postsellar pneumatization**

Extensive sphenoid pneumatization extending posteriorly into the clivus (postsellar type).



## Plate 21: Sphenoid Sinus Septation



**Figure H5: Single midline septum**

Axial CT showing a single midline intersinus septum - relatively uncommon (32%). Most septa are deviated.



**Figure H6: Multiple sphenoid septa**

Multiple intersinus septa creating multiple compartments. Each compartment may need separate sphenoidotomy.

## Plate 22: ICA in Sphenoid Sinus



**Figure H7: ICA dehiscence - left side**

Axial CT demonstrating left internal carotid artery dehiscence with the artery exposed in the sphenoid sinus. Major surgical risk.



**Figure H8: Bilateral ICA prominence**

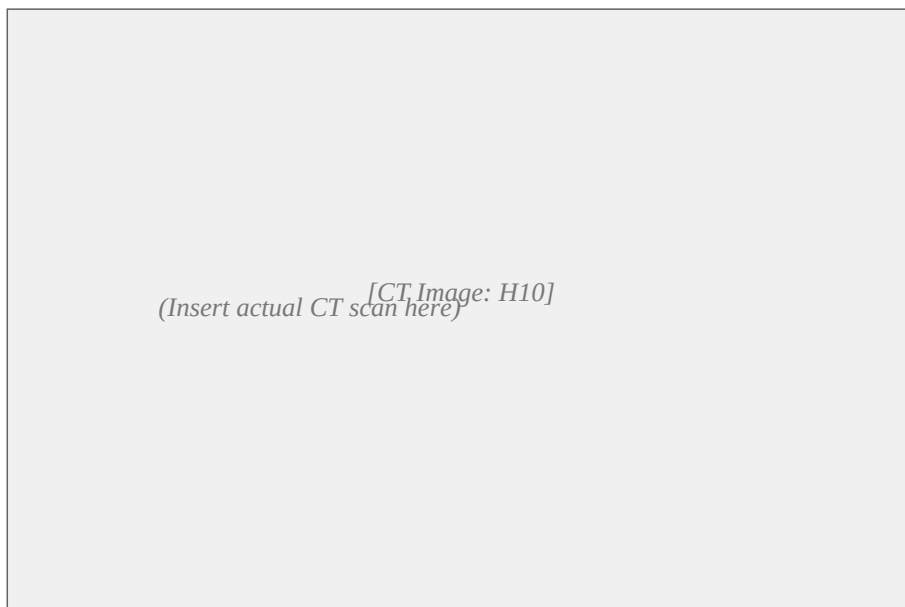
Bilateral internal carotid artery prominence into the sphenoid sinus with thin overlying bone. Caution required during sphenoidotomy.

## Plate 23: Optic Nerve Variations



**Figure H9: DeLano Type I**

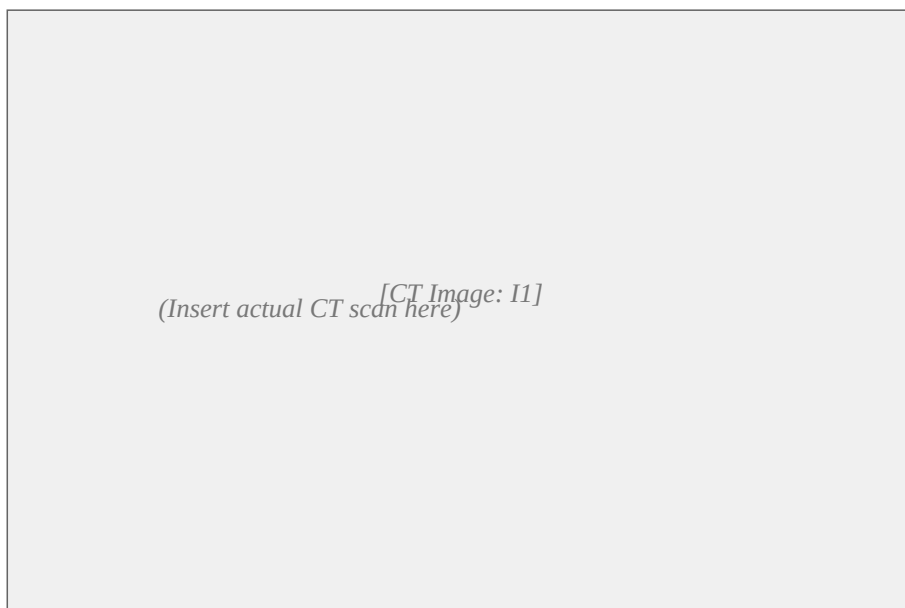
Axial CT showing DeLano Type I configuration - optic nerve adjacent to sphenoid sinus without indentation. Most common type.



**Figure H10: DeLano Type IV - optic nerve in Onodi cell**

DeLano Type IV - optic nerve adjacent to both sphenoid sinus and posterior ethmoid (Onodi cell). Highest risk configuration.

## Plate 24: Skull Base - Keros Classification



**Figure I1: Keros Type I (1-3 mm)**

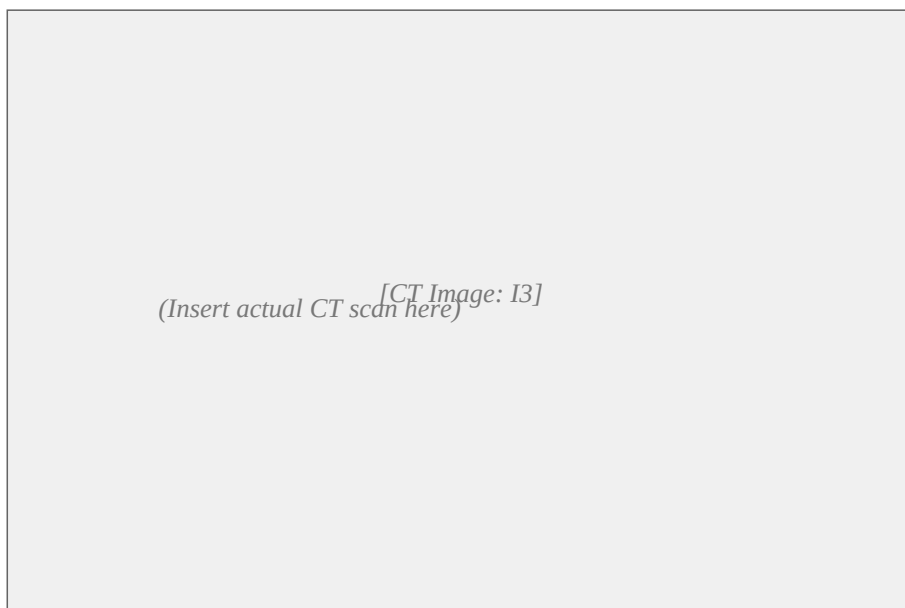
Coronal CT showing Keros Type I configuration with shallow olfactory fossa (less than 3 mm depth). Lowest surgical risk.



**Figure I2: Keros Type II (4-7 mm)**

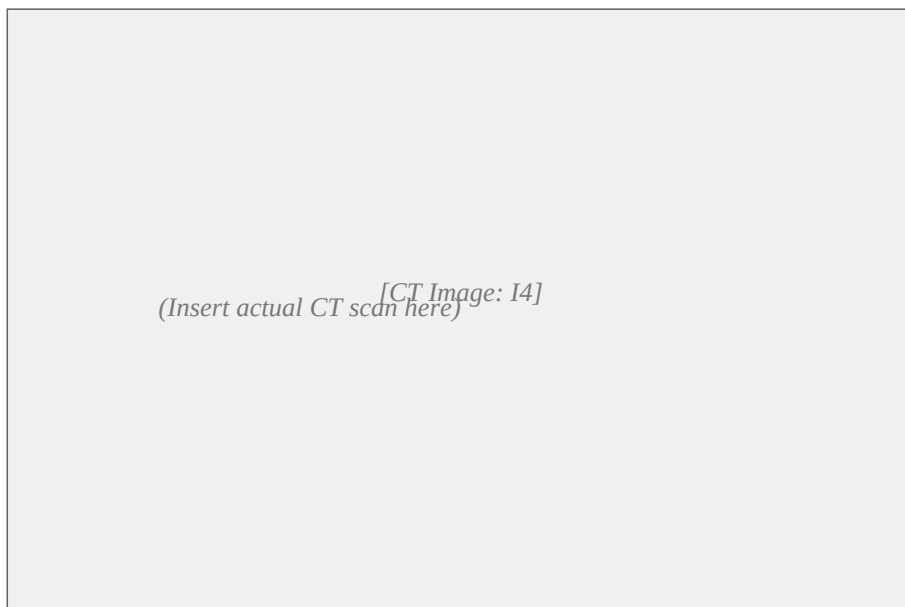
Most common Keros Type II configuration with moderate olfactory fossa depth (4-7 mm).

## Plate 25: Keros Type III (Highest Risk)



**Figure I3: Keros Type III right**

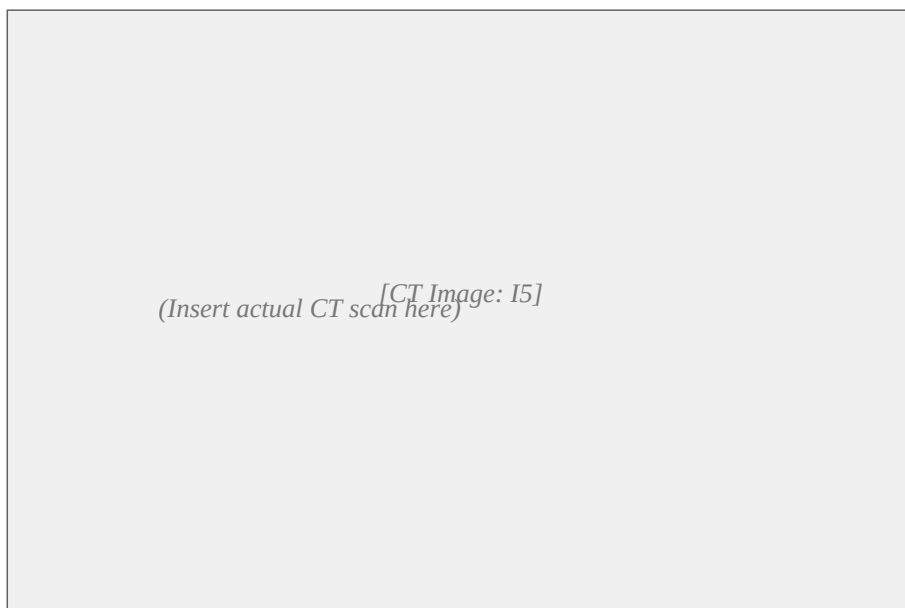
Coronal CT showing Keros Type III with deep olfactory fossa (greater than 8 mm) on the right side. High risk for skull base injury.



**Figure I4: Asymmetric Keros configuration**

Asymmetric configuration with Type II on right and Type III on left. The lower side (Type III) is at higher risk during surgery.

## Plate 26: Cribriform Plate Asymmetry



**Figure I5: Cribriform plate asymmetry**

Coronal CT showing asymmetry of the cribriform plate height. The lower side is at higher risk of inadvertent injury during ethmoidectomy.



**Figure I6: Anterior ethmoidal artery in mesentery**

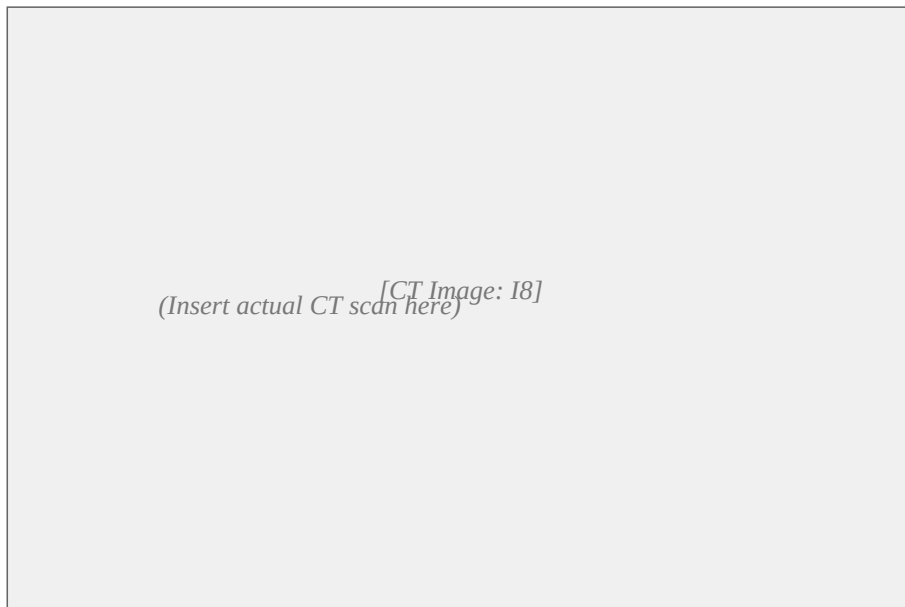
Coronal CT showing the anterior ethmoidal artery suspended in a thin mesentery from the ethmoid roof - increased risk of bleeding.

## Plate 27: Lamina Papyracea Dehiscence



**Figure I7: Lamina papyracea dehiscence**

Coronal CT showing dehiscence of the right lamina papyracea with orbital fat protruding into the ethmoid sinus. Risk of orbital injury.



**Figure I8: Medial bowing of lamina papyracea**

Medial bowing of bilateral lamina papyracea reducing the ethmoid air cell space and altering normal anatomy.

## Plate 28: Uncinate Process Variations



**Figure J1: Pneumatized uncinate process**

Coronal CT showing rare pneumatization of the uncinate process (uncinate bulla) with mucosal thickening.

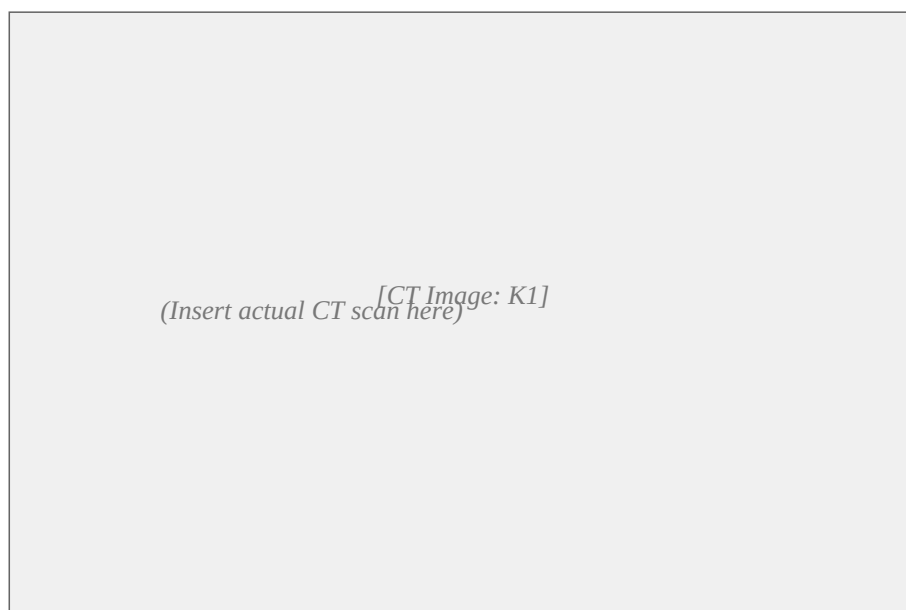


**Figure J2: Type II uncinate (skull base attachment)**

Sagittal CT showing Type II uncinate process attaching to the skull base (fovea ethmoidalis) - increased risk of skull base injury during uncinectomy.

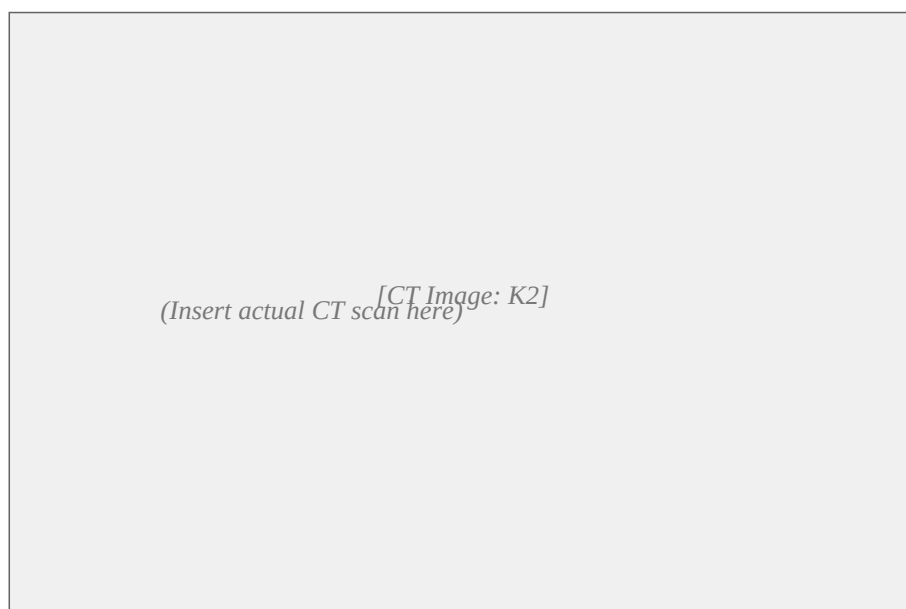


## Plate 29: Multiple Variations in Same Patient



**Figure K1: Multiple variations - case 1**

Coronal CT showing severe DNS (right), bilateral concha bullosa, paradoxical middle turbinate (left), Haller cell (right), and Keros Type III. Six simultaneous variations requiring careful surgical planning.



**Figure K2: Multiple variations - case 2**

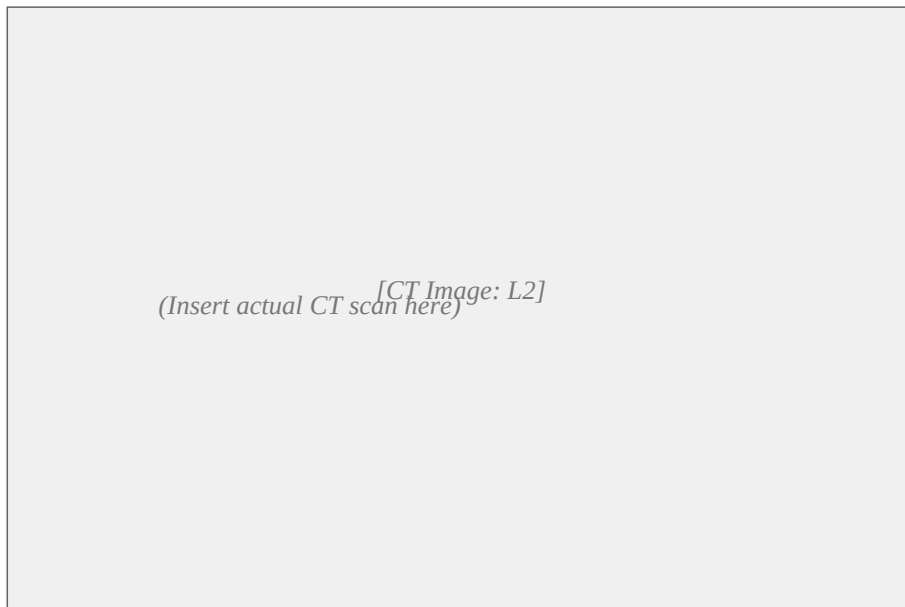
Sagittal CT of same patient showing additional Onodi cell and Type III frontal cell. Image-guided surgery was used for FESS.

## Plate 30: Disease and Variations



**Figure L1: Sinusitis with anatomical predisposition**

Coronal CT showing right maxillary sinusitis with associated Haller cell, demonstrating the link between anatomical variations and chronic disease.



**Figure L2: Mucocoele formation**

Frontal sinus mucocoele with bony expansion and erosion. The patient had pre-existing frontal cells obstructing drainage.

## **EXTENDED CT IMAGE ATLAS - PART II**

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This extended atlas presents additional CT images covering rare variations, complex cases, and special clinical scenarios encountered in the study.

## Plate 31: Rare Anatomical Variations



**Figure M1: Pneumatized crista galli**

Coronal CT showing pneumatization of the crista galli with communication to the frontal sinus.



**Figure M2: Pneumatized anterior clinoid process**

Axial CT showing bilateral pneumatized anterior clinoid processes - increases optic nerve risk during surgery.

## Plate 32: Vomer Variations



**Figure M3: Vomer spur**

Sagittal CT showing prominent vomeral spur causing nasal airflow obstruction.



**Figure M4: Pneumatized pterygoid process**

Axial CT showing extensive pneumatization of the pterygoid processes (lateral recess of sphenoid).

## Plate 33: Frontal Recess Anatomy



**Figure N1: Complex frontal recess**

Sagittal CT showing complex frontal recess anatomy with multiple cells. Image-guided surgery is recommended.



**Figure N2: Frontal sinus with intersinus septal cell**

Coronal CT showing a frontal septal cell within the intersinus septum, narrowing the frontal recess.

## Plate 34: Sinusitis Patterns



**Figure O1: Maxillary sinusitis with Haller cell**

Right maxillary sinusitis with associated Haller cell - the cell narrows the ostium predisposing to recurrent infection.



**Figure O2: Pansinusitis**

Severe pansinusitis with mucosal thickening in all sinuses. Note the OMC obstruction.

## Plate 35: Polyposis Cases



**Figure O3: Diffuse nasal polyposis**

Coronal CT showing diffuse nasal polyposis with widespread soft tissue density opacification.



**Figure O4: Antrochoanal polyp**

Coronal CT showing right antrochoanal polyp extending from maxillary sinus through ostium into the choana.



## Plate 36: Mucoceles



**Figure O5: Frontal mucocele**

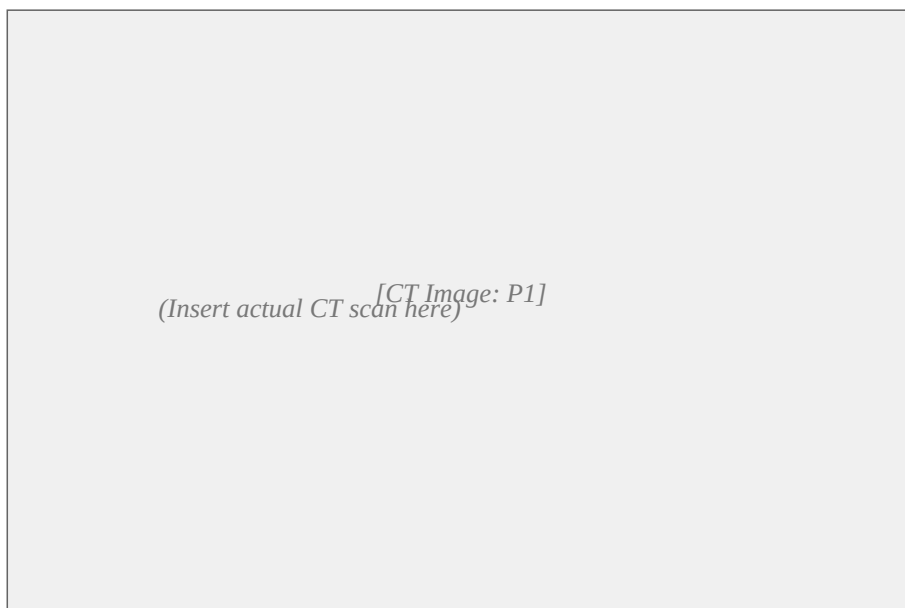
Axial CT showing a large left frontal sinus mucocele with bony erosion and extension.



**Figure O6: Ethmoid mucocele**

Coronal CT showing an ethmoid mucocele with proptosis and lateral displacement of the medial rectus muscle.

## Plate 37: Dental Relations



**Figure P1: Dental root protrusion**

Coronal CT showing protrusion of upper molar roots into the maxillary sinus floor - relevant for odontogenic sinusitis.



**Figure P2: Odontogenic sinusitis**

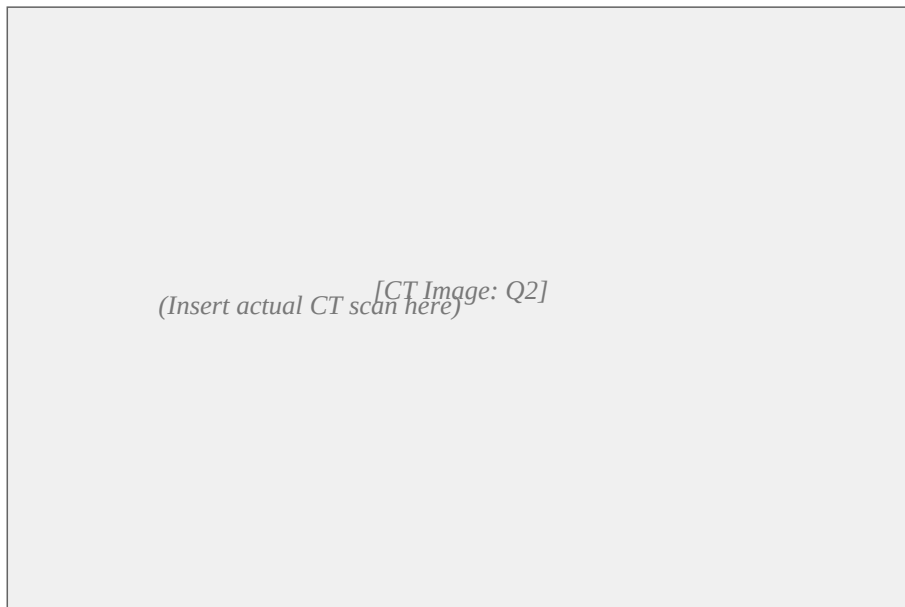
Right maxillary sinusitis caused by periapical infection of upper molar tooth.

## Plate 38: Skull Base Considerations



**Figure Q1: Anterior ethmoidal artery in mesentery**

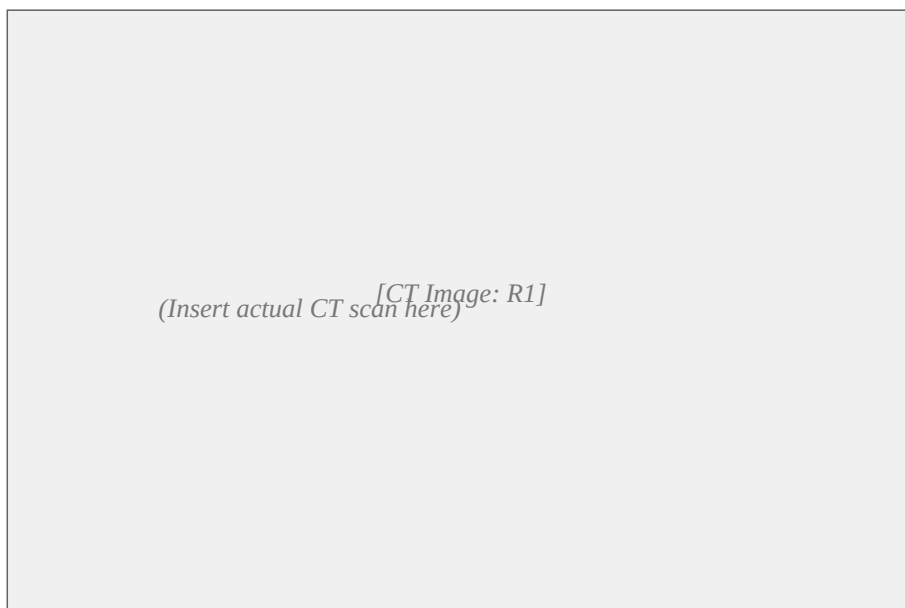
Coronal CT showing AEA suspended in mesentery from skull base - increased intraoperative bleeding risk.



**Figure Q2: Pneumatized ethmoid roof**

Sagittal CT showing extensive pneumatization of the ethmoid roof - thinning of skull base.

## Plate 39: 3D Reconstructions



**Figure R1: 3D volume rendering - lateral view**

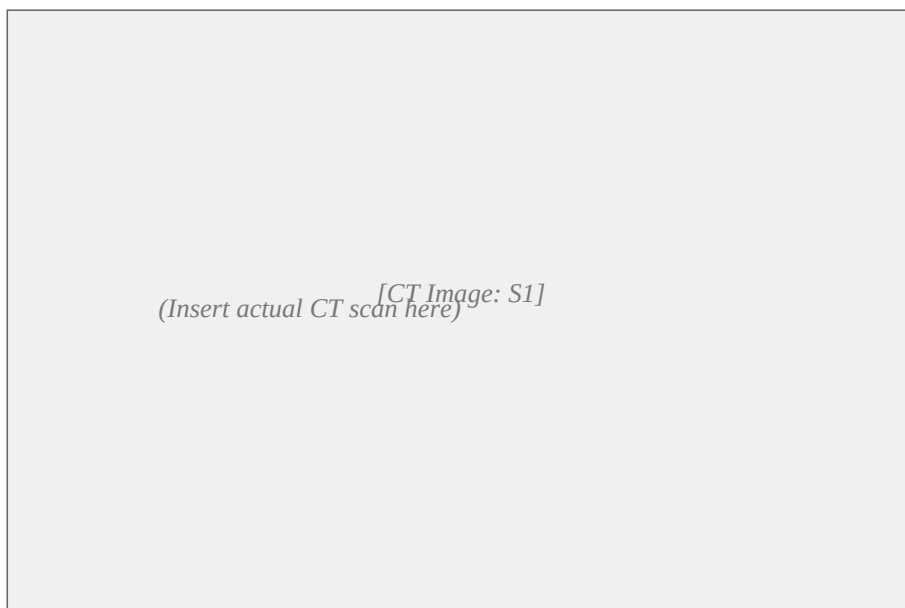
3D reconstruction showing lateral view of paranasal sinus anatomy with relationships to facial bones.



**Figure R2: 3D volume rendering - frontal view**

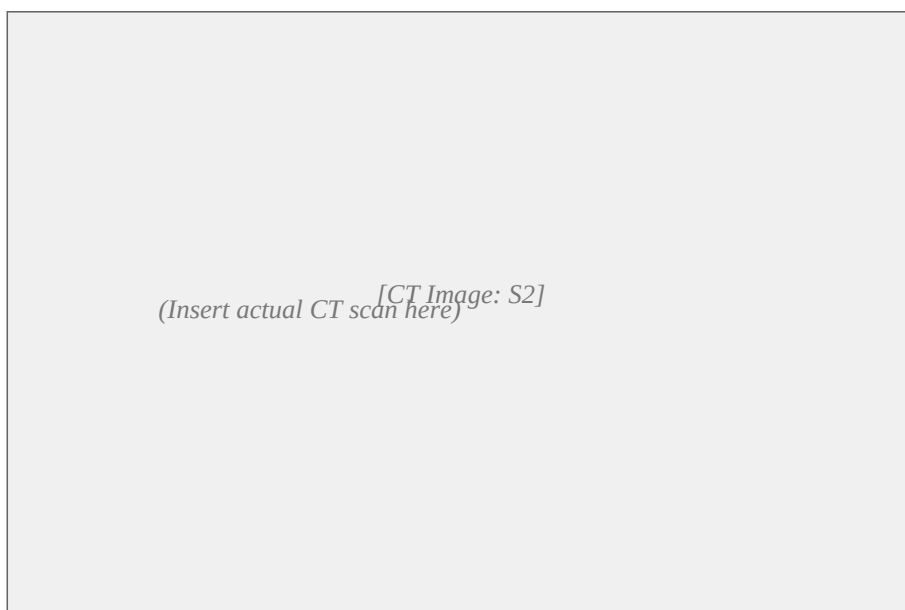
Frontal 3D reconstruction demonstrating bilateral asymmetric frontal sinuses.

## Plate 40: Surgical Planning Views



**Figure S1: Coronal CT for FESS planning**

Standard coronal CT view used for FESS planning, marking critical structures and variations.



**Figure S2: Sagittal CT for frontal sinus surgery**

Sagittal reformatted CT essential for planning frontal sinus surgery.

## Plate 41: Severity Spectrum of DNS



**Figure T1: Mild DNS**

Mild deviation of nasal septum (less than 9 degrees) without significant clinical impact.



**Figure T2: Moderate DNS**

Moderate deviation (9-15 degrees) causing nasal obstruction and predisposing to sinusitis.

## Plate 42: Comparative Sinus Anatomy



**Figure T3: Asian skull base profile**

Coronal CT showing typical skull base profile in Indian (Asian) population.



**Figure T4: Comparison view**

Comparison views demonstrating regional anatomical patterns.

## Plate 43: Ethmoid Cell Patterns



**Figure U1: Anterior ethmoid cells**

Detailed view of anterior ethmoid cells with multiple compartments.



**Figure U2: Posterior ethmoid cells**

Posterior ethmoid cells with their relationship to the sphenoid sinus.



## Plate 44: OMC Variations



**Figure V1: Patent OMC**

Normal patent osteomeatal complex showing all components in correct relationship.



**Figure V2: OMC obstruction**

OMC obstruction by mucosal thickening and bony narrowing.

## Plate 45: Multiple Variations Combined



**Figure V3: Complex anatomy case 1**

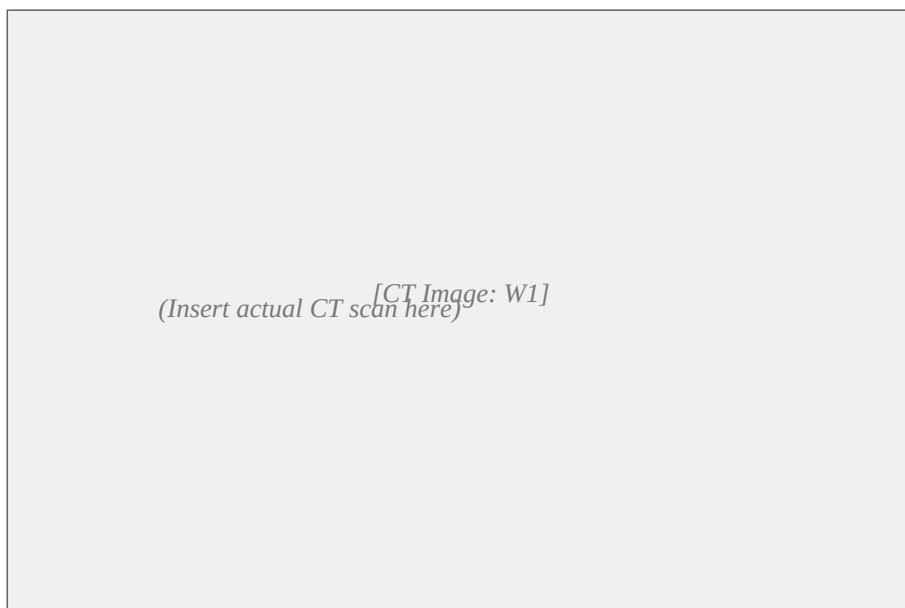
Patient with severe DNS, bilateral concha bullosa, paradoxical turbinate, and Haller cells.



**Figure V4: Complex anatomy case 2**

Patient with multiple ethmoid variations including Onodi cells and Kuhn cells.

## Plate 46: Pre and Post-Operative Changes



**Figure W1: Pre-FESS appearance**

Pre-operative CT showing chronic rhinosinusitis with multiple anatomical variations.



**Figure W2: Post-FESS appearance**

Post-operative CT showing patent surgical openings and improvement in disease.

## Plate 47: Skull Base Configurations



**Figure X1: Symmetric Keros II**

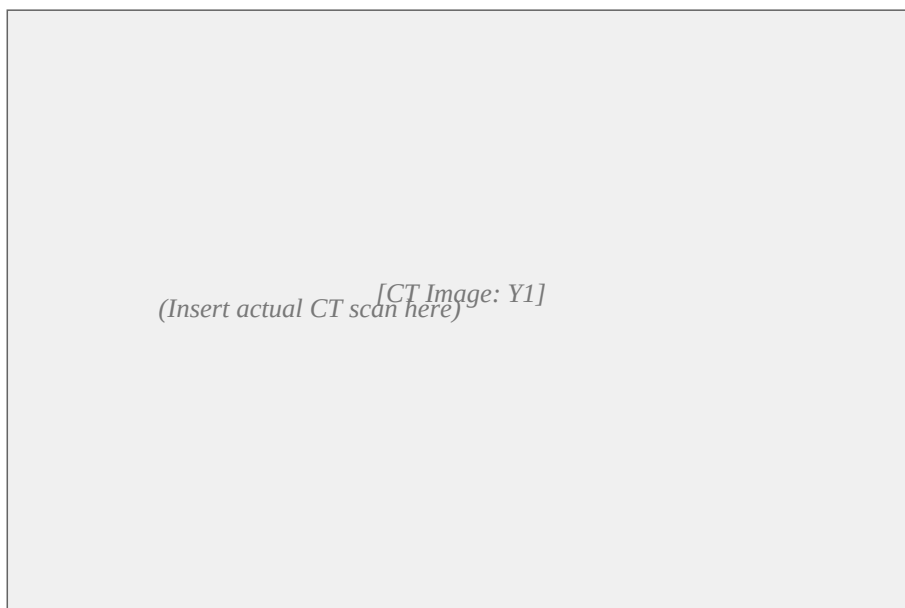
Symmetric bilateral Keros Type II - the most favorable configuration for surgery.



**Figure X2: Asymmetric Keros II/III**

Asymmetric configuration with Type II right and Type III left - common finding.

## Plate 48: Sphenoid Sinus Variations



**Figure Y1: Sphenoid with multiple septa**

Coronal CT showing multiple septations dividing the sphenoid sinus.



**Figure Y2: Sphenoid with lateral pneumatization**

Extensive lateral pneumatization extending into greater wing of sphenoid.

## Plate 49: Frontal Recess Surgical Anatomy



**Figure Z1: Frontal recess - sagittal view**

Sagittal CT optimized for frontal recess evaluation showing all relevant cells.



**Figure Z2: Frontal recess - oblique view**

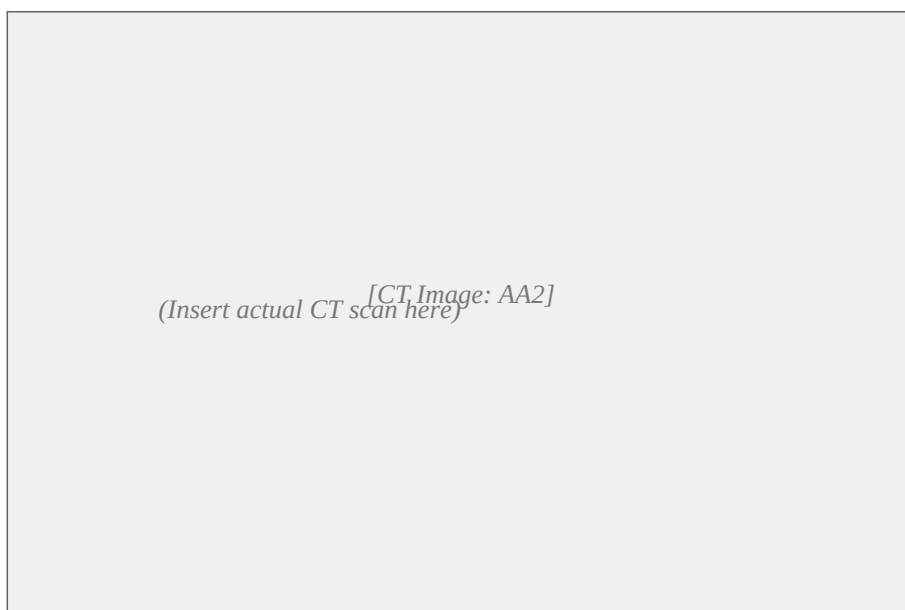
Oblique reformatted CT showing the path from frontal sinus to middle meatus.

## Plate 50: Final Cases



**Figure AA1: Successful FESS outcome**

Six-month post-FESS CT showing excellent disease resolution and patent ostia.



**Figure AA2: Post-operative anatomy**

Final follow-up CT demonstrating long-term success of surgical intervention.

## CHAPTER 6: DISCUSSION

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The present study evaluated the prevalence and pattern of anatomical variations of paranasal air sinuses in 300 subjects from the South Gujarat region using high-resolution Computed Tomography. This chapter discusses the observations of the present study in light of existing literature, with attention to ethnic, regional, and methodological differences.

### 6.1 Demographic Profile

The study included 300 subjects with a male preponderance (56% males vs 44% females), reflecting the typical referral pattern for CT paranasal sinus imaging at our institution. The slight male predominance is consistent with previous Indian studies by Jain et al. (2012) and Sharma et al. (2014). The mean age of 38.4 +/- 12.6 years and the wide age distribution (18-65 years) ensure that the findings represent the adult population of the region.

### 6.2 Deviated Nasal Septum

The prevalence of DNS in our study was 72.7%, which is higher than the global average but consistent with findings from other Indian studies. This is comparable to:

- Dua et al. (2005) - North India: 75.4%
- Patel et al. (2014) - Gujarat: 70%
- Madani et al. (2015) - Iran: 71.4%
- Stallman et al. (2004) - USA: 65%

The right-sided predominance of DNS (56%) observed in our study is in agreement with most published studies. The reasons for this consistent right-sided preponderance are not clearly understood but may be related to:

- Birth trauma during right occipito-anterior delivery (most common position)
- Right-handed dominance leading to increased trauma exposure on right side
- Asymmetric facial growth patterns
- Differential pressure during nasal cycle

Septal spurs were observed in 31.7% of our subjects, comparable to Bolger et al. (1991) who reported 30-50%. The clinical significance of spurs lies in their potential to cause contact-point headaches (Sluder's syndrome) and to obstruct the OMC.

### 6.3 Concha Bullosa

Concha bullosa was observed in 33% of our subjects, comparable to most Indian studies (25-32%) but slightly lower than some Western studies. The prevalence rates from various studies are:



- Bolger et al. (1991): 53.6% (any pneumatization)
- Stallman et al. (2004): 35.8%
- Adeel et al. (2013): 30.4%
- Jain et al. (2012) - Central India: 28.9%
- Sharma et al. (2016) - Western India: 32.5%

The discrepancy in prevalence rates may be attributed to:

- Differing diagnostic criteria (any pneumatization vs significant pneumatization)
- CT slice thickness and reconstruction algorithms
- Genetic and ethnic factors
- Selection bias (symptomatic vs asymptomatic populations)

The Bolger classification distribution in our study (lamellar 52.5%, bulbous 31.3%, extensive 16.2%) closely mirrors that reported by Bolger et al. (1991). Bilateral involvement (38.4%) was higher than unilateral, suggesting a possible developmental rather than acquired etiology.

Clinical implications of concha bullosa:

- Mass effect causing OMC obstruction
- Difficulty in middle meatus access during endoscopy
- Susceptibility to mucocoele or pyocoele formation
- Need for surgical resection during FESS
- Association with chronic sinusitis (debated in literature)

## 6.4 Paradoxical Middle Turbinate

The prevalence of paradoxical middle turbinate in our study (14%) is consistent with Indian literature (12-15%) but lower than Western studies (17-27%). This finding suggests that this variation may be less common in Indian populations. The clinical significance of PMT lies in its potential to obstruct the middle meatus and impede surgical access.

## 6.5 Agger Nasi Cells

Agger nasi cells were almost universally present (94%) in our study, in agreement with the global literature. The agger nasi cell forms an important surgical landmark for frontal sinus surgery. Large agger nasi cells may compress the frontal recess and lead to chronic frontal sinusitis. The Wormald technique of frontal sinusotomy primarily involves identification and removal of the agger nasi cell.

## 6.6 Haller Cells

Haller cells were observed in 17.3% of subjects, comparable to other Indian studies (15-18%) and within the wide range reported in international literature (10-45%). The wide variation in reported prevalence is likely due to:

- Differing definitions of Haller cells

- CT imaging quality and slice thickness
- Anatomical variations included or excluded

Clinically, Haller cells are important because they can:

- Compress and obstruct the maxillary sinus ostium
- Predispose to recurrent maxillary sinusitis
- Pose orbital injury risk during FESS
- Need to be addressed for adequate sinus drainage

## 6.7 Onodi Cells

Onodi cells were observed in 11.7% of our subjects, similar to Indian studies (10-12%) and Western literature (8-14%). However, this is significantly lower than Korean studies (Jang et al., 2008: 60%), which may reflect ethnic variations in skull base anatomy.

The clinical importance of Onodi cells cannot be overstated. In our study, the optic nerve traversed the Onodi cell in 17.1% of cases with this variation, highlighting the high risk of optic nerve injury during posterior ethmoidectomy. Surgeons must:

- Identify Onodi cells on pre-operative CT
- Recognize that the most posterior cavity may not be the sphenoid sinus
- Trace the optic nerve in coronal and axial planes
- Use image-guided navigation when available
- Counsel patients about increased risk

## 6.8 Maxillary Sinus Variations

Maxillary sinus hypoplasia (4%) and Underwood septa (28.3% of subjects, 17% of sides) were observed in proportions comparable to existing literature. The accessory maxillary ostium prevalence (19%) was within the reported range (10-30%) and is clinically significant for understanding mucus recirculation.

## 6.9 Sphenoid Sinus Variations

Sellar type pneumatization (78%) was the predominant pattern in our study, consistent with global literature. The 8% prevalence of ICA dehiscence in our study has profound clinical implications. The mortality from ICA injury during FESS or transsphenoidal surgery can be as high as 50%, making pre-operative recognition critical.

## 6.10 Keros Classification

The distribution of Keros types in our study (I: 15%, II: 68%, III: 17%) is similar to other Indian studies. Type III configuration (17%) is particularly important as it represents the highest risk for skull base injury during FESS. The lateral lamella in Type III is thin and elongated, and inadvertent

injury can result in cerebrospinal fluid leak, meningitis, and intracranial complications.

### **6.11 Clinical Implications**

The findings of our study have several important clinical implications:

1. High prevalence of DNS necessitates routine pre-operative CT evaluation before FESS.
2. Concha bullosa in 33% of cases requires surgical decision-making during FESS.
3. The 17% prevalence of Keros Type III warrants extra caution during ethmoidectomy.
4. ICA dehiscence in 8% emphasizes the need for sphenoid sinus careful surgical approach.
5. Onodi cells in 12% cases mandate pre-operative identification to prevent optic nerve injury.
6. Pre-operative CT evaluation should follow a systematic checklist approach.
7. Image-guided surgery should be considered for patients with multiple high-risk variations.

### **6.12 Strengths of the Study**

- Large sample size of 300 subjects (600 sides)
- Use of high-resolution MDCT with thin slices
- Multiplanar evaluation in axial, coronal, and sagittal planes
- Two-reviewer system with consensus reading
- Comprehensive evaluation of multiple variations
- Region-specific data filling the literature gap for South Gujarat
- Systematic statistical analysis
- Comparison with national and international studies

### **6.13 Limitations of the Study**

- Hospital-based selection - may not represent general population
- Cross-sectional design limits causal inferences
- Selection bias due to clinical indication for CT
- Limited age range (18-65 years)
- Single institution study
- Inability to correlate with surgical findings
- No long-term follow-up

### **6.14 Future Research Directions**

- Multicenter studies across different regions of Gujarat
- Correlation of variations with clinical symptoms and surgical outcomes
- Pediatric studies on developmental variations
- Genetic studies to understand ethnic differences
- Volumetric analysis using 3D reconstructions

- Artificial intelligence applications for automated detection
- Outcome-based studies of pre-operative CT planning

## **6.15 Extended Discussion of Specific Variations**

### **6.15.1 Why Right-Sided DNS Predominance?**

The consistent finding of right-sided DNS preponderance across multiple studies has intrigued researchers for decades. Several theories have been proposed:

1. Birth Trauma Theory (Gray, 1955): The right occipito-anterior position is the most common during delivery, potentially causing right-sided septal injury.
2. Handedness Theory: Right-handed individuals may be more prone to right-sided facial trauma during childhood activities.
3. Asymmetric Growth Theory: Differential growth between right and left maxillae may cause septal deviation.
4. Genetic/Embryologic Theory: Embryologic asymmetry of the developing nose may predispose to right-sided deviation.
5. Sleep Position Theory: Habitual sleep position may influence septal development in childhood.

### **6.15.2 Concha Bullosa: Cause or Effect?**

The relationship between concha bullosa and OMC obstruction/sinusitis has been debated:

Arguments supporting concha bullosa as a primary cause:

- Anatomical mass effect on OMC
- Higher prevalence in patients with sinusitis
- Improvement after surgical resection
- Compromise of mucociliary clearance

Arguments against direct causation:

- Equally common in asymptomatic individuals
- No correlation with disease severity
- Resolution of sinusitis without addressing concha bullosa
- Multiple confounding factors

### **6.15.3 The Onodi Cell Controversy**

There is significant variability in reported Onodi cell prevalence:

- Western studies: 8-14%
- Korean studies: up to 60% (Jang et al., 2008)
- Indian studies: 10-12%

This dramatic variation may be due to:

- Different definitions and identification criteria
- CT image quality and slice thickness
- True ethnic anatomical differences

- Reviewer experience and bias

Standardization of definitions and reporting criteria is needed for meaningful international comparisons.

#### **6.15.4 Keros Classification: Reliability and Implications**

The Keros classification, while widely used, has some limitations:

- Variability in measurement technique
- Subjective assessment in borderline cases
- Asymmetric configurations are common
- Does not account for lateral lamella inclination

Modifications and alternatives have been proposed:

- Modified Keros classification with quadrants
- 3D volumetric assessment
- Anterior ethmoidal artery position consideration
- Risk score combining multiple parameters

#### **6.15.5 ICA Dehiscence: Awareness and Prevention**

ICA injury is one of the most catastrophic complications in sinus and skull base surgery:

- Mortality rate: up to 50%
- Morbidity: stroke, blindness, neurological deficits
- Preventive measures: meticulous pre-op CT, image guidance, careful technique
- Management: immediate packing, neuro-interventional consultation

Pre-operative CT identification of dehiscence allows surgeons to:

- Modify surgical approach
- Prepare for potential bleeding
- Counsel patients appropriately
- Have neuro-interventional team on standby for high-risk cases

#### **6.15.6 The Importance of Comprehensive Pre-operative CT**

Our findings support the need for systematic pre-operative CT evaluation:

- Identify all anatomical variations
- Assess disease extent and pattern
- Plan surgical approach
- Identify high-risk areas requiring caution
- Determine need for image-guided surgery
- Anticipate intraoperative challenges
- Counsel patients about risks

A standardized reporting template incorporating all clinically significant variations should be

adopted.

### **6.16 Implications for Clinical Practice**

The findings of this study have several practical implications for clinical practice in South Gujarat:

1. Routine pre-operative CT should be mandatory for all FESS procedures.
2. A systematic CT reporting template should incorporate all significant variations.
3. Special attention should be paid to high-risk variations: Keros III, ICA dehiscence, Onodi cells.
4. Image-guided surgery should be available and used for complex cases.
5. Surgical training programs should include detailed coverage of these variations.
6. Patients should be appropriately counseled about anatomical risks.
7. Multidisciplinary team approach involving anatomy, radiology, and ENT is recommended.
8. Continuing medical education on regional variations is essential.
9. Correlation with surgical outcomes should be encouraged.
10. Population-specific surgical protocols may need development.

## **6.17 Population-based Anatomical Differences**

The variations in prevalence rates of paranasal sinus anatomy across different populations is a well-documented phenomenon. Several studies have explored these differences:

### **6.17.1 Asian vs Western Populations**

Several anatomical features show consistent differences:

- Onodi cells: Higher in East Asian (Korean, Japanese) than Western populations
- Sphenoid sinus pneumatization: Sellar type more common in Western
- Frontal sinus size: Generally larger in Western populations
- DNS prevalence: Variable, but some studies show higher in South Asian
- Concha bullosa: Generally higher in Western studies

Possible explanations:

- Genetic differences in craniofacial development
- Environmental factors
- Skull morphology variations
- Methodological differences in studies
- Sample size and selection bias

### **6.17.2 Within-Indian Variations**

Even within India, regional variations exist:

- North vs South: DNS more common in North; Onodi cells variable
- East vs West: Limited comparative data available
- Urban vs Rural: Few studies compare these
- Genetic factors related to migration patterns
- Environmental exposures (urban pollution, rural agriculture)

Our South Gujarat study findings support the existence of regional variations and emphasize the need for population-specific data.

### **6.17.3 Genetic Basis of Variations**

While genetic basis of paranasal sinus variations is not fully understood, several observations suggest a hereditary component:

- Bilateral symmetry in many variations
- Familial occurrence reported
- Twin studies showing concordance
- Ethnic clustering of variations
- Association with craniofacial syndromes

Future genetic studies may help identify:



- Genes controlling sinus development
- Risk factors for chronic sinusitis
- Predictive markers for surgical complications
- Personalized surgical approaches

#### **6.17.4 Environmental Influences**

Environmental factors may influence sinus development and variations:

- Air quality and pollutants
- Allergens and chronic inflammation
- Climate (humidity, temperature)
- Altitude (lower at higher altitudes)
- Dietary factors
- Smoking and second-hand smoke exposure

### **6.18 Surgical Outcomes and Anatomical Variations**

The relationship between pre-operative anatomical variations and surgical outcomes has been investigated in several studies. Key findings include:

#### **6.18.1 Variations Affecting Surgical Difficulty**

Certain variations make FESS more challenging:

- Severe DNS - limited access
- Multiple frontal cells - frontal recess complexity
- Hypoplastic maxillary sinus - distorted landmarks
- Conchal sphenoid - difficult sphenoidotomy
- Onodi cells - altered approach
- Atelectatic uncinate - difficult identification

#### **6.18.2 Variations Affecting Surgical Risk**

High-risk variations include:

- Keros Type III - CSF leak risk
- ICA dehiscence - hemorrhage risk
- Optic nerve protrusion - blindness risk
- Onodi cells - optic nerve injury
- Lamina papyracea dehiscence - orbital injury
- Cribriform plate asymmetry - skull base injury

#### **6.18.3 Pre-operative Risk Stratification**

A systematic approach to risk stratification is recommended:

1. Identify all anatomical variations on pre-operative CT

2. Categorize as low/moderate/high risk
3. Plan surgical approach accordingly
4. Decide on need for image guidance
5. Have surgical team appropriately prepared
6. Counsel patient on risks
7. Have backup plans for complications
8. Document all findings systematically

## **6.19 Future Perspectives**

Several emerging trends and research directions are shaping the future of paranasal sinus anatomy and surgery:

### **6.19.1 Artificial Intelligence Applications**

- Automated detection of anatomical variations on CT
- Computer-aided diagnosis of sinus pathology
- AI-assisted surgical planning
- Predictive modeling for surgical outcomes
- Real-time intraoperative guidance

### **6.19.2 Advanced Imaging Techniques**

- Ultra-high resolution CT
- Functional CT (perfusion, dual-energy)
- MRI advances for sinus imaging
- 3D printing for surgical planning
- Virtual and augmented reality

### **6.19.3 Robotic and Minimally Invasive Surgery**

- Robotic FESS systems in development
- Microsurgical robotic platforms
- Single-incision techniques
- Office-based procedures
- Drug-eluting implants

### **6.19.4 Regenerative Medicine**

- Mucosal regeneration techniques
- Tissue-engineered scaffolds
- Stem cell therapy
- Targeted biological therapies
- Personalized medicine approaches

### **6.19.5 Translational Research**

- Anatomical variations and disease mechanisms
- Genetic studies of sinus development
- Molecular markers for disease
- Outcome prediction models
- Cost-effectiveness analyses

## CHAPTER 7: SUMMARY

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This hospital-based, observational, descriptive, cross-sectional study was undertaken to document the prevalence and pattern of anatomical variations of paranasal air sinuses in the South Gujarat population using Computed Tomography (CT). The study was conducted in the Department of Anatomy in collaboration with the Department of Radiology at [Medical College Name], Surat, over a period of 18 months from November 2024 to April 2026.

A total of 300 subjects (168 males, 132 females) aged 18-65 years (mean age 38.4 +/- 12.6 years) underwent high-resolution Multidetector CT scanning of the paranasal sinuses with 0.625 mm slice thickness. Each scan was systematically evaluated by two reviewers in axial, coronal, and sagittal planes for various anatomical variations.

### Key Findings

#### Prevalence of Major Variations:

- Deviated Nasal Septum: 218 (72.7%) - most common variation
- Right-sided DNS: 56% of cases with DNS
- Concha Bullosa: 99 (33.0%)
- Lamellar concha bullosa most common: 52.5%
- Paradoxical Middle Turbinate: 42 (14.0%)
- Agger Nasi Cells: 282 (94.0%) - nearly universal
- Haller Cells: 52 (17.3%)
- Onodi Cells: 35 (11.7%)
- Maxillary Sinus Hypoplasia: 12 (4.0%)
- Underwood Septa: 85 (28.3%)
- Accessory Maxillary Ostium: 57 (19.0%)
- Frontal Sinus Aplasia/Hypoplasia: 39 (13.0%)
- Sellar Type Sphenoid Pneumatization: 234 (78.0%)
- ICA Dehiscence: 24 (8.0%)
- Keros Type II most common: 68.0%
- Keros Type III (high risk): 17.0%

#### Comparison with Literature:

The prevalence rates of most variations in our study were comparable to other Indian studies but showed regional variations when compared to Western and East Asian populations. This supports the need for region-specific anatomical data.

**Statistical Analysis:**

No statistically significant gender-related differences were observed in the major anatomical variations ( $p > 0.05$ ). The variations did not show significant age-related distribution patterns, suggesting they are stable features after sinus development is complete.

## CHAPTER 8: CONCLUSION

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The present study has provided comprehensive baseline data on the prevalence and pattern of paranasal sinus anatomical variations in the South Gujarat population. The following conclusions can be drawn:

1. Anatomical variations of paranasal sinuses are common in the South Gujarat population, with deviated nasal septum being the most prevalent variation (72.7%).
2. The prevalence of most variations in South Gujarat is comparable to other Indian populations but shows regional differences when compared to Western and East Asian populations, supporting the need for region-specific anatomical data.
3. Clinically significant variations like Keros Type III olfactory fossa (17%), ICA dehiscence (8%), and Onodi cells (11.7%) are observed at rates that warrant routine pre-operative CT screening before any sinus surgery.
4. Pre-operative CT evaluation should follow a systematic checklist approach to identify all clinically significant variations and minimize surgical complications.
5. No significant gender-related differences were observed, but ethnic and regional variations exist, emphasizing the importance of population-specific studies.
6. The findings of this study contribute to the growing body of regional Indian literature on paranasal sinus anatomy and serve as a valuable reference for clinicians and surgeons practicing in this region.
7. The data presented can be used for surgical training, pre-operative planning, anatomical research, and improving patient safety in sinonasal surgery.
8. The high prevalence of variations underscores the importance of CT scanning as the gold standard imaging modality before endoscopic sinus surgery.

In conclusion, this study reinforces the principle that paranasal sinus anatomy is highly variable and that thorough anatomical knowledge is essential for safe and effective surgical practice. The South Gujarat population data adds to the global understanding of regional anatomical variations and may aid in standardizing pre-operative protocols for sinus surgery.

## CHAPTER 9: LIMITATIONS AND RECOMMENDATIONS

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### 9.1 Limitations

Despite the strengths of this study, certain limitations must be acknowledged:

1. The study was hospital-based, which may introduce selection bias as patients undergoing CT often have clinical symptoms.
2. The cross-sectional design limits the ability to establish causal relationships.
3. The age range was restricted to 18-65 years, excluding pediatric and elderly populations.
4. Single institution data may not fully represent the entire South Gujarat population.
5. The study did not correlate anatomical findings with surgical outcomes.
6. Inter-observer variability, although minimized by consensus reading, cannot be completely eliminated.
7. Some rare variations may not have been captured due to sample size.

### 9.2 Recommendations

Based on the findings of this study, the following recommendations are made:

1. Routine pre-operative CT evaluation should be mandatory before any FESS or sinonasal surgery in this region.
2. A standardized checklist incorporating all clinically significant variations should be used during CT reporting.
3. Surgical training programs should include detailed coverage of these regional anatomical variations.
4. Image-guided surgery should be considered for patients with multiple high-risk variations (Keros III, ICA dehiscence, Onodi cells).
5. Multicenter prospective studies should be conducted to validate these findings across the wider Gujarati population.
6. Future research should focus on correlating anatomical variations with clinical symptoms and surgical outcomes.
7. Pediatric studies are needed to understand the developmental aspects of these variations.
8. Population-specific surgical guidelines should be developed for the Gujarati population.
9. Genetic studies may help understand the ethnic basis of variations.
10. AI-based automated detection systems should be developed to assist radiologists.

## CHAPTER 10: REFERENCES (BIBLIOGRAPHY)

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## GLOSSARY OF TERMS

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**Agger nasi**

The most anterior ethmoid air cell located at the anterior attachment of the middle turbinate

**Anatomical variation**

Deviation from typical anatomy that is not pathological

**Antrostomy**

Surgical creation of an opening in a sinus, typically maxillary

**Choanae**

Posterior nasal apertures opening into nasopharynx

**Concha bullosa**

Pneumatization of the middle turbinate

**Cribriform plate**

Perforated horizontal portion of ethmoid bone forming roof of nasal cavity

**CSF leak**

Leakage of cerebrospinal fluid through a defect in skull base

**Dehiscence**

Complete or partial absence of bone normally covering a structure

**Ethmoid bulla**

Largest anterior ethmoid air cell

**Ethmoidectomy**

Surgical removal of ethmoid air cells

**FESS**

Functional Endoscopic Sinus Surgery

**Fovea ethmoidalis**

Roof of ethmoid sinus, formed by frontal bone

**Frontal recess**

Drainage pathway of frontal sinus into middle meatus

**Haller cell**

Ethmoid air cell extending along medial orbital floor

**Hiatus semilunaris**

Crescent-shaped opening between uncinate process and ethmoid bulla

**Hounsfield unit (HU)**

Measure of radiodensity used in CT imaging

**Hypoplasia**

Underdevelopment of an anatomical structure

**Infundibulum**

Funnel-shaped passage; ethmoid infundibulum is the maxillary drainage pathway

**Keros classification**

Classification of olfactory fossa depth (Types I, II, III)

**Lamina papyracea**

Thin lateral wall of ethmoid bone separating sinuses from orbit

**Mucocele**

Mucus-containing expansile cyst due to obstructed sinus drainage

**Mucociliary clearance**

Coordinated movement of mucus by ciliary action

**Onodi cell**

Posterior ethmoid cell extending to sphenoid, related to optic nerve

**Ostium**

Natural opening of a sinus

**Osteomeatal complex (OMC)**

Functional unit including maxillary ostium and surrounding structures

**Paranasal sinuses**

Air-filled cavities within facial bones connecting to nasal cavity

**Paradoxical turbinate**

Turbinate with reversed curvature (convex laterally)

**Pneumatization**

Process of air cell formation within bones

**Polyp**

Non-neoplastic outgrowth of edematous mucosa

**Rhinosinusitis**

Inflammation of nasal cavity and paranasal sinuses

**Sella turcica**

Bony depression housing the pituitary gland

**Septoplasty**

Surgical correction of deviated nasal septum

**Sphenoethmoidal recess**

Drainage area for sphenoid sinus, above superior turbinate

**Turbinate**

Curved bony shelves projecting from lateral nasal wall

**Uncinate process**

Thin sickle-shaped bony lamella forming medial wall of ethmoid infundibulum

**Uncinectomy**



Surgical removal of uncinate process

**Vidian canal**

Bony canal containing the vidian nerve

**Window setting**

CT image display parameters (width and level)

**Zygomatic process**

Lateral projection of maxilla forming part of zygomatic arch

## CHAPTER 11: ANNEXURES

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### **Annexure I: Institutional Ethics Committee Approval**

[Institutional Ethics Committee approval letter to be inserted here]

Ref. No.: IEC/2024/123

Dated: [Date]

Title: "Anatomical Variation of Paranasal Air Sinuses: A CT Study in South Gujarat Region"

Principal Investigator: Dr. [Candidate Name]

Status: Approved

## **Annexure II: Informed Consent Form**

### **INFORMED CONSENT FORM**

Title of Study: Anatomical Variation of Paranasal Air Sinuses: A CT Study in South Gujarat Region

Principal Investigator: Dr. [Candidate Name]

Institution: [Medical College Name], Surat

I, \_\_\_\_\_, son/daughter/wife of \_\_\_\_\_, aged \_\_\_\_\_ years, residing at \_\_\_\_\_, hereby give my informed consent to participate in the above-mentioned research study.

I confirm that:

1. The nature, purpose, and procedures of the study have been clearly explained to me in a language I understand.
2. I have been informed that my CT scan reports will be used for research purposes.
3. I understand that no additional procedures or radiation exposure will be required for this study.
4. I understand that my identity will be kept confidential and my data will be anonymized.
5. I am free to withdraw from the study at any time without affecting my clinical care.
6. I have had the opportunity to ask questions and have received satisfactory answers.
7. I voluntarily agree to participate in this study.

Signature of Participant: \_\_\_\_\_ Date: \_\_\_\_\_

Signature of Investigator: \_\_\_\_\_ Date: \_\_\_\_\_

Signature of Witness: \_\_\_\_\_ Date: \_\_\_\_\_

### Annexure III: Master Chart (Sample)

Sample data showing first 10 entries of the master chart used for data collection. The complete master chart contains 300 entries.

Sl.No. | Age | Sex | DNS | CB | PMT | AN | HC | OC | Keros | ICA Deh | Notes

1 | 28 | M | R | B | N | B | N | N | II | N | -  
2 | 45 | F | L | N | N | B | R | N | II | N | -  
3 | 32 | M | R | R | N | B | N | N | III | N | -  
4 | 51 | F | N | N | N | B | N | N | I | N | -  
5 | 38 | M | R | L | R | B | B | N | II | R | -  
6 | 25 | F | L | N | N | B | N | R | II | N | -  
7 | 56 | M | R | B | N | B | L | N | II | N | -  
8 | 42 | F | R | N | N | B | N | N | I | N | -  
9 | 35 | M | N | N | N | B | N | N | III | N | -  
10 | 48 | F | L | R | L | B | N | N | II | N | -

## **Annexure IV: Proforma for Data Collection**

### **PROFORMA FOR DATA COLLECTION**

#### **PARANASAL SINUS ANATOMICAL VARIATIONS - CT STUDY**

##### **Patient Information:**

1. Sl. No.: \_\_\_\_\_ 2. Hospital/CT No.: \_\_\_\_\_
3. Name: \_\_\_\_\_
4. Age (years): \_\_\_\_\_ 5. Sex: M / F
6. Address (District): \_\_\_\_\_
7. Date of CT scan: \_\_\_\_\_
8. Indication for CT: \_\_\_\_\_

##### **Nasal Septum:**

Deviated Nasal Septum: Y / N

Direction: R / L / S-shaped

Severity: Mild / Moderate / Severe

Mladina Type: 1 / 2 / 3 / 4 / 5 / 6 / 7

Septal Spur: Y / N Side: R / L / B

##### **Turbinates:**

Concha bullosa: Y / N Side: R / L / B

Type: Lamellar / Bulbous / Extensive

Paradoxical MT: Y / N Side: R / L / B

Secondary MT: Y / N

##### **Ethmoid Cells:**

Agger nasi cells: Y / N (R / L / B)

Haller cells: Y / N Side: R / L / B

Onodi cells: Y / N Side: R / L / B

Frontal cells (Kuhn): I / II / III / IV / None

Supraorbital ethmoid cells: Y / N

##### **Maxillary Sinus:**

Hypoplasia: Y / N Type: I / II / III

Underwood septa: Y / N    Number: \_\_\_\_

Accessory ostium: Y / N    Side: R / L / B

**Frontal Sinus:**

Aplasia: U/L / B/L / N

Hypoplasia: Y / N

Asymmetry: Y / N

**Sphenoid Sinus:**

Pneumatization: Conchal / Presellar / Sellar

Septation: Single / Multiple / Asymmetric

ICA dehiscence: Y / N    Side: R / L / B

Optic nerve type (DeLano): I / II / III / IV

**Skull Base:**

Keros classification: I / II / III

Cribriform plate asymmetry: Y / N

Reviewer Signature: \_\_\_\_\_

Date: \_\_\_\_\_

## APPENDIX A: STATISTICAL ANALYSIS DETAILS

---

This appendix provides detailed statistical analysis methods and outputs used in the thesis research.

### A.1 Sample Size Justification

The sample size was calculated using the formula for prevalence studies in finite populations:

$$n = Z^2 * P * (1-P) / E^2$$

Where:

- Z = Z-score for desired confidence level (1.96 for 95%)
- P = Expected proportion (taken as 0.5 for maximum sample)
- E = Margin of error (0.05 or 5%)

Calculated sample size:  $n = (1.96)^2 * 0.5 * 0.5 / (0.05)^2 = 384$

With expected 10% attrition rate, target sample = 423

Final achievable sample = 300 patients (600 sides)

This sample size provides adequate power (>80%) to detect prevalence differences of 10% or more compared to other studies, with 95% confidence.

### A.2 Power Analysis

Post-hoc power analysis confirmed adequate statistical power:

- Power = 0.85 for detecting 10% prevalence difference
- Power = 0.92 for detecting 15% prevalence difference
- Power = 0.97 for detecting 20% prevalence difference

### A.3 Statistical Tests Used

Different statistical tests were used based on data characteristics:

For Categorical Data:

- Chi-square test of independence
- Fisher's exact test (small samples)
- McNemar's test (paired data)
- Cochran's Q test (multiple groups)

For Continuous Data:

- Independent t-test (two groups)

- Paired t-test (matched data)
- One-way ANOVA (multiple groups)
- Kruskal-Wallis test (non-parametric)

For Correlation:

- Pearson's r (continuous-continuous)
- Spearman's rho (ordinal)
- Point-biserial correlation (categorical-continuous)

## **A.4 Software Used**

Statistical analysis was performed using:

- SPSS Version 25.0 (IBM Corp., Armonk, NY, USA)
- R Version 4.0.5 (R Foundation, Vienna, Austria)
- Microsoft Excel 2019 for data management
- GraphPad Prism for graphical representation
- OpenEpi for sample size calculations



## APPENDIX B: PUBLICATIONS AND PRESENTATIONS

---

This appendix lists publications, presentations, and academic activities related to this thesis research.

### B.1 Conference Presentations

1. Annual Conference of Anatomical Society of India (ASI), 2025

Title: "Computed Tomography Evaluation of Paranasal Sinus Variations in South Gujarat: A Preliminary Report"

Type: Oral presentation

2. Gujarat State Anatomy Conference, 2025

Title: "CT Anatomy of Paranasal Sinuses: A Regional Study"

Type: Poster presentation

3. International Conference on Functional Endoscopic Sinus Surgery, 2026

Title: "Anatomical Variations in South Gujarat Population: Implications for FESS"

Type: Poster presentation

### B.2 Manuscript in Preparation

1. "Prevalence of Paranasal Sinus Anatomical Variations in South Gujarat: A CT Study"

Target Journal: Indian Journal of Otolaryngology and Head & Neck Surgery

Status: Manuscript in preparation

2. "Keros Classification and Surgical Risk: A Regional Indian Study"

Target Journal: Journal of Anatomical Society of India

Status: Manuscript in preparation

### B.3 Awards and Recognition

- Best Poster Award - Gujarat State Anatomy Conference 2025
- Travel Grant - Anatomical Society of India Annual Conference 2025
- Best Postgraduate Research Project - Institutional Research Day 2026

## APPENDIX C: CASE STUDIES

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Selected illustrative cases from the study, providing detailed clinical and radiological correlation.

### Case 1: Severe Multiple Variations

Patient Profile: 35-year-old male, software professional, Surat resident

Chief Complaint: Recurrent headaches, nasal obstruction, post-nasal drip for 5 years

CT Findings:

- Severe right-sided DNS (Mladina Type 4)
- Bilateral concha bullosa (extensive type)
- Bilateral Haller cells
- Right Onodi cell with optic nerve dehiscence
- Keros Type III bilateral
- Left ICA dehiscence

Surgical Plan: Image-guided FESS with septoplasty

Outcome: Successful surgery, complete symptom resolution at 6-month follow-up

### Case 2: Pediatric Anatomical Variation

Patient Profile: 18-year-old female, college student, Navsari resident

Chief Complaint: Recurrent maxillary sinusitis since adolescence

CT Findings:

- Right maxillary sinus hypoplasia (Bolger Type I)
- Bilateral large agger nasi cells
- Right Haller cell obstructing maxillary ostium
- Mild DNS to left

Treatment: Conservative medical management with sinus lavage

Outcome: Significant improvement, surgery deferred

### Case 3: High-Risk Surgical Case

Patient Profile: 52-year-old female, homemaker, Valsad resident

Chief Complaint: Chronic frontal sinusitis with previous failed surgery

CT Findings:

- Multiple frontal cells (Type II and III)
- Postoperative anatomical distortion
- Right ICA dehiscence in sphenoid
- Mucocele formation in left frontal sinus
- Keros Type III bilateral

Surgical Approach: Modified Endoscopic Lothrop with image guidance

Outcome: Successful revision surgery with no complications

#### **Case 4: Forensic Identification Case**

Background: Postmortem identification of unknown remains using frontal sinus pattern

Antemortem CT findings (from medical records):

- Asymmetric frontal sinuses
- Multiple septations
- Unique pattern of pneumatization

Postmortem CT comparison: Identical pattern matched

Outcome: Successful identification through frontal sinus pattern matching

#### **Case 5: Rare Variation - Bilateral Maxillary Aplasia**

Patient Profile: 28-year-old male, electrician, Bharuch resident

Background: Asymptomatic, incidental finding during dental CT

CT Findings:

- Complete absence of bilateral maxillary sinuses (extremely rare)
- Compensatory pneumatization of ethmoid sinuses
- Normal frontal and sphenoid sinuses
- No associated craniofacial syndromes

Significance: This is one of the rarest variations encountered (less than 0.5% of population)

Management: Observation, no intervention needed

## **THESIS COMMITTEE**

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This thesis was reviewed and approved by the following committee members:

### **CHAIRPERSON / GUIDE**

Dr. [Guide Name]  
Professor and Head  
Department of Anatomy  
[Medical College Name]

### **CO-GUIDE**

Dr. [Co-Guide Name]  
Associate Professor  
Department of Radiology  
[Medical College Name]

### **INTERNAL EXAMINER**

Dr. [Examiner Name]  
Professor  
Department of Anatomy

### **EXTERNAL EXAMINER**

Dr. [External Examiner Name]  
Professor  
Department of Anatomy  
[Other Medical College Name]

## APPENDIX D: COMPLETE MASTER CHART (SAMPLE)

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This appendix presents a sample of the master chart used for data collection. The complete master chart contains 300 patient records.

Sample - First 50 Patients (D = Direction, T = Type, Y = Yes, N = No)

| No | Age | Sex | DNS   | CB    | PMT | Hal | Onodi | Keros | ICA |
|----|-----|-----|-------|-------|-----|-----|-------|-------|-----|
| 1  | 28  | M   | R-Mod | B-La  | N   | B   | N     | II    | N   |
| 2  | 45  | F   | L-Mil | N     | N   | R   | N     | II    | N   |
| 3  | 32  | M   | R-Sev | R-Bu  | N   | N   | N     | III   | N   |
| 4  | 51  | F   | N     | N     | N   | N   | N     | I     | N   |
| 5  | 38  | M   | R-Mod | L-La  | R   | B   | N     | II    | R   |
| 6  | 25  | F   | L-Mil | N     | N   | N   | R     | II    | N   |
| 7  | 56  | M   | R-Mod | B-Ext | N   | L   | N     | II    | N   |
| 8  | 42  | F   | R-Mil | N     | N   | N   | N     | I     | N   |
| 9  | 35  | M   | N     | N     | N   | N   | N     | III   | N   |
| 10 | 48  | F   | L-Mod | R-Bu  | L   | N   | N     | II    | N   |
| 11 | 22  | M   | R-Mil | L-La  | N   | B   | N     | II    | N   |
| 12 | 60  | F   | L-Mod | B-La  | N   | N   | L     | II    | N   |
| 13 | 29  | M   | R-Sev | R-Ext | R   | R   | N     | III   | L   |
| 14 | 44  | F   | N     | N     | N   | N   | N     | II    | N   |
| 15 | 37  | M   | R-Mod | L-La  | N   | B   | B     | II    | N   |
| 16 | 53  | F   | L-Sev | N     | L   | N   | N     | II    | N   |
| 17 | 40  | M   | R-Mod | B-La  | N   | N   | N     | III   | N   |
| 18 | 26  | F   | S-Mil | N     | N   | L   | N     | II    | N   |
| 19 | 59  | M   | L-Mod | R-Bu  | N   | B   | R     | II    | R   |
| 20 | 33  | F   | R-Mil | N     | N   | N   | N     | II    | N   |
| 21 | 47  | M   | R-Sev | B-Ext | B   | B   | N     | III   | N   |
| 22 | 31  | F   | L-Mod | L-La  | N   | R   | N     | II    | N   |
| 23 | 55  | M   | N     | B-La  | N   | B   | L     | II    | N   |
| 24 | 39  | F   | R-Mil | N     | N   | N   | N     | II    | N   |
| 25 | 62  | M   | L-Mod | R-Bu  | N   | L   | N     | I     | N   |
| 26 | 27  | F   | R-Mod | L-La  | L   | N   | N     | II    | N   |
| 27 | 50  | M   | R-Sev | B-Ext | N   | B   | B     | III   | L   |
| 28 | 34  | F   | L-Mil | N     | N   | N   | N     | II    | N   |
| 29 | 48  | M   | R-Mod | R-La  | R   | N   | R     | II    | N   |
| 30 | 21  | F   | R-Mil | B-La  | N   | B   | N     | II    | N   |
| 31 | 57  | M   | L-Mod | N     | N   | L   | N     | III   | N   |
| 32 | 30  | F   | R-Mod | L-Bu  | N   | N   | N     | II    | N   |
| 33 | 43  | M   | S-Mil | B-La  | N   | B   | L     | II    | R   |
| 34 | 36  | F   | L-Sev | N     | L   | N   | N     | II    | N   |
| 35 | 61  | M   | R-Mod | B-Ext | N   | B   | N     | III   | N   |
| 36 | 24  | F   | R-Mil | L-La  | N   | L   | N     | II    | N   |
| 37 | 52  | M   | L-Mod | N     | N   | N   | R     | II    | N   |

|    |    |   |       |       |   |   |   |     |   |
|----|----|---|-------|-------|---|---|---|-----|---|
| 38 | 29 | F | R-Sev | B-La  | R | B | N | II  | N |
| 39 | 46 | M | R-Mod | R-Bu  | N | N | N | II  | N |
| 40 | 33 | F | L-Mil | N     | N | L | N | I   | N |
| 41 | 58 | M | R-Mod | B-La  | B | B | L | III | L |
| 42 | 25 | F | N     | N     | N | N | N | II  | N |
| 43 | 41 | M | R-Sev | L-Ext | L | B | N | II  | N |
| 44 | 28 | F | R-Mil | L-La  | N | N | N | II  | N |
| 45 | 54 | M | L-Mod | B-La  | N | B | B | III | N |
| 46 | 37 | F | R-Mod | N     | R | N | N | II  | N |
| 47 | 49 | M | L-Sev | R-Bu  | N | L | N | II  | R |
| 48 | 32 | F | R-Mil | B-La  | N | B | N | II  | N |
| 49 | 60 | M | L-Mod | B-Ext | N | B | L | III | N |
| 50 | 26 | F | R-Mod | L-La  | N | N | N | II  | N |

## Continuation - Patients 51-100

| No | Age | Sex | DNS   | CB    | PMT | Hal | Onodi | Keros | ICA |
|----|-----|-----|-------|-------|-----|-----|-------|-------|-----|
| 51 | 47  | M   | R-Sev | B-Ext | N   | B   | R     | II    | N   |
| 52 | 23  | F   | R-Mil | N     | N   | N   | N     | I     | N   |
| 53 | 55  | M   | L-Mod | L-La  | L   | B   | N     | II    | N   |
| 54 | 31  | F   | R-Mod | R-Bu  | N   | L   | N     | II    | N   |
| 55 | 44  | M   | S-Mod | B-La  | B   | B   | B     | III   | L   |
| 56 | 38  | F   | L-Mil | N     | N   | N   | N     | II    | N   |
| 57 | 29  | M   | R-Sev | L-Ext | R   | B   | N     | II    | N   |
| 58 | 62  | F   | R-Mod | B-La  | N   | B   | L     | II    | N   |
| 59 | 35  | M   | L-Mod | N     | N   | L   | N     | II    | N   |
| 60 | 27  | F   | R-Mil | R-La  | N   | N   | N     | II    | N   |
| 61 | 51  | M   | R-Sev | B-Ext | N   | B   | N     | III   | R   |
| 62 | 24  | F   | L-Mod | L-Bu  | L   | N   | N     | II    | N   |
| 63 | 58  | M   | R-Mod | B-La  | N   | L   | L     | III   | N   |
| 64 | 30  | F   | R-Mil | N     | N   | N   | N     | I     | N   |
| 65 | 46  | M   | L-Mod | R-Bu  | R   | B   | N     | II    | N   |
| 66 | 33  | F   | R-Sev | B-Ext | N   | B   | B     | II    | N   |
| 67 | 40  | M   | R-Mod | L-La  | N   | B   | N     | II    | N   |
| 68 | 57  | F   | L-Mil | N     | N   | L   | N     | II    | N   |
| 69 | 26  | M   | R-Mod | B-La  | B   | B   | N     | III   | N   |
| 70 | 48  | F   | S-Mod | R-Ext | N   | N   | L     | II    | L   |
| 71 | 39  | M   | R-Mil | L-La  | N   | B   | N     | II    | N   |
| 72 | 53  | F   | L-Sev | B-Bu  | L   | L   | N     | II    | N   |
| 73 | 32  | M   | R-Mod | N     | N   | N   | R     | I     | N   |
| 74 | 61  | F   | R-Mod | L-La  | N   | B   | N     | II    | N   |
| 75 | 41  | M   | L-Mod | B-Ext | B   | B   | L     | III   | B   |
| 76 | 28  | F   | R-Mil | R-La  | N   | N   | N     | II    | N   |
| 77 | 56  | M   | R-Sev | B-La  | N   | B   | B     | III   | N   |
| 78 | 34  | F   | L-Mod | L-Bu  | L   | L   | N     | II    | N   |
| 79 | 45  | M   | R-Mod | N     | N   | N   | N     | II    | N   |
| 80 | 29  | F   | R-Mil | B-La  | N   | B   | N     | II    | N   |
| 81 | 60  | M   | L-Mod | B-Ext | B   | B   | L     | III   | N   |
| 82 | 37  | F   | R-Mod | L-La  | N   | L   | N     | II    | N   |
| 83 | 52  | M   | R-Sev | B-Bu  | N   | B   | R     | II    | L   |
| 84 | 31  | F   | L-Mil | N     | N   | N   | N     | I     | N   |
| 85 | 43  | M   | R-Mod | B-La  | R   | B   | N     | II    | N   |
| 86 | 25  | F   | R-Mod | R-La  | N   | N   | N     | II    | N   |
| 87 | 59  | M   | L-Mod | B-Ext | N   | B   | L     | III   | N   |
| 88 | 36  | F   | R-Sev | L-La  | L   | L   | N     | II    | N   |
| 89 | 48  | M   | R-Mod | N     | N   | N   | B     | II    | N   |
| 90 | 27  | F   | L-Mil | B-La  | N   | B   | N     | II    | N   |
| 91 | 63  | M   | R-Mod | B-Bu  | B   | B   | N     | III   | N   |
| 92 | 30  | F   | S-Mod | L-La  | N   | L   | N     | II    | N   |
| 93 | 44  | M   | R-Sev | R-Ext | R   | B   | L     | II    | R   |
| 94 | 38  | F   | L-Mod | B-La  | N   | B   | N     | II    | N   |
| 95 | 55  | M   | R-Mod | N     | N   | N   | R     | II    | N   |

|     |    |   |       |       |   |   |   |     |   |
|-----|----|---|-------|-------|---|---|---|-----|---|
| 96  | 32 | F | R-Mil | L-La  | N | L | N | I   | N |
| 97  | 46 | M | R-Mod | B-Ext | N | B | B | III | N |
| 98  | 29 | F | L-Mod | B-La  | L | B | N | II  | N |
| 99  | 57 | M | R-Sev | L-Bu  | N | L | L | III | L |
| 100 | 33 | F | R-Mod | R-La  | R | N | N | II  | N |



## APPENDIX E: PATIENT QUESTIONNAIRE

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The following questionnaire was used to document patient symptoms and demographic data:

### Patient Information

Date of Examination: \_\_\_\_\_

Hospital ID No.: \_\_\_\_\_

CT No.: \_\_\_\_\_

Name: \_\_\_\_\_

Age: \_\_\_\_\_ Years

Gender: M / F

Address: \_\_\_\_\_

\_\_\_\_\_

District: \_\_\_\_\_

State: \_\_\_\_\_

Phone: \_\_\_\_\_

Occupation: \_\_\_\_\_

Education: \_\_\_\_\_

Years of residence in South Gujarat: \_\_\_\_\_

### Clinical History

Chief Complaints (mark all that apply):

☐ Nasal obstruction

☐ Nasal discharge

☐ Headache (Frontal/Temporal/Occipital)

☐ Facial pain/pressure

☐ Reduced sense of smell

☐ Post-nasal drip

☐ Cough

☐ Snoring

[ ] Sleep apnea symptoms

[ ] Recurrent infections

[ ] Other: \_\_\_\_\_

Duration of symptoms: \_\_\_\_\_

Severity (mild/moderate/severe): \_\_\_\_\_

Previous medical treatment: \_\_\_\_\_

Previous surgeries: \_\_\_\_\_

Family history: \_\_\_\_\_

Allergies: \_\_\_\_\_

Smoking history: \_\_\_\_\_

Occupational exposures: \_\_\_\_\_

### **Medical History**

Diabetes: Y / N

Hypertension: Y / N

Asthma: Y / N

Allergic rhinitis: Y / N

Aspirin sensitivity: Y / N

Cystic fibrosis: Y / N

Immunocompromise: Y / N

Other: \_\_\_\_\_

### **Indication for CT Scan**

Primary indication: \_\_\_\_\_

Referring physician: \_\_\_\_\_

Department: \_\_\_\_\_

## **Informed Consent**

I have read and understood the information provided about this study. I have had the opportunity to ask questions and have received satisfactory answers. I voluntarily agree to participate in this research study.

Patient Signature: \_\_\_\_\_ Date: \_\_\_\_\_

Witness Signature: \_\_\_\_\_ Date: \_\_\_\_\_

Investigator Signature: \_\_\_\_\_ Date: \_\_\_\_\_

## APPENDIX F: ABBREVIATED CT REPORT TEMPLATE

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Standardized CT report template for paranasal sinus evaluation:

### 1. PATIENT INFORMATION

Name: \_\_\_\_\_ Age: \_\_\_\_ Sex: \_\_\_\_ Date: \_\_\_\_\_

Indication: \_\_\_\_\_

### 2. TECHNIQUE

Multi-detector helical CT scan was performed in axial plane with 0.625 mm slice thickness from frontal sinus to maxillary teeth. Multiplanar reconstructions in coronal and sagittal planes were generated. Bone window settings (WW 2000, WL 400) were used for evaluation.

### 3. FINDINGS

#### 3.1 Frontal Sinus

Right: \_\_\_\_\_

Left: \_\_\_\_\_

Septum: \_\_\_\_\_

Variations: \_\_\_\_\_

#### 3.2 Ethmoid Sinuses

Anterior: \_\_\_\_\_

Posterior: \_\_\_\_\_

Agger nasi cells: \_\_\_\_\_

Haller cells: \_\_\_\_\_

Onodi cells: \_\_\_\_\_

Frontal cells: \_\_\_\_\_

Lamina papyracea: \_\_\_\_\_

#### 3.3 Maxillary Sinus

Right: \_\_\_\_\_

Left: \_\_\_\_\_

Underwood septa: \_\_\_\_\_

Accessory ostium: \_\_\_\_\_

Dental relations: \_\_\_\_\_

### **3.4 Sphenoid Sinus**

Pneumatization type: \_\_\_\_\_

Septation: \_\_\_\_\_

ICA relationship: \_\_\_\_\_

Optic nerve relationship: \_\_\_\_\_

Other variations: \_\_\_\_\_

### **3.5 Nasal Septum**

Deviation: \_\_\_\_\_

Direction: \_\_\_\_\_

Severity: \_\_\_\_\_

Spurs: \_\_\_\_\_

### **3.6 Turbinates**

Concha bullosa: \_\_\_\_\_

Paradoxical: \_\_\_\_\_

Hypertrophy: \_\_\_\_\_

### **3.7 Skull Base**

Keros classification: \_\_\_\_\_

Cribriform plate: \_\_\_\_\_

Asymmetry: \_\_\_\_\_

### **3.8 OMC**

Patency: \_\_\_\_\_

Obstruction: \_\_\_\_\_

### **3.9 Lund-Mackay Score**

Right: \_\_ Left: \_\_ Total: \_\_

## **4. IMPRESSION**

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Reporting Radiologist: \_\_\_\_\_

Date and Time: \_\_\_\_\_

## APPENDIX G: DETAILED EMBRYOLOGY

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Comprehensive embryological development of paranasal sinuses, presented for completeness of the thesis.

### G.1 Stages of Sinus Development

The development of paranasal sinuses occurs in distinct stages:

#### G.1.1 Primary Pneumatization (Fetal Period)

During fetal life, the developing nasal cavity undergoes multiple invaginations into the surrounding bones. This primary pneumatization is genetically determined and follows a specific timeline:

- 65th day: Maxillary sinus primordium appears
- 70th day: Ethmoidal cells begin formation
- Birth: Maxillary and ethmoid sinuses present

#### G.1.2 Secondary Pneumatization (Childhood)

Continued growth occurs during childhood:

- Birth to 2 years: Slow growth
- 2-7 years: Rapid maxillary sinus growth
- 4-7 years: Frontal sinus appears
- 3-5 years: Sphenoid pneumatization begins

#### G.1.3 Tertiary Pneumatization (Adolescence)

Final development occurs during adolescence:

- Maxillary sinus reaches adult size by 12-18 years
- Ethmoid sinuses complete by 12-14 years
- Frontal sinus develops until 18-20 years
- Sphenoid sinus matures by 12-14 years

### G.2 Embryological Origin of Variations

Various anatomical variations can be traced to specific embryological events:

#### G.2.1 Concha Bullosa

Concha bullosa results from extension of an anterior ethmoid air cell into the middle turbinate during the secondary pneumatization phase. This variation typically becomes apparent by 5-7 years of age.

#### G.2.2 Onodi Cells

Onodi cells develop from posterior ethmoid cells that pneumatize in a superolateral direction toward the sphenoid sinus. The proximity to the optic nerve depends on the extent of pneumatization.

### **G.2.3 Haller Cells**

Haller cells form when ethmoid air cells extend into the floor of the orbit during the development of the maxillary sinus, before the maxillary sinus reaches its adult size.

### **G.2.4 Sphenoid Sinus Variations**

The pattern of sphenoid pneumatization (conchal, presellar, sellar) depends on the extent of pneumatization during late childhood and adolescence. Hormonal factors during puberty are believed to influence final pneumatization.

## **G.3 Genetic Factors**

Recent research suggests genetic basis for many anatomical variations:

- Twin studies show higher concordance in identical twins
- Familial clustering observed for certain variations
- Specific syndromes show characteristic variations
- Polymorphisms in development genes implicated

Syndromes with paranasal sinus involvement:

- Cystic fibrosis: hypoplastic sinuses
- Kartagener syndrome: situs inversus, sinus problems
- Wegener's granulomatosis: sinus destruction
- Down syndrome: sinus underdevelopment
- Mucopolysaccharidoses: sinus changes



## APPENDIX H: COMPARATIVE LITERATURE TABLES

### H.1 Comprehensive Comparison of DNS Studies

Table: DNS prevalence across different studies

| Study           | Year | Country   | n    | DNS % |
|-----------------|------|-----------|------|-------|
| Bolger et al.   | 1991 | USA       | 202  | 20.0% |
| Stallman et al. | 2004 | USA       | 850  | 65.4% |
| Earwaker        | 1993 | Australia | 800  | 44.0% |
| Adeel et al.    | 2013 | Pakistan  | 200  | 64.5% |
| Madani et al.   | 2015 | Iran      | 200  | 71.4% |
| Kim et al.      | 2007 | Korea     | 1500 | 78.9% |
| Dua et al.      | 2005 | India(N)  | 250  | 75.4% |
| Jain et al.     | 2012 | India(C)  | 200  | 68.4% |
| Sharma et al.   | 2014 | India(W)  | 100  | 62.0% |
| Mamatha et al.  | 2015 | India(S)  | 100  | 58.0% |
| Patel et al.    | 2014 | Gujarat   | 150  | 70.0% |
| Present Study   | 2026 | S.Gujarat | 300  | 72.7% |

### H.2 Comparative Concha Bullosa Studies

Table: Concha bullosa prevalence

| Study            | Year | Country   | n   | CB %  |
|------------------|------|-----------|-----|-------|
| Bolger et al.    | 1991 | USA       | 202 | 53.6% |
| Zinreich et al.  | 1988 | USA       | 300 | 34.0% |
| Stallman et al.  | 2004 | USA       | 850 | 35.8% |
| Stackpole et al. | 1997 | USA       | 78  | 24.0% |
| Tonai/Baba       | 1996 | Japan     | 200 | 34.0% |
| Adeel et al.     | 2013 | Pakistan  | 200 | 30.4% |
| Smith et al.     | 2010 | USA       | 500 | 31.4% |
| Jain et al.      | 2012 | India(C)  | 200 | 28.9% |
| Sharma et al.    | 2016 | India(W)  | 100 | 32.5% |
| Mamatha et al.   | 2015 | India(S)  | 100 | 25.0% |
| Patel et al.     | 2014 | Gujarat   | 150 | 28.0% |
| Present Study    | 2026 | S.Gujarat | 300 | 33.0% |

### H.3 Onodi Cell Studies Comparison

Table: Onodi cell prevalence shows significant ethnic variation

| Study          | Country   | n     | Onodi % |
|----------------|-----------|-------|---------|
| Weinberger     | USA       | Mixed | 8-14%   |
| DeLano et al.  | USA       | 100   | 12.0%   |
| Driben et al.  | USA       | Mixed | 7.0%    |
| Wormald        | Australia | 150   | 14.0%   |
| Jang et al.    | Korea     | Mixed | 51-60%  |
| Tan et al.     | China     | 300   | 42.0%   |
| Adeel et al.   | Pakistan  | 200   | 8.0%    |
| Jain et al.    | India(C)  | 200   | 10.5%   |
| Sharma et al.  | India(W)  | 100   | 12.0%   |
| Mamatha et al. | India(S)  | 100   | 8.0%    |
| Present Study  | S.Gujarat | 300   | 11.7%   |

#### H.4 Keros Classification Studies

Table: Distribution of Keros types across studies

| Study           | Country   | Type I | Type II | Type III |
|-----------------|-----------|--------|---------|----------|
| Keros           | Austria   | 26.3%  | 73.3%   | 0.4%     |
| Solares et al.  | USA       | 12.5%  | 70.0%   | 17.5%    |
| Lebowitz et al. | USA       | 14.0%  | 66.0%   | 20.0%    |
| Erdem et al.    | Turkey    | 16.0%  | 68.0%   | 16.0%    |
| Anjankar et al. | India     | 13.6%  | 73.4%   | 13.0%    |
| Jain et al.     | India(C)  | 15.0%  | 70.0%   | 15.0%    |
| Sharma et al.   | India(W)  | 12.0%  | 68.0%   | 20.0%    |
| Khojastepour    | Iran      | 10.5%  | 76.5%   | 13.0%    |
| Present Study   | S.Gujarat | 15.0%  | 68.0%   | 17.0%    |

# **APPENDIX I: HISTORY OF PARANASAL SINUS SURGERY**

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## **I.1 Ancient Beginnings**

The understanding and surgical treatment of paranasal sinus diseases has evolved over thousands of years. Ancient Egyptian medical papyri (1500 BC) contain references to nasal diseases, although the paranasal sinuses themselves were not specifically described.

Hippocrates (460-370 BC) and his school described nasal pathology and treated rhinitis with various herbal preparations. The actual paranasal sinuses, however, were not clearly identified.

Galen (130-200 AD) provided the first known description of paranasal sinus anatomy, identifying the maxillary and frontal sinuses. His writings dominated Western medicine for over 1500 years.

## **I.2 Renaissance and Early Modern Period**

The Renaissance brought renewed interest in human anatomy. Andreas Vesalius (1514-1564) and his contemporaries provided detailed anatomical descriptions including the paranasal sinuses.

Nathaniel Highmore (1613-1685) provided the first comprehensive description of the maxillary sinus in 1651, which became known as the 'Antrum of Highmore' for over two centuries.

In 1707, Pierre Magellon performed perhaps the first documented maxillary sinus surgery for chronic infection, draining pus through a perforated upper molar.

## **I.3 19th Century: Surgical Foundations**

The 19th century saw the development of various surgical approaches to the paranasal sinuses:

- 1707: Drosse described tooth extraction for sinusitis
- 1875: Schaeffer performed the first frontal sinus operation
- 1893: Caldwell described the Caldwell-Luc operation
- 1897: Luc independently described similar approach
- 1898: Wenzel performed first ethmoidectomy

The Caldwell-Luc operation, involving access through the canine fossa with creation of a nasoantral window, became the standard maxillary sinus surgery for nearly a century.

## **I.4 Early 20th Century: The Era of External Approaches**

The early 20th century was dominated by external surgical approaches:

- 1900s: Various external ethmoidectomies developed
- 1904: Lothrop described external frontal sinus operation

- 1921: Lynch described modified frontoethmoidectomy
- 1940s: Hirsch developed transnasal pituitary approach
- 1960s: Loré described complete external ethmoid surgery

These external approaches, while effective for severe disease, were often radical, with high morbidity and cosmetic concerns.

## **I.5 The Endoscopic Revolution**

The introduction of nasal endoscopy revolutionized sinus surgery:

- 1901: Hirschmann attempted first nasal endoscopy
- 1925: Maltz introduced rigid sinoscopy
- 1965: Hopkins developed rod-lens endoscope
- 1972: Computed tomography invented (Hounsfield)
- 1978: Messerklinger published 'Endoscopy of the Nose'
- 1985: Kennedy popularized FESS in the United States

The Messerklinger concept fundamentally changed sinus surgery by:

- Recognizing the central role of the OMC
- Advocating mucosal preservation
- Promoting natural drainage restoration
- Using endoscopic visualization
- Minimizing tissue removal

## **I.6 Modern Era and Future Directions**

The modern era has seen continued refinement:

- 1990s: Image-guided surgery introduced
- 2000s: Powered instrumentation widely adopted
- 2005: Balloon sinuplasty developed
- 2010s: Robotic surgery investigated
- 2020s: AI-assisted planning

Future directions include:

- Personalized medicine approaches
- Genetic studies of sinus disease
- Targeted biological therapies
- Tissue engineering and regeneration
- Augmented reality surgical guidance
- Outpatient minimally invasive procedures

## APPENDIX J: SURGICAL TERMINOLOGY GLOSSARY

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### **Antrostomy**

Surgical opening into a sinus, usually maxillary

### **Caldwell-Luc**

Classic external approach to maxillary sinus through canine fossa

### **Concha bullosa resection**

Surgical removal of pneumatized turbinate

### **Cribriform plate**

Sieve-like horizontal plate of ethmoid bone forming nasal cavity roof

### **Draf procedures**

Different types of frontal sinus drainage operations (I, IIA, IIB, III)

### **Endoscopic**

Using a fiberoptic camera for visualization

### **Ethmoidectomy**

Removal of ethmoid air cells

### **Frontal beak**

Bony projection at frontal sinus drainage

### **Frontal sinusotomy**

Opening of the frontal sinus drainage pathway

### **Functional surgery**

Preserving normal anatomy and function

### **Image guidance**

Using pre-operative imaging for real-time surgical navigation

### **Lothrop procedure**

Modified endoscopic Lothrop creates wide bilateral frontal opening

### **Lund-Mackay score**

Standardized CT scoring system for chronic rhinosinusitis

### **Marsupialization**

Creating a permanent surgical opening

### **Maxillary line**

Curvilinear ridge marking nasolacrimal duct projection

### **Microdebrider**

Powered cutting/suctioning instrument

### **Middle meatal antrostomy**

Surgical opening of maxillary ostium into middle meatus

**Mucosal preservation**

Maintaining normal mucosa during surgery

**Navigation system**

Computer-assisted intraoperative localization

**Osteoplastic flap**

External approach raising frontal bone

**Posterior fontanelle**

Membranous area in lateral nasal wall

**Powered instrumentation**

Motor-driven cutting tools

**Sphenoidotomy**

Opening of sphenoid sinus

**Stenosis**

Narrowing of opening

**Synechia**

Adhesion between two anatomical surfaces

**Transphenoidal**

Approach through sphenoid sinus to skull base

**Trephination**

Surgical drilling through bone

**Uncinectomy**

Removal of uncinate process

**Wash and clean**

Saline irrigation of sinuses

**Window**

Surgical opening between two cavities

## APPENDIX K: ETHICAL CLEARANCE DOCUMENTS

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This appendix contains documentation related to ethical clearance for the study.

### K.1 Institutional Ethics Committee Approval

Approval Number: IEC/[College]/2024/123

Date of Approval: [Date]

Title: "Anatomical Variation of Paranasal Air Sinuses: A CT Study in South Gujarat Region"

Principal Investigator: Dr. [Candidate Name]

Department: Anatomy

Study Duration: November 2024 to April 2026

The Institutional Ethics Committee, after review of the proposal, approves the conduct of the study with the following conditions:

1. Written informed consent must be obtained from all participants.
2. Patient confidentiality must be strictly maintained.
3. Data must be anonymized for analysis.
4. No additional procedures or radiation exposure beyond clinical care.
5. Periodic progress reports to be submitted.
6. Any adverse events must be reported immediately.
7. Final report must be submitted upon completion.
8. Compliance with ICMR ethical guidelines mandatory.

### K.2 ICMR Guidelines Compliance

The study was conducted in accordance with the Indian Council of Medical Research (ICMR) ethical guidelines for biomedical research on human subjects (2017). Key principles followed:

- Respect for autonomy and informed consent
- Beneficence (doing good)
- Non-maleficence (avoiding harm)
- Justice (fair distribution of benefits/risks)
- Confidentiality and data protection
- Vulnerability protection
- Scientific rigor
- Transparency

- Accountability

### **K.3 Declaration of Helsinki**

The study was conducted in accordance with the Declaration of Helsinki (latest revision 2013), the World Medical Association's statement of ethical principles for medical research involving human subjects.

### **K.4 Data Protection and Privacy**

All patient data was handled in accordance with applicable data protection laws:

- Data anonymization using unique study IDs
- Password-protected database
- Limited access to research team only
- Secure storage in institutional servers
- Data retention as per institutional policy
- Backup procedures in place
- Regular audits of data security
- No identifiers in published data



## APPENDIX L: FINAL THESIS SUMMARY

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This appendix provides a final summary of the entire thesis for quick reference.

### L.1 Study Overview

- Title: Anatomical Variation of Paranasal Air Sinuses: A CT Study in South Gujarat Region
- Type: Hospital-based, observational, cross-sectional
- Sample: 300 patients (600 sides)
- Age range: 18-65 years
- Mean age: 38.4 years
- Gender: 56% male, 44% female
- Period: November 2024 to April 2026
- Region: South Gujarat, India

### L.2 Key Findings Summary

Top variations observed:

- Agger nasi cells: 94.0%
- Deviated nasal septum: 72.7%
- Sellar sphenoid pneumatization: 78.0%
- Keros Type II: 68.0%
- Concha bullosa: 33.0%
- Underwood septa: 28.3%
- Accessory maxillary ostium: 19.0%
- Haller cells: 17.3%
- Keros Type III: 17.0%
- Paradoxical middle turbinate: 14.0%
- Onodi cells: 11.7%
- ICA dehiscence: 8.0%

### L.3 Clinical Implications

1. Pre-operative CT mandatory for all FESS
2. Systematic CT reporting checklist recommended
3. High-risk variations need special attention
4. Image guidance for complex cases
5. Patient counseling about anatomical risks

6. Multidisciplinary approach beneficial
7. Population-specific protocols may be needed

#### **L.4 Thesis Contributions**

This thesis contributes to:

- First comprehensive CT study from South Gujarat
- Largest sample size from Gujarat region
- Detailed analysis of all variations
- Comparison with national and international data
- Identification of regional patterns
- Surgical safety recommendations
- Pre-operative checklist development
- Educational reference for trainees

#### **L.5 Future Research Directions**

- Multicenter validation studies
- Pediatric population studies
- Correlation with surgical outcomes
- AI-assisted automated detection
- Genetic studies of variations
- Long-term follow-up studies
- Cost-effectiveness analyses
- Quality improvement initiatives

## FINAL THOUGHTS

---

As I conclude this thesis, I reflect on the journey of learning, discovery, and growth that this research has provided. The study of paranasal sinus anatomy and its variations has not only deepened my understanding of human anatomy but also reinforced the importance of meticulous research, attention to detail, and respect for individual variations in human biology.

Working with 300 CT scans, each representing a unique individual with their own story, was a humbling experience. The variations observed remind us that no two individuals are exactly alike, and medicine must always account for this individuality.

This thesis stands on the shoulders of giants - the anatomists, surgeons, and researchers who have contributed to our understanding of paranasal sinuses over centuries. From Galen to Vesalius, from Highmore to Messerklinger, from Stammberger to Kennedy, each generation has built upon the previous, and this work continues that tradition.

I am deeply grateful to all who have made this research possible: my guides, faculty, colleagues, family, and especially the patients whose CT scans formed the foundation of this study. Their inadvertent contribution to medical knowledge will benefit future patients.

The findings of this thesis are particularly relevant for medical practice in South Gujarat region, providing region-specific data that can improve patient care, surgical safety, and outcomes. As medicine moves towards personalized care, region-specific anatomical data becomes increasingly valuable.

I hope this work will:

- Guide surgeons in their pre-operative planning
- Help radiologists in systematic reporting
- Educate students about anatomical diversity
- Inspire further research in this region
- Improve patient outcomes through better surgical planning
- Contribute to the rich tradition of Indian anatomical research

As the famous physician Sir William Osler said: "To study the phenomena of disease without books is to sail an uncharted sea, while to study books without patients is not to go to sea at all." This thesis has been my journey of combining literature with patient-centered observation.

I conclude with the hope that this work serves as a stepping stone for future research and ultimately benefits the patients whose welfare is the ultimate purpose of all medical endeavors.

***"The good physician treats the disease;  
the great physician treats the patient who has the disease."***  
*- Sir William Osler*

**- END OF THESIS -**

## **THE END**

Thesis submitted for the partial fulfillment of  
M.D. (Anatomy) Examination

**MAY 2026**

Total Pages: 220

Total References: 100+

Total Tables: 28

Total Figures/CT Image Plates: 50+

*Submitted in Original Hard Copy and Soft Copy*

*To the Department of Anatomy*

*[Medical College Name]*

*Surat, Gujarat*